It Was Already Dobbs:
Abortion Bans and Maternal Health

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Clinicians warned that the *Dobbs* abortion bans would harm maternal health, but ban states had restricted abortion for over a decade, possibly limiting a full ban's effect. Using claims on nearly 1 million insured pregnancies and a stacked difference-in-differences design, I find suggestive evidence that the bans raised fertility and the live-birth share of pregnancies. More deliveries brought more prenatal visits and a small rise in adverse pregnancy outcomes, with estimates ruling out increases above 4.9 percent. Other health and care outcomes changed little. Restoring the federal right to abortion alone may not meaningfully change maternal health for insured women.

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Between September 2021 and August 2023, fifteen U.S. states banned abortion at six weeks of gestation or earlier (Curhan, 2026). The Supreme Court enabled these laws with its June 2022 decision in *Dobbs v. Jackson Women's Health Organization*, which overturned the constitutional right to abortion recognized in *Roe v. Wade*. The decision was the largest change to abortion law in half a century, what Dench, Pineda-Torres and Myers (2024) call "the most profound transformation of the landscape of abortion access in 50 years." As the bans took effect, clinicians, medical societies, and journalists warned that the harm would reach beyond women seeking an elective abortion for reasons unrelated to their health (American College of Obstetricians and Gynecologists, 2022; Harris, 2022; Arey et al., 2022; MacDonald, Gershengorn and Ashana, 2022; Simmons-Duffin, 2022).

Three features of abortion care and of the bans drove this concern. First, the medications and procedures used to end a pregnancy are also the standard treatment for miscarriage, ectopic pregnancy, and several severe complications of pregnancy, so their provision could be mistaken for an abortion. Second, the line between an "elective" and a "medically necessary" abortion is often unclear. For instance, a pregnancy with a severe fetal anomaly may pose no imminent risk to the woman when the anomaly is diagnosed but can become riskier as it continues (Nambiar et al., 2025). Third, although the bans include exceptions for the woman's life and sometimes her health (Felix, Sobel and Salganicoff, 2024), they say little about how and when a physician should act in these ambiguous cases, and bans threaten physicians who violate them with severe penalties (Felix, Sobel and Salganicoff, 2025). Investigative reporting has since documented women with serious pregnancy complications sent home from emergency departments until their condition deteriorated enough to meet an exception (Grossman et al., 2023; Seitz, 2024; Presser and Surana, 2024). The prevailing expectation was therefore that the bans would mark a sharp break in the care pregnant women received and in their health.

However, the abortion bans enabled by *Dobbs* followed a decade of tightening restrictions on abortion in the same states that would ultimately go on to ban the procedure. By the start of 2022, many states that would later ban abortion already imposed waiting periods and regulations targeting abortion providers that closed clinics and lengthened travel distances (Guttmacher Institute, 2022, 2021*b*). In 2021 alone, states enacted 108 new restrictions, more than in any earlier year since *Roe* (Nash, 2021), and some of these rules already carried criminal penalties for physicians. Further, many forms of insurance had stopped paying for most abortions long before any formal ban. The Hyde Amendment has barred federal Medicaid funds from paying for abortion in most circumstances since 1977 (Salganicoff et al., 2024), and many of these states also limited coverage in private plans and in plans sold on the Affordable Care Act Marketplace (Thompson, 2019; Ranji

et al., 2025). Even before *Dobbs*, women in ban states paid outside insurance for most of the abortions they did obtain (Roberts et al., 2014), and increasingly obtained them out of state or by telehealth (Guttmacher Institute, 2023; Society of Family Planning, 2025). In these states, access to abortion likely already resembled a post-*Dobbs* environment well before *Dobbs*.

How the largest change in abortion law in half a century changed the maternal health landscape is therefore an empirical question of great interest and significance. This paper provides the first comprehensive evidence on this question by estimating the effect of abortion bans on maternal health, the timing and intensity of maternal health care, and pregnancy outcomes. From 350 million insurance claim lines, I construct pregnancy episodes that follow each woman from a year before conception through the postpartum period and record the diagnoses, procedures, and providers for services billed to her insurance. My core sample follows the nearly 1 million pregnancies of women continuously insured from 2020 through 2024. Using a stacked difference-in-differences design, I compare pregnancies in ban states with pregnancies in states that continuously protected abortion, before and after each state imposed its ban. I study three core sets of outcomes. First, outcomes that describe the ending of a pregnancy, the shares ending in a live birth, a miscarriage, or an abortion billed to insurance, show whether the bans changed how pregnancies ended. Second, maternal health outcomes (a composite of adverse pregnancy outcomes, severe maternal morbidity, hypertensive disorders of pregnancy, and preterm birth) test the warnings of harm to health directly. Third, timing and intensity of care outcomes (the days to the first visit, whether a delivery stay ran past the federal minimum, the length of pregnancy, and the number of visits) register any delayed or deferred care before it reached health. Finally, to understand why these outcomes moved as they did, I examine whether the bans changed fertility and the pre-pregnancy health of the women who became pregnant.

My stacked difference-in-differences estimates show that the bans caused little change in maternal health and in the timing and intensity of care, while modestly raising fertility and the share of pregnancies ending in a live birth. First, the live-birth share rose a suggestive 2.7 percent relative to protected states. Second, this shift accounts for the one health outcome that rises significantly. A composite of common adverse pregnancy outcomes, such as infection, preterm birth, and gestational diabetes, is 2.8 percent higher ($p = 0.01$), but because these outcomes are far more common in deliveries than in losses, the added live births alone account for most of this rise, and among live births the composite changes by -1.2 percent, ruling out increases larger than 0.6 percent. The most common complications moved even less, as hypertensive disorders change by 1.0 percent and preterm birth among live births by -0.9 percent, with intervals that rule out increases larger

than 3.4 and 2.6 percent. Changes in severe maternal morbidity, a rare but serious outcome at the center of the warnings, are also functionally null, at −2.9 percent, ruling out increases larger than 5.6 percent. Lastly, the timing of care changed little, as the days to the first visit change by 1.4 percent, with an interval that rules out delays longer than 3.8 percent. However, these estimates describe the average insured pregnancy, most of which are routine pregnancies carried to delivery or lost early. Among losses, the small group the warnings most concerned, severe maternal morbidity changes by 11.3 percent, but its interval reaches 33.4 percent, so I cannot rule out harm concentrated among the pregnancies whose complications call for urgent care or a termination.

These data also document a pre-ban landscape that helps explain why maternal health changed so little, and two facts stand out. First, abortion had already largely left the insurance system in these states. Before any formal ban, insurance-billed abortion ended about 1 percent of ban-state pregnancies, against 14 percent of protected-state pregnancies in 2020. Second, maternal health in ban states was already worse, as ban-state pregnancies more often carried a hypertensive disorder or an adverse pregnancy outcome such as infection or gestational diabetes, and even among live births more often ended preterm. These gaps persist after adjusting for women's demographic and health characteristics. To my knowledge, these patterns are new facts about pregnancy care and outcomes in the years leading up to *Dobbs*, and they are visible here because claims follow a pregnancy from its start to its ending, while the sources that commonly measure pregnancy outcomes, such as birth certificates and abortion surveillance records, do not.

Turning to the channels through which the bans could have acted, I find suggestive evidence that fertility rose. The birth rate among enrolled women is 3.7 percent higher relative to protected states ($p = 0.02$), with positive estimates in every subgroup, including 5.5 percent among women with a high-risk condition, although the subgroup intervals overlap. The pregnancy rate changes little, so the added births appear to come from more pregnancies reaching a live birth rather than from more conceptions, in line with the rise in the live-birth share. Despite this rise in births, the women who became pregnant after the bans were not measurably less healthy, as the share entering pregnancy with a chronic condition changed by 0.4 percent, with an interval that rules out increases larger than 2.6 percent.

I contribute to two active streams of literature. The first studies how abortion restrictions shape maternal health. Studies of the legalization of abortion around the time of *Roe* find that it lowered maternal mortality, by 30 to 50 percent among nonwhite women (Farin, Hoehn-Velasco and Pesko, 2024), and studies of later restrictions find that they worsened maternal health. Regulations targeting abortion providers raised hypertensive disorders of pregnancy (Gardner, 2024), sepsis

and critical illness rose among high-risk hospitalized pregnancies in Texas after its 2021 ban, in before-and-after comparisons within the state (Presser et al., 2025; Chen et al., 2026), and blood transfusions for pregnancy loss rose after its 2022 total ban (Nagle et al., 2025). Women denied a wanted abortion also report worse physical health and more financial distress in the years that follow (Ralph et al., 2019; Miller, Wherry and Foster, 2023). In Mexico, by contrast, harsher penalties in states where abortion was already illegal had small or null effects on women's health (Clarke and Mühlrad, 2021). Early evaluations of *Dobbs* find that pregnancy-associated mortality may have increased, although the estimates have wide intervals (Bell et al., 2026; Abraha et al., 2026; Stevenson and Root, 2024; Nuss et al., 2025), and that infant deaths rose, largely from congenital anomalies (Gemmill et al., 2024; Singh and Gallo, 2024; Gemmill et al., 2025) and most among Black infants in Texas (Caraher, 2024). I extend this literature to a broad set of health outcomes and pregnancy endings, measured in detailed claims that describe care from conception through the postpartum period. To my knowledge, this is the first comprehensive evidence on the effect of the post-*Dobbs* bans on a broad range of maternal health outcomes in a large sample of insured pregnancies. However, in contrast to some of this work, I find that the bans enabled by *Dobbs* changed maternal health little, a difference that my broader sample and my focus on the later, *Dobbs*-era restrictions may explain.

The second literature measures the fertility response to abortion restrictions. It generally finds that restrictions raise births modestly, and that the bans after *Dobbs* raised them by about 2 percent (Fischer, Royer and White, 2018; Venator and Fletcher, 2021; Myers, 2024; Dench, Myers and Pineda-Torres, 2025; Bell, Stuart and Gemmill, 2023; Caraher, 2024; Dench, Pineda-Torres and Myers, 2024; Bell et al., 2025). Restrictions short of a ban had already lowered abortion rates, most of all those aimed at the women seeking care rather than at providers (Caraher, 2026). My population, unlike the birth records these studies use, excludes uninsured women and women whose coverage changed between 2020 and 2024. In it, I find modest and suggestive evidence of a rise in births that partly continues earlier trends. Because claims also record pregnancies that do not end in a birth, I can measure the pregnancy rate, which vital statistics cannot, and separate pregnancies carried to term from added conceptions. Natality records often omit a woman's health before conception, but claims record it, so I can test whether the bans changed who became pregnant, as a ban can (Pop-Eleches, 2006).

I confront several threats to the validity of my design. First, women enter and leave my data for many reasons, some of which may correlate with treatment. Therefore, I restrict my analysis sample to women continuously enrolled from 2020 through 2024, to ensure a stable denominator

and follow women before and after pregnancy (Section II.D). However, this requirement may select women better able to circumvent a ban, for example by traveling, and such selection would lead the estimates to understate rather than overstate the effect of the bans. To check this restriction, in Supplemental Appendix A4 I use a weaker enrollment requirement and find similar health estimates, although the shift in endings there runs toward miscarriages rather than live births. I further show that the pregnancies in my sample broadly resemble all U.S. births on several demographic measures (Supplemental Appendix Table A7). Second, protected states expanded access to abortion over the same years that ban states continued to restrict it, so their paths before the bans could differ from those of ban states. To address this concern, I reweight protected states to match the pre-ban paths of ban states with synthetic difference-in-differences (Supplemental Appendix A8), and in Supplemental Appendix A9 I compare women near and far from protected-state care within ban states, a comparison that leaves out protected states entirely. Neither points toward harm to maternal health. Lastly, by raising fertility, the bans may change who becomes pregnant, toward women at higher or lower baseline risk. Because claims record each woman's health before conception, I measure this directly, and I find little evidence that a change in composition masks harm (Section VI.B).

Taken together, my results place the *Dobbs* bans in the environment of restrictions that preceded them. Maternal health in ban states was already worse than in protected states before the bans and changed little after them, while fertility and the share of pregnancies ending in a live birth rose modestly. These patterns are consistent with the earlier restrictions having already done much of what a ban could do to maternal health. For instance, Gardner (2024) finds that targeted regulations of abortion providers, which most later ban states had adopted, raised hypertensive disorders of pregnancy by 11 to 16 percent, while the 95 percent confidence interval for the effect of the bans on the same outcome runs from -1.5 to 3.4 percent. My design cannot fully separate this explanation from another, that insured women in ban states got around the bans by traveling or by telehealth, but under either explanation a ban added little to the restrictions these women already faced. However, my estimates do not rule out harm. Because they average over every insured pregnancy, they cannot exclude severe harm concentrated among the few percent of pregnancies with the complications the bans made hardest to treat, but they do bound how common such harm could have been. Finally, these findings suggest that returning the federal right to abortion alone may not meaningfully change the maternal health and care landscape for insured women.

The paper proceeds as follows. Section I describes the history of restrictions before and after *Dobbs* and how abortion restrictions could affect maternal health. Section II describes the data,

samples, and outcomes, Section III the pre-ban landscape, and Section IV the empirical strategy. Section V presents the effects on pregnancy endings, maternal health, and the timing and intensity of care, and Section VI the fertility channel. Section VII concludes.

I. Background and Expected Effects

A. Abortion Policy Before and After Dobbs

The history of abortion restrictions in states that enacted post-*Dobbs* abortion bans began long before *Dobbs*, and the restrictions grew tighter in the years leading up to the decision (Nash, 2021). By the start of 2022, every one of the fifteen states that would later ban abortion already imposed a gestational limit on abortion and required a parent's consent or notice before a minor could obtain an abortion, and all but Tennessee made women wait 18 to 72 hours between counseling and the procedure (Guttmacher Institute, 2022). Eleven states enforced targeted regulations of abortion providers, rules that go beyond what patient safety requires, often holding clinics to the standards of ambulatory surgical centers (Guttmacher Institute, 2021*b*). Studies of these rules find that, together, they closed clinics and lengthened the distances women traveled to reach a provider (Lindo et al., 2020; Smith et al., 2022; Lindo and Pineda-Torres, 2021; Myers and Ladd, 2020). Some rules already carried criminal penalties for physicians, as in Texas, where performing an abortion without the sonogram the state required had been a crime since 2011 (Texas State Law Library, 2026*a*,*b*).

Insurance coverage was minimal in states that ultimately banned abortion as well. Beyond the federal Hyde Amendment's limits on Medicaid (Salganicoff et al., 2024), none of the fifteen states with post-*Dobbs* bans used their own funds to pay for most medically necessary abortions for women on Medicaid (Guttmacher Institute, 2022), fourteen of the fifteen also restricted abortion coverage in plans sold on the health insurance exchanges, and six restricted it in every private plan written in the state (Guttmacher Institute, 2021*a*). As a result, insurance pays for only a minority of abortions in the United States (Roberts et al., 2014). Coverage also matters for access, because studies show that cost is a barrier to receiving an abortion, and in 2020 a patient paying out of pocket faced a median charge of $560 for a medication abortion (Upadhyay et al., 2022). Previous work has found that state restrictions on Medicaid funding reduced abortions, with the response concentrated among low-income women (Levine, Trainor and Zimmerman, 1996), and a review of this literature concludes that about one in four women who would have had a Medicaid-funded abortion give birth when that funding is unavailable (Henshaw et al., 2009).

In addition, several of these states had already passed near-total or early gestational bans, only to

see them enjoined under *Roe*, so the legal status of abortion in these states shifted from year to year, making the practice environment already potentially unstable and stress-inducing. For example, Georgia's six-week LIFE Act passed in 2019 and stayed blocked until the Eleventh Circuit allowed it to take effect weeks after *Dobbs* (*SisterSong v. Governor of Georgia*, 2022), and courts blocked similar bans in Arkansas, Kentucky, and Ohio before or shortly after they took effect (Nash et al., 2019; DeMillo, 2021).

Against this unsettled background, Texas became the first state to enforce an early abortion ban under *Roe*. Texas Senate Bill 8 (SB8) took effect on September 1, 2021, nearly ten months before *Dobbs*, and banned abortion after about six weeks of gestation. Because private lawsuits rather than state officials enforced the law, there was no official a federal court could enjoin to stop it. The law nonetheless stood on uncertain legal footing. When the Supreme Court declined to block it when it took effect, the Court noted "serious questions regarding the constitutionality of the Texas law" and stated that its order was "not based on any conclusion about the constitutionality" of the law (*Whole Woman's Health v. Jackson*, 2021). After *Dobbs*, Texas's ban came to rest on the decision itself, because the state's trigger law, which banned nearly all abortions once *Roe* was overturned, took effect on August 25, 2022 (Supplemental Appendix A1 gives the legal sources).[1]

Then, in June 2022, the Supreme Court held in *Dobbs v. Jackson Women's Health Organization*, a challenge to Mississippi's 15-week ban, that the Constitution confers no right to abortion (*Dobbs v. Jackson Women's Health Organization*, 2022). Trigger laws and previously enjoined bans then took effect in eleven states (Alabama, Arkansas, Georgia, Idaho, Kentucky, Louisiana, Mississippi, Missouri, Oklahoma, South Dakota, and Tennessee) between June and August 2022 (Nash and Guarnieri, 2022). West Virginia passed a total ban in September 2022, and Indiana and South Carolina enforced bans in August 2023 after litigation (Curhan, 2026).

Every ban came with potential health exceptions, but their reach was unclear. Each preserves an exception to prevent the death of the pregnant woman, and some also for a serious risk to her health (Felix, Sobel and Salganicoff, 2024). However, physicians describe the language of these exceptions as too vague to apply at the bedside (Sabbath et al., 2024; Landay et al., 2026), and guessing wrong can mean license revocation, civil liability, or prison for providers (Felix, Sobel and Salganicoff, 2025). More than 60 percent of obstetrician-gynecologists in ban states report concern about their own legal risk when deciding whether an abortion is medically necessary (Frederiksen et al., 2023), and qualitative work documents clinical decisions that departed from the prior standard of care (Sabbath et al., 2024). However, this uncertainty did not begin with *Dobbs*. Long before the

[1]Given this background, I treat SB8 as a ban enabled by *Dobbs*, but one enacted earlier.

decision, ethics committees at Catholic-owned hospitals denied approval to evacuate the uterus of a miscarrying patient while fetal heart tones were present, forcing physicians to delay care or transfer the patient elsewhere (Freedman, Landy and Steinauer, 2008). In Texas, clinicians interviewed before *Dobbs*, some of them before SB8, described narrow hospital criteria for abortion that limited care for patients whose health was at risk (Arey et al., 2023).

B. How Abortion Restrictions Could Affect Maternal Health

Restricting access to abortion, and leaving unclear when exceptions apply, can have potentially wide-reaching effects on maternal health during pregnancy. An early abortion uses either medication, mifepristone followed by misoprostol, or a procedure that empties the uterus, such as uterine aspiration (American College of Obstetricians and Gynecologists and Society of Family Planning, 2020). However, the medications and procedures that make up an abortion are also the standard of care for several conditions unrelated to elective abortion. For example, an ectopic pregnancy implants outside the uterus and cannot survive, and physicians treat it with methotrexate or surgery (American College of Obstetricians and Gynecologists, 2018*a*). Physicians treat an incomplete, retained, or septic miscarriage with dilation and curettage or with the same medications used for abortion (American College of Obstetricians and Gynecologists, 2018*b*). In previable rupture of membranes, preeclampsia with severe features, a hypertensive disorder, or intra-amniotic infection, ending the pregnancy is sometimes the only treatment that prevents sepsis, hemorrhage, or organ failure (Arey et al., 2023). Additionally, terminations for a lethal fetal anomaly, including ones that may risk the mother's health if continued, use the same procedures and are ambiguously elective (Grossman and Chasen, 2020).

Further, a pregnancy that does not end in a live birth is common, and its management is routine obstetric care, so many women encounter the procedures a ban restricts. In 2019, an estimated 66.9 percent of U.S. pregnancies ended in a live birth, 13.1 percent in an induced abortion, and 20.0 percent in a miscarriage, ectopic pregnancy, or stillbirth (Rossen et al., 2023). Further, even before any formalized abortion ban, 12 percent of women obtaining an abortion cited concerns about their own health and 13 percent possible problems with the health of the fetus (Finer et al., 2005). Therefore, restrictions on abortion potentially have a large scope to impact pregnant women's experiences in health care.

One potential channel through which these restrictions could harm maternal health is delayed management of complications. Because the procedures overlap, a physician unsure whether an exception applies may wait until a miscarriage becomes septic or an ectopic pregnancy ruptures

(Sabbath et al., 2024; Arey et al., 2023). Clinicians practicing in environments where abortion is restricted describe waiting to intervene in ruptured membranes, infection, and severe preeclampsia until the fetus has no cardiac activity or the patient has deteriorated enough to meet a narrow reading of the exception (Sabbath et al., 2024; Arey et al., 2023; Lift Louisiana et al., 2024). Waiting can be costly. In Texas hospital discharge records, sepsis among women hospitalized with a second-trimester loss rose from 2.9 to 4.5 percent after SB8 (Presser et al., 2025), and reporting describes ectopic pregnancies that ruptured before treatment (Seitz, 2024).

A second way restrictions on abortion can harm maternal health is through reduced access to terminations for medical reasons. These can follow the diagnosis of a fetal anomaly or maternal condition, usually later in pregnancy than medication abortion can address (Grossman and Chasen, 2020). A woman who cannot end such a pregnancy carries it longer, and carrying it is not without risk. In a before-and-after comparison at one Texas center, with no comparison group, maternal morbidity among patients with a life-limiting fetal condition was 72 percent in 2022 and 2023, when the bans left nearly all of them to expectant management, against 60 percent with expectant management and 15 percent with termination in 2017 and 2018 (Nambiar et al., 2025).

A separate channel for harm is the timing of care. Some practices declined to see pregnant patients before 12 weeks to avoid managing early losses (Lift Louisiana et al., 2024). If practices stop seeing women early in pregnancy, care could deteriorate for every pregnancy.

Further, abortion restrictions may change average maternal health solely through who is pregnant, as I test in Section VI.B. If restrictions prevent women who would have ended a pregnancy from ending it, and these women are on average at higher baseline risk than women already carrying pregnancies, average outcomes would worsen even if no woman's care changed. If instead they are at lower risk, average outcomes could improve and hide a deterioration in care.

II. Data

A. Insurance Claims

To study the effects of the bans on maternal health, care, and pregnancy outcomes, I use Inovalon closed claims real-world data, which are comprised of post-adjudication medical and pharmacy claims received directly from more than 160 primary payer sources, including national, regional, and local health plans across all 50 states, the District of Columbia, and Puerto Rico. The dataset includes more than 246 million covered lives over a 10-year lookback across commercial, Medicare Advantage, and Medicaid lines of business. Patient identity is resolved before de-identification using a conservative deterministic matching approach, producing a persistent identifier that follows each

individual across health plans and over time. Only birth year, gender, and three-digit ZIP code carry into the de-identified dataset. Data are de-identified under the HIPAA Expert Determination standard (45 CFR §164.514(b)(1)). Available elements include claim type, place of service, service and admission and discharge dates, diagnosis and procedure codes, DRG, HCPCS, NDC, proxy cost data, enrollment and continuous enrollment, insurance type, ACA indicators, and a provider's National Provider Identifier (NPI), taxonomy, specialty, and facility type.

Within these data, the claims file records each woman's health care, and the enrollment file records her demographics, state of residence, and payer each month. I also merge in the National Plan and Provider Enumeration System (NPPES), the federal NPI registry, for each provider's practice address by year (Centers for Medicare & Medicaid Services, 2026).

These data provide unusually rich detail on the health care of millions of insured women. In my analysis sample, I record the services billed to a woman's insurance from a year before conception through 42 days after her pregnancy ends. However, one feature of these data is that they are a convenience sample, meaning that insurers contribute data under contracts with the data provider that begin and end over time. Because the data carry no plan identifier, I cannot distinguish a woman who leaves her plan from one whose plan stopped contracting with the data provider. Additionally, because insurance plans can be regional, such as Medicaid managed care plans that each state contracts for separately (Medicaid and CHIP Payment and Access Commission, 2019), an insurer's entry or exit can add or remove a whole population and potentially shift the risk profile of a state's members. I address this by restricting my analysis sample to women continuously enrolled from 2020 through 2024 (Section II.D).

From these files I construct two analysis samples: the pregnancy sample, the core of the analysis, with one observation per pregnancy episode of a continuously enrolled woman, and a panel of continuously enrolled women aged 15 to 45 by state and quarter for state pregnancy and birth rates (Section VI.A).

B. Ban and Protected States

I group thirteen treated states into three cohorts: Texas; the eleven states that banned abortion directly after *Dobbs*; and West Virginia. I compare them with 24 states and the District of Columbia where abortion remained legal through at least 22 weeks of gestation throughout the period (Curhan, 2026). I exclude from my analysis eleven states whose bans were enjoined, partial, or changing during the window; these states do not enter the analysis at all, and the protected states enter only as never-treated controls. Because Indiana and South Carolina banned abortion

in August 2023 and the data end in December 2024, I cannot follow a complete post-ban window of their pregnancies to its ending, so I also exclude them from the pregnancy estimates. Supplemental Appendix Table A2 further describes the rationale behind the choice of treatment, control, and excluded states, and Table A1 lists each state with its effective date and statute.

C. Constructing Pregnancy Episodes

I build pregnancy episodes following validated claims-based algorithms (Ailes et al., 2016; Matcho et al., 2018; Moll et al., 2021). I begin by using pregnancy-specific codes to identify each pregnancy and date its start and its ending.[2] I then gather the woman's complete claim history around these dates. Each episode has three windows. The pre-pregnancy window covers the 365 days before the pregnancy start and supplies the measures of health before pregnancy, the pregnancy window runs from the start to the ending, and the postpartum window runs from the day after the ending through 42 days. This process captures all of the insured care a woman receives during pregnancy, not just pregnancy-specific care. This matters because providers bill some care with no code indicating pregnancy. For example, an emergency department may bill sepsis with only the presenting condition, and outcomes measured only on pregnancy-coded claims would miss it.

Endings

I identify each ending from diagnosis, procedure, and diagnosis-related-group codes for a live birth, stillbirth, molar pregnancy, ectopic pregnancy, induced abortion, or spontaneous abortion, and where an ending carries codes for more than one type, I apply the hierarchy that Moll et al. (2021) validated against physician-adjudicated records. Because the abortion codes include the billing codes for mifepristone and misoprostol, which also treat miscarriage, the classification errs toward counting too many abortions rather than too few. Supplemental Appendix A2 describes how I link ending codes and exclude endings that imply a pregnancy too close to the previous one.

Pregnancy start

I date each pregnancy from the first day of the estimated last menstrual period (American College of Obstetricians and Gynecologists, 2017), and call this date the pregnancy start. I date most pregnancies from Z3A diagnosis codes, which record the gestational week at a visit, using a validated algorithm with a median absolute error of four days (Chomistek et al., 2023). Dating is

[2]The pregnancy and ending codes draw on the Maternal and Infant Health code lists that the Centers for Medicare & Medicaid Services publish for identifying pregnant and postpartum beneficiaries (Centers for Medicare & Medicaid Services, 2023), with the ending hierarchy of Moll et al. (2021). Supplemental Appendix Table A22 lists the codes by outcome.

least accurate for early losses. Supplemental Appendix A2 gives the full procedure and the share of pregnancies dated by each source.

Pregnancies without an observed ending

Some pregnancies carry pregnancy care but no ending code. In my main analysis sample, they are 6.2 percent of ban-state pregnancies conceived in the six quarters before each state's ban (Section II.D). I build these episodes from clusters of pregnancy-indicating claims, such as prenatal visits and gestational-age codes, that fall outside every other observed pregnancy for a woman, as Sarayani et al. (2020) do for pregnancies with unknown outcomes, and end each on the date of its last such claim (Supplemental Appendix A2).

I keep pregnancies without an observed ending rather than excluding them, because their share is itself informative. For a woman who is insured throughout, a pregnancy with prenatal care and no billed ending is most plausibly one that ended in care insurance did not pay for, such as an abortion paid out of pocket, obtained out of state, or obtained by mail, or a loss that received no care. I exclude them from the outcomes that require an observed ending, namely the live-birth, miscarriage, and insurance-billed abortion shares, preterm birth, the length of pregnancy, and the days to the first visit.

D. Analysis Samples

In the core analysis sample, the *continuous-enrollment sample*, I keep only the pregnancies of women continuously enrolled from January 2020 through December 2024. This requirement lets me observe, for nearly every pregnancy, the full year preceding it and the full postpartum period.[3] Holding the same women in the data throughout also keeps the sample's membership from shifting with other changes in who appears in the claims over the years the bans took effect, such as the end, on March 31, 2023, of the pandemic-era Medicaid continuous-enrollment provision that had kept enrollees covered without annual eligibility checks since March 2020 (Tolbert and Ammula, 2023).

While the continuous-enrollment sample keeps a constant group of women at risk of pregnancy, each observed in detail, it is not without trade-offs. First, it retains only 19 percent of the pregnancies that pass every other restriction when I impose no enrollment requirement (995,377 of 5,325,806). Second, it may select certain kinds of women. For example, women with stable coverage of either kind are plausibly better able than most to travel around a ban, and nationally, continuous

[3]Supplemental Appendix A3 reports how often I fully observe the pre-pregnancy year for pregnancies that start in 2020.

coverage is more common among higher-income and married women (Daw et al., 2017). Despite this, the pregnancies in my sample broadly resemble all U.S. births on a number of demographic features (Supplemental Appendix Table A7). For example, 22.7 percent of sample pregnancies are to women under 25, against 21.1 percent of U.S. births, and 15.1 percent are to Black women, against 13.7 percent. The sample also includes more Medicaid-covered pregnancies than the nation does, as Medicaid covers 50.7 percent of sample pregnancies against 41.5 percent of U.S. births (Osterman et al., 2025). If the bans harmed Medicaid-covered women more, this would push the estimates toward larger effects rather than smaller ones.

In addition, my enrollment requirement excludes, by construction, every woman who died before December 2024, because a woman who dies loses her insurance enrollment.[4] The main sample therefore lacks pregnancies that ended in or led to a maternal death, the most severe outcomes, and so tilts slightly toward healthier pregnancies. If the bans raised maternal deaths, the excluded deaths would come disproportionately from ban states after the bans, and the estimates for severe maternal morbidity would understate the true effect. However, maternal death, at 17.9 per 100,000 live births in 2024 (Hoyert, 2026), is more than a hundred times rarer than severe maternal morbidity, at 2.6 percent of ban-state pregnancies in my sample before the bans, so the understatement is small.

Additional restrictions

I keep pregnancies in the 15 ban states (the 13 treated states, Indiana, and South Carolina) and 25 protected jurisdictions that started from January 2020 through February 2024, so that I can follow a typical 40-week pregnancy to its ending, and I exclude observed endings with an implied gestation shorter than 28 or longer than 315 days, likely coding or dating errors. The continuous-enrollment sample contains 995,377 pregnancies, and Supplemental Appendix A3 reports counts after each step.

Demographics, health, and treatment

A woman's demographics for each pregnancy come from the enrollment file in the month the pregnancy started. I control for age and its square;[5] race and ethnicity, in six mutually exclusive groups (Hispanic of any race, and non-Hispanic White, Black, Asian, other or multiple races, and missing);[6] Medicaid coverage; indicators for a first observed pregnancy and a multiple gestation;

[4]The data also do not let me identify deaths directly, because the data provider withholds them to protect privacy.

[5]Age enters as a quadratic because the risk of severe maternal morbidity rises nonlinearly with maternal age, low from the early twenties to the mid-thirties and rising steeply after about 39 (Lisonkova et al., 2017).

[6]The data provider imputes race and ethnicity using a proprietary algorithm, and they can be unknown.

and a pre-pregnancy comorbidity score. I measure this score from available claims in the 365 days before the pregnancy start, using the conditions of the obstetric comorbidity index of Leonard et al. (2020), a validated predictor of severe maternal morbidity. Supplemental Appendix A3 lists the conditions and their weights. One limitation of this measure is that a condition appears in the claims only if the woman sought care for it, so women who rarely see a doctor may appear healthier than they are. However, every woman in my sample is insured throughout the year before conception, and the index's authors developed and validated it on administrative hospital records that share this limitation (Leonard et al., 2020).

I assign treatment by the state of residence in the enrollment file rather than by where care was delivered, because claims locate care only through the registered address of the provider's NPI (Supplemental Appendix A3). The enrollment file records one state per woman, so a woman who moved from a ban state to a protected state would count as untreated, biasing the estimates toward zero. To address this in part, in the within-ban-state comparison (Supplemental Appendix A9) I assign location from the providers a woman saw before conception.

Several of these demographic and health characteristics define the subgroups in Section V.D. Medicaid-covered pregnancies are those with a Medicaid payer in the month the pregnancy started, minority women are those with a Black race code or a Hispanic ethnicity code, and high-risk women are those with at least one condition in the obstetric comorbidity index, about 42 percent of ban-state pregnancies before the bans.

E. Outcomes

Each channel in Section I.B predicts a pattern of maternal harm from abortion restrictions that may be observable in health care claims. I therefore define twelve outcomes across three groups that capture the parts of pregnancy which abortion bans are most likely to affect. Supplemental Appendix Table A22 provides code-level definitions and sources for each.

Maternal health

Severe maternal morbidity measures the life-threatening complications at the center of the warnings, such as sepsis and hemorrhage, and would rise if delayed management let a miscarriage become septic or an ectopic pregnancy rupture. To define it, I use the Centers for Disease Control and Prevention's definition (Centers for Disease Control and Prevention, 2024), as specified in ICD-10 diagnosis codes by Leonard et al. (2020, Appendix 1), and count a pregnancy as having severe maternal morbidity if any of its 16 diagnosis indicators, among them sepsis, shock, eclampsia, and

acute renal failure, appears on any claim from the pregnancy start through 42 days after the ending.[7] Because severe morbidity is rare, affecting about 2.6 percent of ban-state pregnancies before the bans, I also measure a composite of more common *adverse pregnancy outcomes*, adapted from Crump, Sundquist and Sundquist (2024). It captures the harms of a pregnancy carried longer or managed later. If terminations for medical reasons became harder to obtain, pregnancies that would have ended in a managed termination would instead end in a complicated birth or a loss, and a delayed miscarriage can become infected. It equals one if a pregnancy has a diagnosis of infection or sepsis, preterm birth, gestational diabetes, poor fetal growth, or placental abruption.

I also study two of the most common serious complications of pregnancy. *Hypertensive disorders of pregnancy* are the most common, affecting 21 percent of ban-state pregnancies before the bans. They are disorders of high blood pressure that arise in pregnancy, among them gestational hypertension and preeclampsia, and in their severe forms they can damage the kidneys, liver, and brain. Delivery or termination is the treatment for these severe forms, so delayed management could let them progress, and they are the outcome earlier restrictions on abortion providers affected (Gardner, 2024), so they give the sharpest comparison with prior work.

Finally, I measure *preterm birth* only among live births. Because the bans may themselves change the mix of pregnancy endings, this outcome isolates harm to pregnancies carried to delivery, where lost access to terminations for medical reasons may show up, as a pregnancy with a severe anomaly or complication that would have been ended instead continues until an early delivery.

Timing and intensity of care

The second group of outcomes asks whether the timing and intensity of care changed. Delayed management or deferred early care would appear here first, before or without a change in health. The *days to the first visit*, the log number of days from the pregnancy start to the woman's first claim of any kind during the pregnancy, tests whether practices deferred early care, as some practices in ban states said they would to avoid managing early losses (Lift Louisiana et al., 2024). A *prolonged delivery stay*, a live birth whose delivery admission ran past the federal minimum stay of two days after a vaginal delivery and four after a cesarean (42 U.S.C. 300gg-25), tests for delayed management of complications. Such delays would lengthen hospital stays at delivery. The log *length of pregnancy* and the log *number of visits* describe how long a pregnancy lasted and how much care it received. However, I expect these two to rise mechanically when fewer pregnancies

[7] Supplemental Appendix A10 explains why I omit the definition's five procedure indicators.

end early, so I read them alongside the endings rather than as measures of the quality of care.[8]

Pregnancy endings

The third group of outcomes records how each pregnancy ended. A ban that prevented terminations should change how pregnancies end, with fewer insurance-billed abortions and more pregnancies ending in a live birth, a miscarriage, or an unknown ending.[9] I study four outcomes, whether a pregnancy ended in a *live birth*, a *miscarriage*, or an *insurance-billed abortion*, an induced abortion identified from any diagnosis, procedure, or medication code for one, and whether its *ending was not observed* (Section II.C). The first three are shares of pregnancies with an observed ending and the fourth is a share of all pregnancies, so the four do not sum to one. Insurance-billed abortion is the most direct measure of the policy for insured women, although it captures only a fraction of all abortions. As Section II.C explains, this measure errs toward counting too many abortions rather than too few.

III. Descriptive Patterns

Table 1 reports means in the continuous-enrollment sample for ban and protected states before and after the bans; the protected-state post-period begins with pregnancies that started in or after June 2022. Before the bans, ban-state mothers were younger than protected-state mothers (29.0 against 30.4 years), and the racial and ethnic composition of insured pregnancies differed between the two groups of states. Medicaid covered 42.4 percent of ban-state pregnancies before the bans and 53.4 percent in protected states, and ban-state women entered pregnancy with a higher mean comorbidity score, 8.7 against 8.4. Supplemental Appendix Table A6 reports normalized differences for these characteristics in the pre-period (Imbens and Wooldridge, 2009).

In addition to these baseline differences in characteristics, the pregnancies themselves differed, and the gaps persist in the rows of Table 1 adjusted for characteristics. First, the difference in how pregnancies ended is striking. Before the bans, insurance-billed abortion ended about 1 percent of ban-state and 14 percent of protected-state pregnancies with an observed ending in 2020, and a live birth ended a correspondingly larger share of ban-state pregnancies. Some of this gap likely reflects ban-state pregnancies that ended in an abortion paid outside insurance and never appear in the claims, and some likely reflects differences in the insured populations the data cover in each state.

[8] I log these outcomes because their distributions have a long right tail, and in levels a few extreme pregnancies would drive the estimates. I winsorize counts at the 99th percentile.

[9] Ectopic, stillbirth, and molar endings, 2.5, 0.9, and 0.1 percent of observed endings, are too rare to estimate.

Second, ban-state pregnancies fared worse, particularly in the states that banned at *Dobbs* rather than in Texas (Figure 1). They more often involved an adverse pregnancy outcome and a hypertensive disorder. Part of this gap perhaps follows from more ban-state pregnancies continuing to term, and so facing the complications of later pregnancy, but even among live births, and adjusted for characteristics, preterm birth was more common in ban states before the bans. Finally, ban-state pregnancies lasted longer, with about as many visits and prolonged delivery stays, while the first visit came at about the same point in both groups of states.

Figure 1 shows the raw series for the twelve outcomes over calendar time for Texas, the states that banned at *Dobbs*, and protected states. The gaps were largely stable, and in some cases widening, before the bans. For maternal health and for the timing and intensity of care, the ban-state and protected-state series move roughly in parallel, apart from the length of pregnancy and the number of visits. However, there are clear divergences in the pregnancy endings. The ban-state series for insured abortion and live birth are nearly flat, while in protected states the live-birth share fell and insured abortion rose, as several states widened protections and coverage (Supplemental Appendix A1) and access to abortion by telehealth expanded after the COVID-19 pandemic (Society of Family Planning, 2025). No maternal health or care series shows a sharp break at either ban, and Section V tests whether the smaller movements in the adverse-outcome composite and preterm birth exceed the changes in protected states.

IV. Empirical Strategy

In this section I describe my core empirical design. I compare pregnancies in ban states with pregnancies in protected states, before and after each ban. The unit of observation is a pregnancy p in state s conceived in quarter q. Because the bans took effect at different times, I estimate the stacked difference-in-differences design of Wing, Freedman and Hollingsworth (2024), which formalizes the stacked event studies of Cengiz et al. (2019) and Deshpande and Li (2019).[10] For each adoption quarter a, the first fully treated quarter of one or more ban states (the fourth quarter of 2021 for Texas, the third quarter of 2022 for the *Dobbs* cohort, and the fourth quarter of 2022 for West Virginia),[11] I build a sub-experiment containing those states and, as controls, the protected states. A ban state never serves as a control. Event time is $e_{pa} = q - a$, with a window of

[10]Stacking compares each treated cohort only with never-treated states and avoids the comparisons with already-treated states that bias two-way fixed effects estimates when treatment is staggered and effects vary over time (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak, Jaravel and Spiess, 2024). Supplemental Appendix Figure A4 compares the stacked event studies with two-way fixed effects.

[11]The third quarter of 2022 is only partly treated for Georgia, Idaho, and Mississippi, whose bans took effect in July and August, and for Kentucky and Louisiana, where state courts blocked enforcement for much of July, so the estimate at $h = 0$ may be attenuated for these five states. The later event-study coefficients, in fully treated quarters, show whether an effect appears once the bans were fully in force.

$e_{pa} \in [-6, 4]$. A sub-experiment enters the stack only if its full window lies inside the data and it has at least one clean control, so I estimate every event-time coefficient on the same cohorts. Supplemental Appendix A8 gives each cohort's window, and Supplemental Appendix Table A14 the pregnancies in each sub-experiment. I then estimate

$$Y_{psqa} \;=\; \sum_{h=-6}^{4} \delta_h \cdot \mathbf{1}\{e_{pa} = h\} \cdot D_{sa} \;+\; \sum_{h} \lambda_h \cdot \mathbf{1}\{e_{pa} = h\} \;+\; X_p'\gamma \;+\; \phi_{sa} \;+\; \varepsilon_{psqa},$$

where Y_{psqa} is an outcome of pregnancy p, conceived in quarter q by a woman living in state s, in sub-experiment a, and h indexes event time. D_{sa} indicates the treated states of sub-experiment a, λ_h are event-time effects shared by treated and control states, ϕ_{sa} are sub-experiment by state fixed effects, and X_p contains the individual controls described in Section II.D (age and its square, race and ethnicity, Medicaid coverage, first pregnancy, multiple gestation, and comorbidity score). The sub-experiment by state fixed effects hold fixed each state's average outcome within each sub-experiment, so the estimates come from changes within a state over its own window rather than from the gaps between ban and protected states in Section III. The event-time effects, common to every sub-experiment, absorb changes that ban and protected states share, so δ_h measures how far ban-state pregnancies departed from that shared path at event time h. Rather than omitting quarter -1, I normalize the coefficients so that the six pre-ban coefficients average zero, $\sum_{h=-6}^{-1} \delta_h = 0$, so I measure each coefficient, δ_{-1} included, against the pre-ban average, with its own standard error. The joint pre-trend test, that δ_{-6} through δ_{-1} are all zero, is the same under either normalization. I weight control observations in sub-experiment a by $(N_a^D/N^D)/(N_a^C/N^C)$, the sub-experiment's share of treated observations divided by its share of control observations; without these weights a stacked regression does not identify a convex combination of cohort-specific effects, and with them δ_h is the trimmed aggregate average treatment effect on the treated at event time h (Wing, Freedman and Hollingsworth, 2024). The weights give each sub-experiment's controls the same share of the weighted control group that its ban states have of the treated group, and I count them in each regression's sample. For the pooled estimate I replace all the event-time interactions with one indicator $D_{sa} \cdot \mathbf{1}\{e_{pa} \geq 0\}$, on the same stacked sample and with the same weights, so the pooled and event-study estimates use the same pregnancies.

I compare the post-ban average with the pre-ban average, rather than with the last pre-ban quarter as in Wing, Freedman and Hollingsworth (2024), because quarter -1 is likely partly treated. The main reason is that a pregnancy carried to term lasts about 40 weeks, so a pregnancy conceived in quarter -1 or -2 is still under way when the ban takes effect and those quarters are partly

exposed to it. Physicians and patients may also have acted in anticipation of a ban (Section I.A), and because pregnancy start dates in claims are noisy, most of all for early losses, some pregnancies assigned to quarter -1 may have been conceived after the ban (Supplemental Appendix A2). Averaging over all six pre-ban quarters dilutes these partly exposed quarters rather than letting one of them set the reference. Supplemental Appendix Table A15 and Figure A5 report estimates against quarter -1. I cluster standard errors by state rather than by sub-experiment and state, as the same control states appear in several sub-experiments. I report two specifications. The Raw specification omits X_p. The Controls specification is the equation above, and it is my preferred specification: it measures the effect of a ban among observably similar women.

I measure event time in quarters, because conception dates are much less accurate for early losses and each quarterly coefficient rests on more pregnancies (Supplemental Appendix A2).

A causal reading of these estimates requires that, absent the bans, outcomes in each treated cohort would have followed the path of its controls. Several features of the setting strain this assumption. The first is that for some outcomes the trends clearly do not move together before the bans. This is most true for insurance-billed abortion, whose protected-state series rises well before the bans (Section III), consistent with the argument that it was already *Dobbs* in ban states, so I read its estimate as describing ongoing patterns. The live-birth and miscarriage shares also move before the bans, though less, so I read their estimates as suggestive. However, the maternal health outcomes and most of the timing and intensity outcomes move together before the bans, apart from the length of pregnancy and the number of visits, both rising with the share of pregnancies carried to term, so I read their estimates around the ban dates as more causal in nature.

An additional strain falls on the stable unit treatment value assumption: the bans must not change outcomes in the comparison states. Protected states absorbed travel from ban states and expanded access to abortion after *Dobbs*. Expanded access would bias the estimates toward finding harm and crowding would bias them toward zero, and the within-ban-state comparison in Supplemental Appendix A9 avoids protected states. Synthetic difference-in-differences (Arkhangelsky et al., 2021) reweights protected states and pre-ban quarters to match ban states' pre-ban paths, in case protected states are a poor comparison (Supplemental Appendix A8).

V. The Effects of the Bans

The bans changed maternal health and the timing and intensity of care little, while modestly shifting pregnancies toward a live birth. I report pooled estimates from the Controls specification in the continuous-enrollment sample, each scaled by the pre-ban mean, with the quarterly event

studies (Figure 2). Table 2 also reports the Raw specification and the p-value of a joint test that the pre-period coefficients are zero.

A. Pregnancy Endings

If the bans prevented terminations, the most direct place to see it is abortion billed to insurance, although it is an imperfect proxy for the abortions a ban prevents. However, my raw data and event studies show no fall in ban states. The pooled estimate is -0.025 (standard error 0.006), or -237.1 percent of the pre-ban mean of 1.06 percent, a magnitude larger than the mean that is possible only because I measure the estimate against protected states. The raw series show that the protected-state side accounts for most of this (Figure 1). In ban states, an insurance-billed abortion ended between 0.8 and 1.2 percent of pregnancies with an observed ending in every quarter from 2020 through 2023, while in protected states the share rose from 14 percent in 2020 to 17 percent in the quarter of the decision and 20 percent by the end of 2023. The event study traces the same divergence and rejects parallel pre-trends (p <0.001), so I read this estimate as describing ongoing patterns rather than an effect of the bans.

Additionally, if the bans moved terminations outside the insurance system, to other states or to medication by mail, those pregnancies may appear as a higher share of pregnancies with care but no recorded ending, because in the continuous-enrollment sample every woman is insured and enrolled throughout. In Texas this share was 4.3 percent among pregnancies conceived from 2020 through the first quarter of 2021, rose to 8.4 percent among pregnancies conceived in the three quarters around SB8, and returned to 4.4 percent afterward (Figure 1). Over the same quarters it moved from 4.7 to 5.2 percent in protected states and from 6.2 to 7.0 percent in the states that banned at *Dobbs*, so states whose laws had not yet changed share part of the Texas rise. No comparable rise follows *Dobbs*, perhaps because by mid-2022 women who meant to end a pregnancy could do so before any prenatal care was billed, or after a single visit, not enough to form an episode, increasingly through telehealth (Society of Family Planning, 2025). The pooled estimate for this share is -13.3 percent ($p = 0.08$), and its 95 percent confidence interval rules out increases larger than 1.6 percent, although its pre-period coefficients jointly differ from zero ($p = 0.001$).

The share of pregnancies with an observed ending that ended in a live birth rises 2.7 percent (p <0.001; 95 percent confidence interval 1.7 to 3.8 percent). The miscarriage share would rise if pregnancies that would have been terminated instead ended in a loss, and it changes by 3.0 percent, with an interval from -0.9 to 6.9 percent. The live-birth estimate is weaker than its p-value suggests, because its event study sits 1 percent of the pre-ban mean below the pre-ban

average from six to four quarters before the bans and 1 percent above it from three quarters before (joint p <0.001), and without Texas the estimate falls to 2.3 percent ($p = 0.005$). I therefore read the rise in live births as suggestive. This share describes how pregnancies ended, not how many women gave birth, so Section VI.A turns to birth and pregnancy rates among all enrolled women.

B. Maternal Health

The only suggestion of harm in the maternal health estimates is in the composite of common adverse pregnancy outcomes. The estimate is 0.008 (standard error 0.003), or 2.8 percent of the pre-ban mean, significant at the 5 percent level ($p = 0.01$), and 3.2 percent ($p = 0.007$) in the Raw specification. Its event study is flat before the bans and positive in every quarter after (Figure 2), and the 95 percent confidence interval rules out increases larger than 4.9 percent. This rise follows from the shift in endings rather than from worse outcomes for each pregnancy. Before the bans, 36.9 percent of live births had one of these outcomes, against 2.2 percent of losses, so the 2.2-point rise in the live-birth share would by itself raise the composite by about 0.75 percentage points at pre-ban rates, most of the 0.79-point estimate. Within live births, the composite changes by −1.2 percent, with an interval that rules out increases larger than 0.6 percent (Supplemental Appendix Table A10). Against quarter −2, the estimate falls to 1.8 percent (Supplemental Appendix Table A15). None of its five components rises significantly on its own. Poor fetal growth, a diagnosis made only in pregnancies that continue, rises 7.4 percent ($p = 0.12$), while infection or sepsis, the complication delayed management would produce, falls 6.8 percent, although its pre-period coefficients jointly differ from zero ($p = 0.03$; Supplemental Appendix Table A11).

Severe maternal morbidity, a much rarer outcome, has a less precise but still null estimate. It changes by −7.6 per 10,000 pregnancies (standard error 11.1), or −2.9 percent of a pre-ban rate of 264.5 per 10,000, and the 95 percent confidence interval rules out increases larger than 5.6 percent. Its post-period coefficients scatter around zero. Hypertensive disorders of pregnancy, the most common serious complication and a well-studied outcome, show a flat null. The estimate is 1.0 percent ($p = 0.42$), and the 95 percent confidence interval rules out increases larger than 3.4 percent, below the 11 to 16 percent rise that earlier restrictions on abortion providers produced (Gardner, 2024). Hypertensive disorders rose in both groups of states over the period (Figure 1), and the bans added little to that trend in ban states relative to protected states. Preterm birth among live births changes by −0.9 percent, and its interval, from −4.5 to 2.6 percent, likewise bounds harm through a common adverse outcome among live births.

C. *Timing and Intensity of Care*

The raw series for the timing and intensity of care show no clear break at either ban (Figure 1), and where the gaps with protected states widen, they widen gradually, mostly because the protected-state series fall. The days to the first visit would lengthen if practices deferred early care. They change by 1.4 percent ($p = 0.27$), with pre-period coefficients that do not jointly differ from zero (joint $p = 0.12$), and the interval rules out delays longer than 3.8 percent.

Hospital care at delivery did not lengthen. The share of live births whose delivery stay ran past the federal minimum falls 4.6 percent ($p = 0.03$; 95 percent confidence interval -8.6 to -0.6 percent), but its pre-period coefficients jointly differ from zero ($p = 0.03$), so I do not read the decline as an effect of the bans. It gives no sign that physicians managed complications later.

The length of pregnancy rises 3.0 percent ($p < 0.001$), as expected when fewer pregnancies end early, and the number of visits is 3.3 percent higher ($p = 0.03$; 95 percent confidence interval 0.4 to 6.2 percent). Both rise before the bans as live births do (Section V.A; joint $p < 0.001$ for the length of pregnancy and < 0.001 for visits), so, as anticipated in Section II.E, I read them as part of the shift in endings rather than as evidence about the quality of care. None points to a change in how physicians managed pregnancies on average.

D. *Heterogeneity Analysis*

A small average effect can hide large effects in particular groups, so I define five subgroups whose effects have reason to differ from the overall average. I first look at Medicaid-covered pregnancies, because women with low incomes may be the least able to travel for abortion or pregnancy care or to pay for an abortion out of pocket (Jerman, Jones and Onda, 2016; Hill et al., 2024). I next look at minority women, Black and Hispanic women reported together, whose outcomes were worse before the bans. Black women in the United States die of maternal causes at about three times the rate of White women (Hoyert, 2026) and experience severe maternal morbidity at nearly twice the rate (Leonard et al., 2019). I then look directly at women who enter pregnancy with a high-risk condition, who are the most likely to need potentially lifesaving care, such as a termination for medical reasons, that the bans may have called into question. The opposite concern is that the bans led many young, healthy women to carry pregnancies to term. Such births could mask worse outcomes for everyone else, so I look at pregnancies to women under 25. Finally, I exclude Texas, whose ban came first, was enforced through private lawsuits, and was in place longest, to ensure the results are consistent without it.

For each group, I rerun the stacked design and report the percent change over the group's own

pre-ban mean, so each estimate is the change for that group rather than its contribution to the pooled effect (Figure 3 and Supplemental Appendix Table A9). The estimates vary little across these groups. For each of the twelve pregnancy outcomes, every subgroup's 95 percent confidence interval contains the pooled point estimate, apart from the under-25 interval for the adverse-outcome composite, which starts at 2.8 percent. The composite rises 4.2 percent among Medicaid pregnancies ($p = 0.02$), 4.0 percent among high-risk women ($p = 0.04$), and 6.3 percent among women under 25 ($p < 0.001$), the only subgroup estimates for maternal health whose intervals exclude zero. Severe maternal morbidity among high-risk women, the group of most concern, changes by -2.2 percent ($p = 0.64$; 95 percent confidence interval -11.4 to 7.1 percent).

Because the bans changed how pregnancies ended, I also estimate the health and care outcomes within live births and within losses (miscarriage, stillbirth, and ectopic pregnancy; Supplemental Appendix Table A10). These estimates condition on an ending the bans changed, so they describe the pooled estimates rather than isolate effects of their own. Within live births, the length of pregnancy changes by -0.1 percent, and the number of visits by -2.0 percent, so the pooled rises in these outcomes mostly reflect more pregnancies carried to delivery. Losses, the small group the warnings most concerned, yield imprecise estimates. Among them, severe maternal morbidity changes by 11.3 percent, with an interval that does not rule out a rise of 33.4 percent, and the first visit comes 2.6 percent later ($p = 0.05$). These intervals cannot rule out that more losses ended in a severe complication after the bans.

VI. The Fertility Channel

Beyond the outcomes in Section V, the bans could also change how many women carried pregnancies to term and who became pregnant. I show below that births rose suggestively while the health of the women who became pregnant changed little, consistent with the small effects above.

A. Fertility

A ban that prevented abortions, insured or not, should raise births, as earlier restrictions to abortion access do (Fischer, Royer and White, 2018; Venator and Fletcher, 2021). Conceptions may also fall if women use more contraception after a ban (Xu et al., 2024; Ellison, Brown-Podgorski and Morgan, 2024), so the net effect on births is ambiguous. I test this prediction by estimating the stacked design on the panel of enrolled women aged 15 to 45 by state and quarter, weighted by enrollment. The outcome is live births or pregnancies per 1,000 enrolled women per year, counted by the month the pregnancy started, and I report each estimate in percent of its pre-ban mean

(Supplemental Appendix Table A12). The birth rate is 3.7 percent higher relative to protected states ($p = 0.02$; 95 percent confidence interval 0.7 to 6.7 percent), the same sign as the 2.3 percent increase that Dench, Pineda-Torres and Myers (2024) estimate from birth certificates. I read this rise as suggestive, because its pre-ban coefficients jointly differ from zero ($p < 0.001$) and synthetic difference-in-differences gives 6.2 percent with a wider interval (Supplemental Appendix Table A17). The raw rates in panels (c) and (d) of Figure 4 show where that gap comes from. Both the pregnancy rate and the birth rate fall over the period in both groups of states, as the women in a continuously enrolled panel age, and birth rates in ban and protected states move together through 2022. From 2023 on they separate, as the ban-state birth rate holds near its 2022 level while the protected-state rate keeps falling, so relative to protected states, births held up in ban states.

The pregnancy rate counts conceptions, not births. It changes by 0.2 percent ($p = 0.86$; 95 percent confidence interval -2.2 to 2.6 percent). Comparing the two rates shows where any added births came from. Together with the rise in the live-birth share of observed endings (Section V.A), these estimates suggest that about as many women became pregnant after the bans but more of their pregnancies ended in a live birth. Across the subgroups in Supplemental Appendix Table A12, the birth-rate estimate is always positive. Among women with a high-risk condition, the birth rate rises 5.5 percent ($p = 0.002$) but their pregnancy rate only 1.6 percent ($p = 0.21$), so they carried more of their pregnancies to term rather than more of them conceiving (Section VI.B).

B. Who Became Pregnant

The bans may have changed who was pregnant if they changed how pregnancies ended (Section I.B). Because the claims record women's health before they conceive, I test this directly by estimating the stacked design with each pregnancy's pre-pregnancy characteristics as the outcome. Because a ban cannot change any one woman's health before conception, a nonzero estimate means that the set of women who became pregnant changed.

The pregnant population's health before pregnancy barely moved (Supplemental Appendix Table A13). After the bans, the comorbidity score changes by 0.4 percent ($p = 0.81$), the number of comorbid conditions by 0.7 percent, and the share with any condition by 0.4 percent ($p = 0.70$), and the last, whose pre-period is closest to flat, rules out increases larger than 2.6 percent. The Medicaid share of pregnancies, the most direct test of whether the added pregnancies came from women with low incomes, changes by 2.2 percent ($p = 0.28$), with an interval from -1.9 to 6.4 percent. The share of pregnancies to women under 25 rises by 11.2 percent ($p = 0.004$), a share

that was already rising in ban states relative to protected states before the bans.

Holding these characteristics fixed also changes the health estimates little: they move by no more than a few tenths of a percent between the Raw and Controls columns of Table 2, so a change in who became pregnant would not explain them.

VII. Discussion

The *Dobbs* decision ended a constitutional right for women that had stood for half a century, and many expected the bans that followed to harm maternal health broadly, by delaying the management of complications and ending access to terminations for medical reasons. However, two features of the pre-*Dobbs* environment, both documented in outside work, are consistent with effects this much smaller than predicted. First, abortion care in states that subsequently banned abortion was already highly restricted, and those earlier restrictions had already caused harm, as restrictions on abortion providers raised hypertensive disorders of pregnancy (Gardner, 2024). Potentially, then, the marginal effect of the bans was small. Second, even as restrictions came rapidly and repeatedly in the years leading up to *Dobbs*, the national environment loosened in some ways, as telehealth provision of abortion expanded after the COVID-19 pandemic (Society of Family Planning, 2025). In line with this work, insurance-billed abortion was already rare in ban states before the bans and continued to be rare after them (Section V.A). For insured women in ban states, abortion had left the insurance system before the bans.

After the bans, ban-state residents obtained abortions across state lines (Odouard et al., 2024) and through telehealth providers mailing medication under shield laws (Society of Family Planning, 2025), and I find that the bans changed maternal health less than the warnings assumed, likely because much of what a ban restricts had already been restricted or could be obtained elsewhere.

How do these estimates compare with prior work? My interval for severe morbidity rules out increases larger than 5.6 percent, while the credible interval for pregnancy-associated mortality, a far rarer outcome, reaches 20.7 percent in Bell et al. (2026), so claims can exclude large harms that death records cannot.

To make the magnitudes of my estimates concrete, consider 10,000 insured pregnancies in a ban state. Before the bans, about 264 of them would have involved severe maternal morbidity. The point estimate changes this by about -7.6, and the upper end of the 95 percent interval allows about 15 more. Without Texas, whose six-week ban came first, the interval reaches 11.8 percent, or about 34 more per 10,000 (Supplemental Appendix Table A9). About 2,080 of the same pregnancies would have had a hypertensive disorder, and the interval allows at most about 71 more. For both

outcomes, even the upper end is small next to how common they were before the bans.

Prior work found large increases in severe complications within narrow groups. The rise in sepsis among Texas women hospitalized with a second-trimester loss after SB8, described in Section I.B, is an increase of 55 percent (Presser et al., 2025), and, among high-risk hospitalized pregnancies, a Texas time series with no comparison state associates the law with sustained increases in critical illness, sepsis, and infection, an association the same study does not detect across all hospitalized pregnancies (Chen et al., 2026). Because these groups are a small share of pregnancies, an effect of that size within them would move severe maternal morbidity among all insured pregnancies by far less than my bound. For the Texas rise in sepsis, 1.6 percentage points among women hospitalized with a second-trimester loss, to add the 15 cases per 10,000 pregnancies at the upper end of my interval, about 9 percent of pregnancies would need to end in a second-trimester loss, more than half of the 16 percent of ban-state pregnancies with an observed ending that ended in a miscarriage before the bans.

These averages do not imply that the bans were harmless for every woman they reached. A pooled estimate spreads any harm across every insured pregnancy in a ban state. Spread instead over the 2.5 percent of pregnancy endings that are ectopic, or over the smaller share that end in a second-trimester loss, the same bound would allow the risk of severe morbidity in those groups to rise several times over. Estimated among losses directly, severe morbidity changes by 11.3 percent, with an interval that reaches 33.4 percent (Section V.D). Because losses are a small share of pregnancies and carry a lower rate of severe morbidity than deliveries, a rise of that size among them would move the pooled rate by only a few percent, inside its interval. The deaths and near-deaths documented in reporting and clinician interviews (Presser and Surana, 2024; Seitz, 2024; Sabbath et al., 2024) can be consistent with the null aggregate estimates I provide. My estimates speak to how common such harm was across a wide population, not to whether it occurred at all.

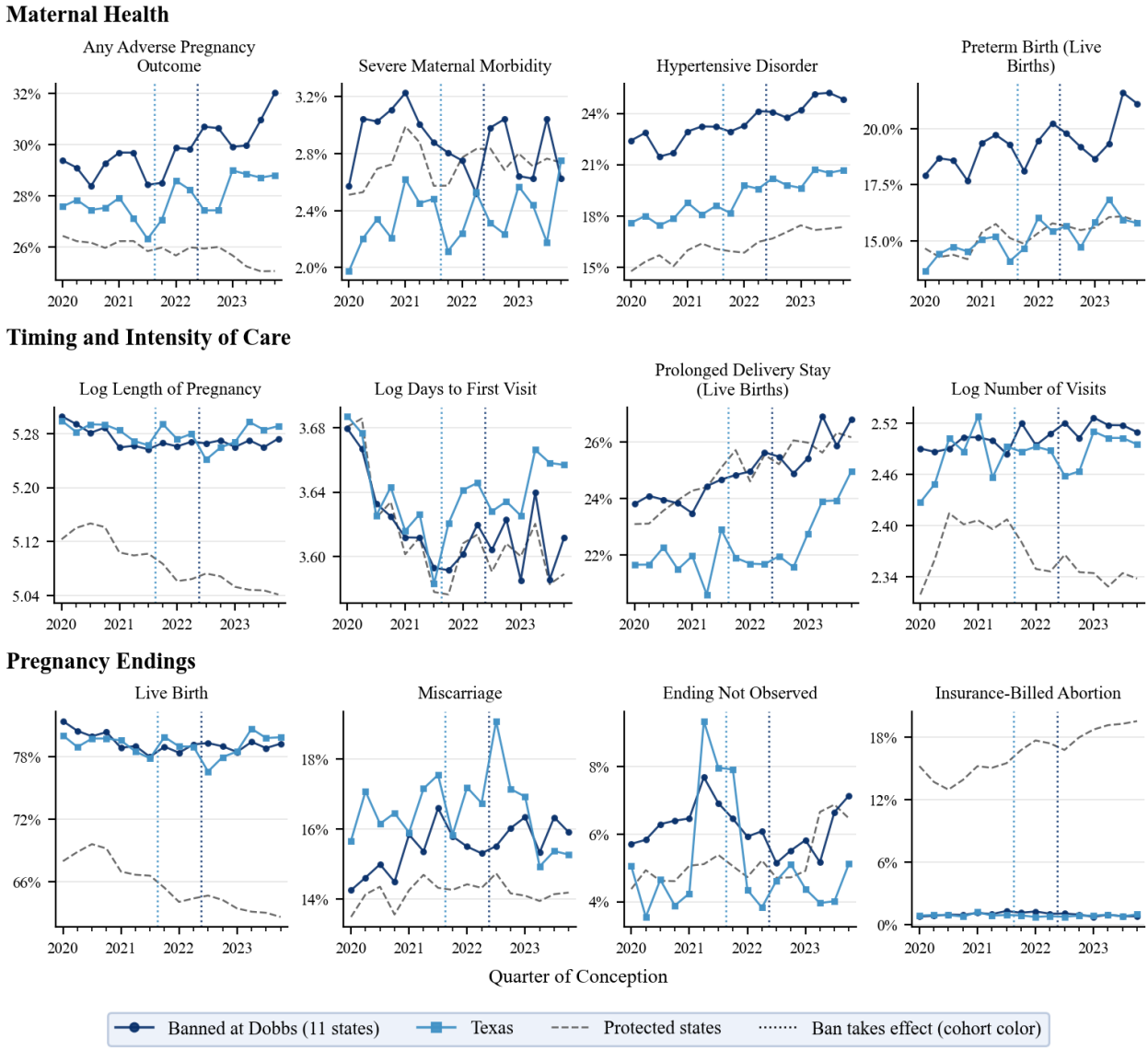
The results, taken in line with previous work, carry a stark lesson for policy makers and maternal health advocates. The debate over *Dobbs* has centered on the constitutional right to abortion, but for the women studied here, the restrictions that limited access came earlier. Restoring a federal right to abortion would lift the bans, but it would not by itself restore insurance coverage of abortion or the clinics those earlier restrictions closed. On its own, it may not meaningfully change the maternal health and care landscape for insured women in states that banned abortion.

Table 1—Characteristics of Pregnancies by State Policy and Period

	Ban States		Protected States	
	Before	**After**	**Before**	**After**
Pregnancies	140,836	121,531	437,625	295,385
Demographics, % Unless Noted				
Age, mean (SD)	29.01 (6.40)	28.84 (6.82)	30.35 (6.23)	30.18 (6.57)
Under 25	26.7	31.0	19.3	22.6
Medicaid	42.4	37.1	53.4	56.3
Commercial	57.5	62.8	46.4	43.5
White	41.6	41.2	45.9	46.4
Black	19.2	16.2	14.0	14.4
Asian	1.9	1.8	3.9	3.7
Hispanic	8.7	14.0	23.8	26.5
First observed pregnancy	25.7	27.2	25.7	25.4
Health Before Pregnancy, % Unless Noted				
Hypertension, obesity, or substance use	23.6	25.5	24.0	27.5
Obstetric comorbidity score, mean (SD)	8.74 (14.34)	9.14 (14.78)	8.37 (14.30)	9.48 (15.27)
Maternal Health, %				
Any adverse pregnancy outcome	28.6	29.2	26.1	25.5
Adjusted for characteristics	[29.4]	[30.1]	[25.7]	[25.1]
Severe maternal morbidity	2.8	2.6	2.7	2.8
Adjusted for characteristics	[2.6]	[2.4]	[2.4]	[2.4]
Hypertensive disorder	21.3	22.1	15.7	17.2
Adjusted for characteristics	[20.4]	[21.4]	[15.1]	[16.4]
Preterm birth (live births)	17.3	17.6	14.9	15.9
Adjusted for characteristics	[16.7]	[17.0]	[14.3]	[14.9]
Timing and Intensity of Care, % Unless Noted				
Length of pregnancy, days, mean (SD)	225.0 (83.6)	224.2 (84.1)	201.7 (96.8)	193.9 (99.2)
Adjusted for characteristics	[226.7]	[226.6]	[204.3]	[198.3]
Days to first visit, mean (SD)	46.0 (33.4)	45.7 (33.1)	45.3 (31.9)	44.3 (31.1)
Adjusted for characteristics	[51.9]	[52.5]	[48.6]	[48.0]
Prolonged delivery stay (live births)	23.3	24.2	24.3	25.9
Adjusted for characteristics	[23.5]	[23.9]	[24.8]	[26.1]
Number of visits, mean (SD)	16.2 (10.9)	16.3 (10.8)	16.3 (12.5)	16.3 (13.1)
Adjusted for characteristics	[16.1]	[16.1]	[15.9]	[15.9]
Pregnancy Endings, % of Observed Endings Unless Noted				
Live birth	79.3	79.0	67.1	63.3
Adjusted for characteristics	[80.0]	[80.0]	[68.2]	[65.4]
Miscarriage	15.7	16.3	14.2	14.3
Adjusted for characteristics	[14.1]	[14.2]	[12.8]	[12.6]
Ending not observed (percent of all pregnancies)	6.4	5.7	4.9	5.8
Adjusted for characteristics	[5.2]	[4.6]	[4.8]	[5.6]
Insurance-billed abortion	1.2	0.9	15.2	18.6
Adjusted for characteristics	[2.3]	[2.1]	[15.7]	[18.6]

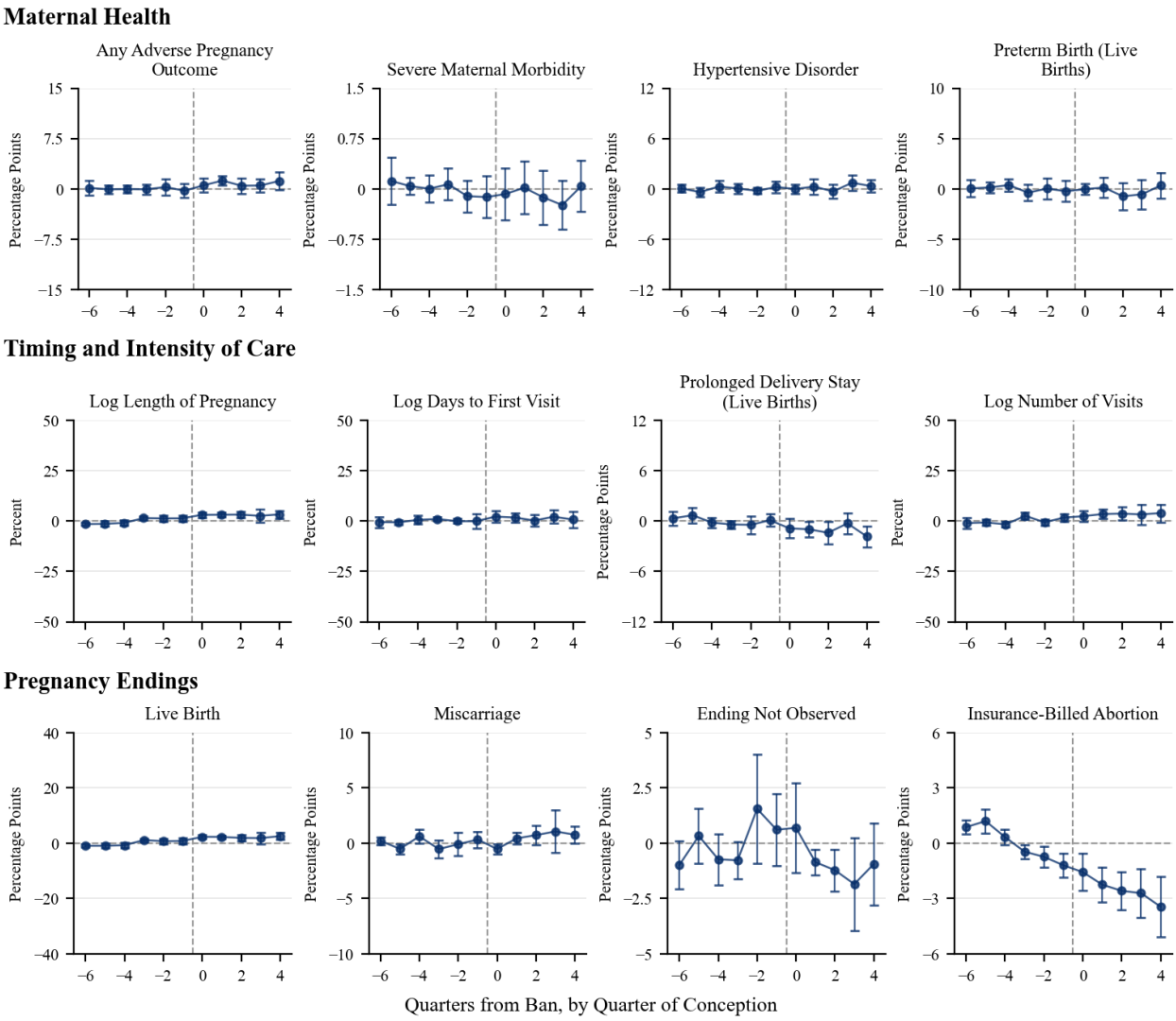
Notes: Pregnancies started January 2020 through February 2024 in the continuous-enrollment sample. Ban states enacted a strict abortion ban; protected states never did. Before and after refer to pregnancies that began before or after the ban took effect in the woman's state, and after begins in June 2022 for protected states. Shares are percentages; age, the obstetric comorbidity score, the length of pregnancy, days to first visit, and number of visits are means with standard deviations (SD) in parentheses. Preterm birth and a prolonged delivery stay, a delivery admission longer than the federal minimum of two days after a vaginal and four after a cesarean delivery (42 U.S.C. 300gg-25), are percentages of live births; live birth, miscarriage, and insurance-billed abortion are percentages of observed endings; ending not observed is a percentage of all pregnancies. Hypertension, obesity, or substance use is a diagnosis of any of the three in the year before pregnancy. Each bracketed row adjusts the outcome for the individual controls of the main specification (the residual from a regression on them, plus the overall mean), so gaps in that row are not due to those characteristics. Table A6 reports normalized differences.

Figure 1. Raw Means of the Pregnancy Outcomes by Quarter of Conception



Notes: Means by calendar quarter of conception in the continuous-enrollment sample for the eleven states that banned at *Dobbs* (circles; first treated in the third quarter of 2022), Texas (squares; fourth quarter of 2021), and protected states (dashed line). A dotted vertical line for each cohort falls between its last untreated and first treated quarter, the earlier one for Texas. West Virginia, whose quarterly counts are small, and Indiana and South Carolina, whose post-ban window runs past the data, are omitted (Supplemental Appendix Table A5). Live birth, miscarriage, and insurance-billed abortion are percentages of pregnancies with an observed ending, preterm birth and a prolonged delivery stay of live births, and the other shares of all pregnancies; the length of pregnancy, the days to the first visit, and the number of visits are in logs. Quarters after the fourth quarter of 2023 are omitted because their pregnancies cannot all be followed to their ending.

Figure 2. Event Studies for the Pregnancy Outcomes



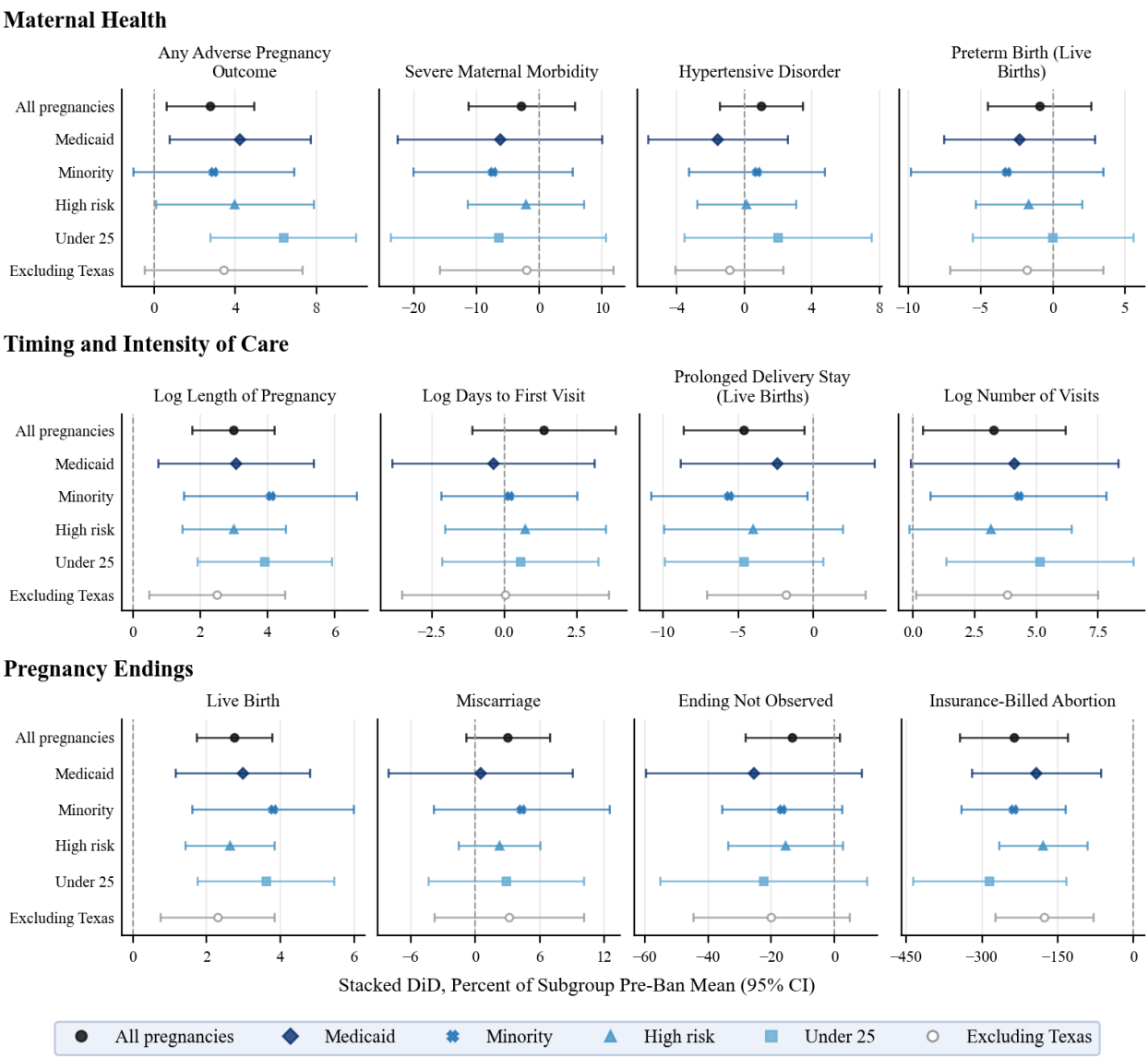
Notes: Stacked difference-in-differences event-study coefficients by quarter of conception relative to the ban, quarters −6 to 4, with 95 percent confidence intervals from standard errors clustered by state, in the Controls specification and the continuous-enrollment sample. Coefficients are in percentage points for shares and in percent (100 times the coefficient) for the outcomes in logs. Each ban cohort is compared only with states untreated through the end of its window, using the weights of Wing, Freedman and Hollingsworth (2024). The pre-ban coefficients average zero, and the dashed vertical line falls between quarters −1 and 0. Each vertical axis spans at least 50 percent of the outcome's ban-state pre-ban mean on either side of zero (50 percent for outcomes in logs), wider where an interval runs past it. Table 2 reports pre-ban means and joint pre-trend tests.

Table 2—The Effect of Abortion Bans on the Pregnancy Outcomes

Outcome	Pre-Ban Mean (1)	Controls				Raw	
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Pre-Trend p-Value (5)	Estimate (SE) (6)	Percent of Mean (7)
Maternal Health							
Any adverse pregnancy outcome	0.285	0.008 (0.003)	2.8	[0.6, 4.9]	0.98	0.009 (0.003)	3.2
Severe maternal morbidity	0.026	−0.001 (0.001)	−2.9	[−11.4, 5.6]	0.86	−0.001 (0.001)	−2.4
Hypertensive disorder	0.208	0.002 (0.003)	1.0	[−1.5, 3.4]	0.48	0.003 (0.002)	1.4
Preterm birth (live births)	0.171	−0.002 (0.003)	−0.9	[−4.5, 2.6]	0.29	−0.001 (0.003)	−0.8
Timing and Intensity of Care							
Log length of pregnancy	5.272	0.030 (0.006)	3.0	[1.8, 4.2]	<0.001	0.033 (0.008)	3.3
Log days to first visit	3.616	0.014 (0.012)	1.4	[−1.1, 3.8]	0.12	0.014 (0.013)	1.4
Prolonged delivery stay (live births)	0.233	−0.011 (0.005)	−4.6	[−8.6, −0.6]	0.03	−0.007 (0.004)	−3.2
Log number of visits	2.496	0.033 (0.014)	3.3	[0.4, 6.2]	<0.001	0.038 (0.016)	3.8
Pregnancy Endings							
Live birth	0.789	0.022 (0.004)	2.7	[1.7, 3.8]	<0.001	0.024 (0.005)	3.0
Miscarriage	0.162	0.005 (0.003)	3.0	[−0.9, 6.9]	<0.001	0.003 (0.004)	1.7
Ending not observed	0.062	−0.008 (0.005)	−13.3	[−28.3, 1.6]	0.001	−0.009 (0.005)	−14.1
Insurance-billed abortion	0.011	−0.025 (0.006)	−237.1	[−344.5, −129.6]	<0.001	−0.024 (0.005)	−232.0

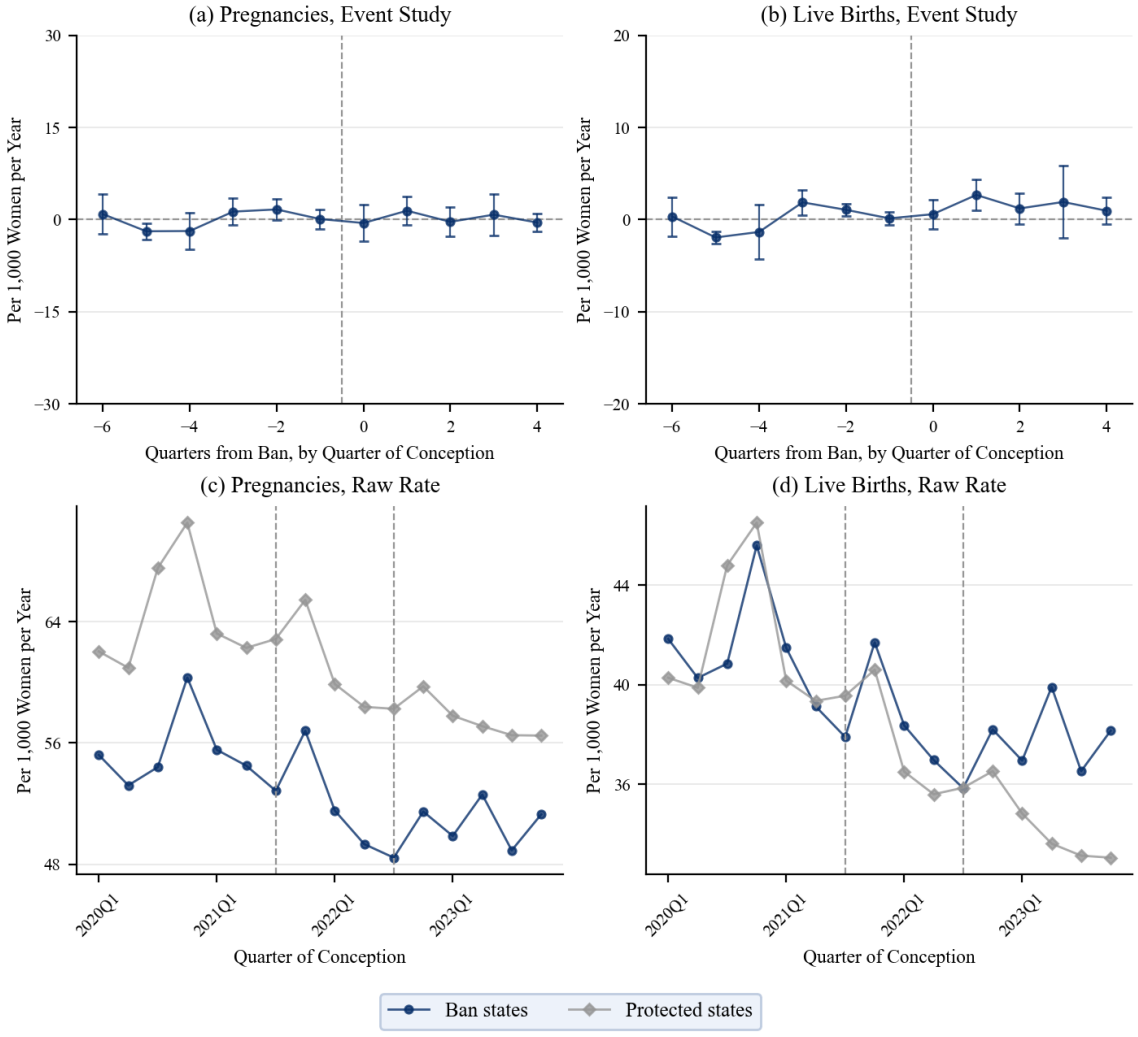
Notes: Event-study sample of 818,245 pregnancies in the continuous-enrollment sample (fewer for outcomes defined for observed endings or live births). Column 1 is the ban-state mean in event quarters −6 to −1. For severe maternal morbidity, column 2 is −7.6 (11.1) per 10,000 pregnancies. Columns 1, 2, and 6 are in each outcome's units (proportions or logs); preterm birth and a prolonged delivery stay (Table 1) are shares of live births. Under Controls, column 2 is the stacked difference-in-differences estimate of the post-ban quarters against the average of quarters −6 to −1, with event-time and sub-experiment by state fixed effects, the weights of Wing, Freedman and Hollingsworth (2024), and the individual controls (age and its square, six race and ethnicity groups, Medicaid, first observed pregnancy, multiple gestation, and the comorbidity score of Leonard et al. (2020)), with its standard error (SE), clustered by state, in parentheses; column 3 is that estimate as a percent of the pre-ban mean (100 times the coefficient for outcomes in logs), column 4 its 95 percent confidence interval (CI) on the same scale, and column 5 the p-value of the joint test that the pre-period event-study coefficients are zero. Under Raw, columns 6 and 7 repeat columns 2 and 3 without the individual controls. The insurance-billed abortion estimate exceeds 100 percent of its mean because insured abortion rose in protected states (Figure 1).

Figure 3. Heterogeneity: Percent of Each Subgroup's Pre-Ban Mean



Notes: Each marker is a separate stacked difference-in-differences (DiD) regression in the named group, as a percent of that group's pre-ban mean (100 times the coefficient for outcomes in logs), with a 95 percent confidence interval (CI) from standard errors clustered by state, in the Controls specification and the continuous-enrollment sample. The all-pregnancies row repeats Table 2, and each panel has its own scale. Medicaid pregnancies have a Medicaid payer in the month the pregnancy started; minority women have a Black race code or a Hispanic ethnicity code; high-risk women have at least one condition in the obstetric comorbidity index before pregnancy; and excluding Texas removes the Texas cohort. The dashed vertical line marks zero, and Supplemental Appendix Table A9 reports the estimates.

Figure 4. Fertility: Event Studies and Raw Rates



Notes: Panels (a) and (b) plot stacked difference-in-differences event-study coefficients for the pregnancy and live-birth rates on the panel of enrolled women aged 15 to 45 by state and quarter, with event-time and sub-experiment by state fixed effects, weights for enrolled women, and 95 percent confidence intervals from standard errors clustered by state; the pre-ban coefficients average zero, and the dashed vertical line falls between quarters −1 and 0. Their vertical axes span at least 50 percent of the ban-state pre-ban rate on either side of zero. Panels (c) and (d) plot the raw rates by quarter of conception in ban states (circles) and protected states (diamonds) in the continuous-enrollment sample, monthly rates averaged to quarters; dashed lines mark the third quarters of 2021 (Texas Senate Bill 8) and 2022 (*Dobbs*). All rates are per 1,000 enrolled women per year, and Supplemental Appendix Figure A1 shows the episode-enrolled sample.

REFERENCES

Abraha, Hiluf Ebuy, Jeffrey Buzas, Marta Bornstein, and Nansi S. Boghossian. 2026. "US Abortion Bans and Pregnancy-Associated Mortality." *JAMA Network Open*, 9(4): e264801.

Ailes, Elizabeth C., Regina M. Simeone, April L. Dawson, Emily E. Petersen, and Suzanne M. Gilboa. 2016. "Using Insurance Claims Data to Identify and Estimate Critical Periods in Pregnancy: An Application to Antidepressants." *Birth Defects Research Part A: Clinical and Molecular Teratology*, 106(11): 927–934.

American College of Obstetricians and Gynecologists. 2017. "Committee Opinion No. 700: Methods for Estimating the Due Date." *Obstetrics and Gynecology*, 129(5): e150–e154.

American College of Obstetricians and Gynecologists. 2018*a*. "ACOG Practice Bulletin No. 193: Tubal Ectopic Pregnancy." *Obstetrics and Gynecology*, 131(3): e91–e103.

American College of Obstetricians and Gynecologists. 2018*b*. "ACOG Practice Bulletin No. 200: Early Pregnancy Loss." *Obstetrics and Gynecology*, 132(5): e197–e207.

American College of Obstetricians and Gynecologists. 2022. "ACOG Statement on the Decision in *Dobbs v. Jackson*." https://www.acog.org/news/news-releases/2022/06/acog-statement-on-the-decision-in-dobbs-v-jackson, Accessed September 26, 2026.

American College of Obstetricians and Gynecologists, and Society of Family Planning. 2020. "Medication Abortion up to 70 Days of Gestation: ACOG Practice Bulletin No. 225." *Obstetrics and Gynecology*, 136(4): e31–e47.

Arey, Whitney, Klaira Lerma, Anitra Beasley, Lorie Harper, Ghazaleh Moayedi, and Kari White. 2022. "A Preview of the Dangerous Future of Abortion Bans: Texas Senate Bill 8." *New England Journal of Medicine*, 387(5): 388–390.

Arey, Whitney, Klaira Lerma, Emma Carpenter, Ghazaleh Moayedi, Lorie Harper, Anitra Beasley, Tony Ogburn, and Kari White. 2023. "Abortion Access and Medically Complex Pregnancies Before and After Texas Senate Bill 8." *Obstetrics and Gynecology*, 141(5): 995–1003.

Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. "Synthetic Difference-in-Differences." *American Economic Review*, 111(12): 4088–4118.

Bell, Suzanne O., Alexander M. Franks, Amy Ozinsky, Selena Anjur-Dietrich, Claire E. Margerison, Elizabeth A. Stuart, Avi Feller, and Alison Gemmill. 2026. "Abortion Bans and Maternal, Pregnancy-Related, and Pregnancy-Associated Mortality in 14 US States, 2016–2023: Estimated Impacts amid Substantial Measurement Challenges." *American Journal of Public Health*, 116(6): 819–828.

Bell, Suzanne O., Alexander M. Franks, David Arbour, Selena Anjur-Dietrich, Elizabeth A. Stuart, Eli Ben-Michael, Avi Feller, and Alison Gemmill. 2025. "US Abortion Bans and Fertility." *JAMA*, 333(15): 1324–1332.

Bell, Suzanne O., Elizabeth A. Stuart, and Alison Gemmill. 2023. "Texas' 2021 Ban on Abortion in Early Pregnancy and Changes in Live Births." *JAMA*, 330(3): 281–282.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. "Revisiting Event-Study Designs: Robust and Efficient Estimation." *The Review of Economic Studies*, 91(6): 3253–3285.

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225(2): 200–230.

Caraher, Raymond. 2024. "Do Abortion Bans Affect Reproductive and Infant Health? Evidence from Texas's 2021 Ban and Its Impact on Health Disparities." Political Economy Research Institute, University of Massachusetts Amherst Working Paper 606.

Caraher, Raymond. 2026. "Reproductive Injustice? A County-Level Analysis of the Impact of Abortion Restrictions on Abortion Rates." *American Journal of Health Economics*. Published online July 15, 2026.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics*, 134(3): 1405–1454.

Centers for Disease Control and Prevention. 2024. "Severe Maternal Morbidity." https://www.cdc.gov/maternal-infant-health/php/severe-maternal-morbidity/index.html, Accessed September 7, 2026.

Centers for Medicare & Medicaid Services. 2023. "Identifying Pregnant and Postpartum Beneficiaries in Medicaid and CHIP Administrative Data: Technical Specifications." *Maternal and Infant Health (MIH) Technical Specifications and Reference Codes, Center for Medicaid and CHIP Services.* https://www.medicaid.gov/medicaid/data-and-systems/downloads/macbis/mih_techspecs.pdf.

Centers for Medicare & Medicaid Services. 2026. "NPPES Downloadable File." https://download.cms.gov/nppes/NPI_Files.html, Accessed September 26, 2026.

Chen, Catherine, Deepshikha C. Ashana, Kevin Callison, Courtney Baker, Ralph Nicholas Burns, Jing Cao, Li Yuan, and Hayley B. Gershengorn. 2026. "Critical Illness Outcomes of Hospitalized Pregnant Women Following a Texas Abortion Ban." *American Journal of Respiratory and Critical Care Medicine*, 212(8): 1730–1739.

Chomistek, Andrea K., Kelesitse Phiri, Michael C. Doherty, Jenna F. Calderbank, Stephanie E. Chiuve, Brenda Hinman McIlroy, Michael C. Snabes, Cheryl Enger, and John D. Seeger. 2023. "Development and Validation of ICD-10-CM-Based Algorithms for Date of Last Menstrual Period, Pregnancy Outcomes, and Infant Outcomes." *Drug Safety*, 46(2): 209–222.

Clarke, Damian, and Hanna Mühlrad. 2021. "Abortion Laws and Women's Health." *Journal of Health Economics*, 76: 102413. Corrigendum in Journal of Health Economics 80 (2021): 102535.

Crump, Casey, Jan Sundquist, and Kristina Sundquist. 2024. "Adverse Pregnancy Outcomes and Long-Term Mortality in Women." *JAMA Internal Medicine*, 184(6): 631–640.

Curhan, Talia. 2026. "State Bans on Abortion Throughout Pregnancy." *Guttmacher Institute.* https://www.guttmacher.org/state-policy/explore/state-policies-abortion-bans.

Daw, Jamie R., Laura A. Hatfield, Katherine Swartz, and Benjamin D. Sommers. 2017. "Women in the United States Experience High Rates of Coverage "Churn" in Months before and after Childbirth." *Health Affairs*, 36(4): 598–606.

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review*, 110(9): 2964–2996.

DeMillo, Andrew. 2021. "Federal Judge Blocks Arkansas Law Banning Most Abortions." *Associated Press, via PBS NewsHour.* https://www.pbs.org/newshour/nation/federal-judge-blocks-arkansas-law-banning-most-abortions.

Dench, Daniel L., Caitlin K. Myers, and Mayra Pineda-Torres. 2025. "The Road Not Taken: How Driving Distance and Appointment Availability Shape the Effects of Abortion Bans." National Bureau of Economic Research Working Paper 33548.

Dench, Daniel, Mayra Pineda-Torres, and Caitlin Myers. 2024. "The Effects of Post-Dobbs Abortion Bans

on Fertility." *Journal of Public Economics*, 234: 105124.

Deshpande, Manasi, and Yue Li. 2019. "Who Is Screened Out? Application Costs and the Targeting of Disability Programs." *American Economic Journal: Economic Policy*, 11(4): 213–248.

Dobbs v. Jackson Women's Health Organization. 2022. *597 U.S. 215 (Supreme Court of the United States).*

Ellison, Jacqueline E., Brittany L. Brown-Podgorski, and Jake R. Morgan. 2024. "Changes in Permanent Contraception Procedures among Young Adults Following the Dobbs Decision." *JAMA Health Forum*, 5(4): e240424.

Farin, Sherajum Monira, Lauren Hoehn-Velasco, and Michael F. Pesko. 2024. "The Impact of Legal Abortion on Maternal Mortality." *American Economic Journal: Economic Policy*, 16(3): 174–216.

Felix, Mabel, Laurie Sobel, and Alina Salganicoff. 2024. "A Review of Exceptions in State Abortion Bans: Implications for the Provision of Abortion Services." *KFF*. https://www.kff.org/womens-health-policy/a-review-of-exceptions-in-state-abortions-bans-implications-for-the-provision-of-abortion-services/.

Felix, Mabel, Laurie Sobel, and Alina Salganicoff. 2025. "Criminal Penalties for Physicians in State Abortion Bans." *KFF*. https://www.kff.org/womens-health-policy/criminal-penalties-for-physicians-in-state-abortion-bans/.

Finer, Lawrence B., Lori F. Frohwirth, Lindsay A. Dauphinee, Susheela Singh, and Ann M. Moore. 2005. "Reasons U.S. Women Have Abortions: Quantitative and Qualitative Perspectives." *Perspectives on Sexual and Reproductive Health*, 37(3): 110–118.

Fischer, Stefanie, Heather Royer, and Corey White. 2018. "The Impacts of Reduced Access to Abortion and Family Planning Services on Abortions, Births, and Contraceptive Purchases." *Journal of Public Economics*, 167: 43–68.

Frederiksen, Brittni, Usha Ranji, Ivette Gomez, and Alina Salganicoff. 2023. "A National Survey of OBGYNs' Experiences After Dobbs." *KFF*. https://www.kff.org/womens-health-policy/a-national-survey-of-obgyns-experiences-after-dobbs/.

Freedman, Lori R., Uta Landy, and Jody Steinauer. 2008. "When There's a Heartbeat: Miscarriage Management in Catholic-owned Hospitals." *American Journal of Public Health*, 98(10): 1774–1778.

Gardner, Graham. 2024. "The Maternal and Infant Health Consequences of Restricted Access to Abortion in the United States." *Journal of Health Economics*, 98: 102938.

Gemmill, Alison, Alexander M. Franks, Selena Anjur-Dietrich, Amy Ozinsky, David Arbour, Elizabeth A. Stuart, Eli Ben-Michael, Avi Feller, and Suzanne O. Bell. 2025. "US Abortion Bans and Infant Mortality." *JAMA*, 333(15): 1315–1323.

Gemmill, Alison, Claire E. Margerison, Elizabeth A. Stuart, and Suzanne O. Bell. 2024. "Infant Deaths after Texas' 2021 Ban on Abortion in Early Pregnancy." *JAMA Pediatrics*, 178(8): 784–791.

Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics*, 225(2): 254–277.

Grossman, Daniel, Carole Joffe, Shelly Kaller, Katrina Kimport, Elizabeth Kinsey, Klaira Lerma, Natalie Morris, and Kari White. 2023. "Care Post-Roe: Documenting Cases of Poor-quality Care Since the Dobbs Decision." Advancing New Standards in Reproductive Health (ANSIRH), University of California, San Francisco. Preliminary findings. https://www.ansirh.org/sites/default/files/2023-05/Care%20Post-Roe%20P

reliminary%20Findings.pdf.

Grossman, Tracy B., and Stephen T. Chasen. 2020. "Abortion for Fetal Genetic Abnormalities: Type of Abnormality and Gestational Age at Diagnosis." *AJP Reports*, 10(1): e87–e92.

Guttmacher Institute. 2021*a*. "Regulating Insurance Coverage of Abortion (as of November 1, 2021)." *Guttmacher Institute, State Laws and Policies. Archived December 16, 2021.* http://web.archive.org/web/20211216042906/https://www.guttmacher.org/state-policy/explore/regulating-insurance-coverage-abortion.

Guttmacher Institute. 2021*b*. "Targeted Regulation of Abortion Providers (as of November 1, 2021)." *Guttmacher Institute, State Laws and Policies. Archived January 2, 2022.* http://web.archive.org/web/20220102054820/https://www.guttmacher.org/state-policy/explore/targeted-regulation-abortion-providers.

Guttmacher Institute. 2022. "An Overview of Abortion Laws (as of January 1, 2022)." *Guttmacher Institute, State Laws and Policies. Archived January 7, 2022.* http://web.archive.org/web/20220107211004/https://www.guttmacher.org/state-policy/explore/overview-abortion-laws.

Guttmacher Institute. 2023. "The High Toll of US Abortion Bans: Nearly One in Five Patients Now Traveling Out of State for Abortion Care." https://www.guttmacher.org/2023/12/high-toll-us-abortion-bans-nearly-one-five-patients-now-traveling-out-state-abortion-care.

Harris, Lisa H. 2022. "Navigating Loss of Abortion Services: A Large Academic Medical Center Prepares for the Overturn of Roe v. Wade." *New England Journal of Medicine*, 386(22): 2061–2064.

Henshaw, Stanley K., Theodore J. Joyce, Amanda Dennis, Lawrence B. Finer, and Kelly Blanchard. 2009. "Restrictions on Medicaid Funding for Abortions: A Literature Review." Guttmacher Institute, New York. https://www.guttmacher.org/report/restrictions-medicaid-funding-abortions-literature-review.

Hill, Latoya, Samantha Artiga, Usha Ranji, Ivette Gomez, and Nambi Ndugga. 2024. "What Are the Implications of the Dobbs Ruling for Racial Disparities?" *KFF.* https://www.kff.org/womens-health-policy/what-are-the-implications-of-the-dobbs-ruling-for-racial-disparities/.

Hoyert, Donna L. 2026. "Maternal Mortality Rates in the United States, 2024." National Center for Health Statistics NCHS Health E-Stat 113.

Imbens, Guido W., and Jeffrey M. Wooldridge. 2009. "Recent Developments in the Econometrics of Program Evaluation." *Journal of Economic Literature*, 47(1): 5–86.

Jerman, Jenna, Rachel K. Jones, and Tsuyoshi Onda. 2016. "Characteristics of U.S. Abortion Patients in 2014 and Changes since 2008." Guttmacher Institute Report. https://www.guttmacher.org/report/characteristics-us-abortion-patients-2014.

Landay, Sophia, Emily Newton-Hoe, Subasri Narasimhan, Anitra Beasley, Jessica Adkins, and Sara Neill. 2026. "Abortion Bans and Pregnancy-Related Care Across Physician Specialties: A Qualitative Study." *JAMA Network Open*, 9(6): e2619644.

Leonard, Stephanie A., Chris J. Kennedy, Suzan L. Carmichael, Deirdre J. Lyell, and Elliott K. Main. 2020. "An Expanded Obstetric Comorbidity Scoring System for Predicting Severe Maternal Morbidity." *Obstetrics and Gynecology*, 136(3): 440–449.

Leonard, Stephanie A., Elliott K. Main, Karen A. Scott, Jochen Profit, and Suzan L. Carmichael. 2019. "Racial and Ethnic Disparities in Severe Maternal Morbidity Prevalence and Trends." *Annals of Epidemiology*, 33: 30–36.

Levine, Phillip B., Amy B. Trainor, and David J. Zimmerman. 1996. "The Effect of Medicaid Abortion Funding Restrictions on Abortions, Pregnancies and Births." *Journal of Health Economics*, 15(5): 555–578.

Lift Louisiana, Physicians for Human Rights, RH Impact, and Center for Reproductive Rights. 2024. "Criminalized Care: How Louisiana's Abortion Bans Endanger Patients and Clinicians." *Joint report of Lift Louisiana, Physicians for Human Rights, Reproductive Health Impact, and Center for Reproductive Rights, March 19.* https://phr.org/our-work/resources/louisiana-abortion-bans/.

Lindo, Jason M., and Mayra Pineda-Torres. 2021. "New Evidence on the Effects of Mandatory Waiting Periods for Abortion." *Journal of Health Economics*, 80: 102533.

Lindo, Jason M., Caitlin Knowles Myers, Andrea Schlosser, and Scott Cunningham. 2020. "How Far Is Too Far? New Evidence on Abortion Clinic Closures, Access, and Abortions." *Journal of Human Resources*, 55(4): 1137–1160.

Lisonkova, Sarka, Jayson Potts, Giulia M. Muraca, Neda Razaz, Yasser Sabr, Wee-Shian Chan, and Michael S. Kramer. 2017. "Maternal Age and Severe Maternal Morbidity: A Population-Based Retrospective Cohort Study." *PLOS Medicine*, 14(5): e1002307.

MacDonald, Andrea, Hayley B. Gershengorn, and Deepshikha Charan Ashana. 2022. "The Challenge of Emergency Abortion Care Following the Dobbs Ruling." *JAMA*, 328(17): 1691–1692.

Matcho, Amy, Patrick Ryan, Daniel Fife, Dina Gifkins, Chris Knoll, and Andrew Friedman. 2018. "Inferring Pregnancy Episodes and Outcomes within a Network of Observational Databases." *PLOS ONE*, 13(2): e0192033.

Medicaid and CHIP Payment and Access Commission. 2019. "Managed Care Overview." *MACPAC.* https://www.macpac.gov/subtopic/managed-care-overview/.

Miller, Sarah, Laura R. Wherry, and Diana Greene Foster. 2023. "The Economic Consequences of Being Denied an Abortion." *American Economic Journal: Economic Policy*, 15(1): 394–437.

Moll, Keran, Hui Lee Wong, Kathryn Fingar, Shayan Hobbi, Minya Sheng, Timothy A. Burrell, Linda O. Eckert, Flor M. Munoz, Bethany Baer, Azadeh Shoaibi, and Steven Anderson. 2021. "Validating Claims-Based Algorithms Determining Pregnancy Outcomes and Gestational Age Using a Linked Claims-Electronic Medical Record Database." *Drug Safety*, 44(11): 1151–1164.

Myers, Caitlin. 2024. "Forecasts for a Post-Roe America: The Effects of Increased Travel Distance on Abortions and Births." *Journal of Policy Analysis and Management*, 43(1): 39–62.

Myers, Caitlin, and Daniel Ladd. 2020. "Did Parental Involvement Laws Grow Teeth? The Effects of State Restrictions on Minors' Access to Abortion." *Journal of Health Economics*, 71: 102302.

Nagle, Amanda, Goleen Samari, Lauren Thaxton, and Alison Gemmill. 2025. "Blood Transfusions for Pregnancy Loss in Texas Before and After Abortion Bans, 2017–2023." *American Journal of Public Health*, 115(11): 1887–1894.

Nambiar, Anjali, Elaine L. Duryea, Lindsay R. Thiele, Patricia Santiago-Munoz, David B. Nelson, Catherine Y. Spong, and Christina C. Baker. 2025. "Maternal Morbidity with Expectant Management of Life-Limiting Fetal Conditions." *JAMA Network Open*, 8(7): e2521883.

Nash, Elizabeth. 2021. "State Policy Trends 2021: The Worst Year for Abortion Rights in Almost Half a Century." *Guttmacher Institute.* https://www.guttmacher.org/article/2021/12/state-policy-trends-2021-worst-year-abortio

n-rights-almost-half-century.

Nash, Elizabeth, and Isabel Guarnieri. 2022. "13 States Have Abortion Trigger Bans: Here's What Happens When Roe Is Overturned." *Guttmacher Institute.* https://www.guttmacher.org/article/2022/06/13-states-have-abortion-trigger-bans-heres-what-happens-when-roe-overturned.

Nash, Elizabeth, Lizamarie Mohammed, Olivia Cappello, and Sophia Naide. 2019. "State Policy Trends 2019: A Wave of Abortion Bans, but Some States Are Fighting Back." *Guttmacher Institute.* https://www.guttmacher.org/article/2019/12/state-policy-trends-2019-wave-abortion-bans-some-states-are-fighting-back.

Nuss, Emily, Mari Iwasaki, Lindsay Robbins, Peggy Ye, Misa Hayasaka, George Saade, and Tetsuya Kawakita. 2025. "Maternal Mortality According to State Abortion Legislative Climate Following the US Supreme Court's Dobbs v. Jackson Women's Health Organization Ruling." *Pregnancy*, 1(6): e70128.

Odouard, Ilina C., Jenny S. Guadamuz, Apurba Chakraborty, G. Caleb Alexander, and Dima M. Qato. 2024. "Induced Abortion and Out-of-State Travel among Insured Women before and after the Dobbs v. Jackson Women's Health Organization Decision." *O&G Open*, 1(2): 16.

Osterman, Michelle J. K., Brady E. Hamilton, Joyce A. Martin, Anne K. Driscoll, and Claudia P. Valenzuela. 2025. "Births: Final Data for 2023." National Center for Health Statistics National Vital Statistics Reports 74(1).

Pop-Eleches, Cristian. 2006. "The Impact of an Abortion Ban on Socioeconomic Outcomes of Children: Evidence from Romania." *Journal of Political Economy*, 114(4): 744–773.

Presser, Lizzie, and Kavitha Surana. 2024. "A Third Woman Has Died under Texas' Abortion Ban as Doctors Reach for Riskier Miscarriage Treatments." *ProPublica and The Texas Tribune, November 27.* https://www.texastribune.org/2024/11/27/texas-abortion-death-porsha-ngumezi/.

Presser, Lizzie, Andrea Suozzo, Sophie Chou, and Kavitha Surana. 2025. "Texas Banned Abortion. Then Sepsis Rates Soared." *ProPublica, February 20.* https://www.propublica.org/article/texas-abortion-ban-sepsis-maternal-mortality-analysis.

Ralph, Lauren J., Eleanor Bimla Schwarz, Daniel Grossman, and Diana Greene Foster. 2019. "Self-Reported Physical Health of Women Who Did and Did Not Terminate Pregnancy after Seeking Abortion Services." *Annals of Internal Medicine*, 171(4): 238–247.

Ranji, Usha, Alina Salganicoff, Jennifer Tolbert, Brittni Frederiksen, and Ivette Gomez. 2025. "5 Key Facts About Medicaid and Pregnancy." *KFF.* https://www.kff.org/medicaid/5-key-facts-about-medicaid-and-pregnancy/.

Roberts, Sarah C. M., Heather Gould, Katrina Kimport, Tracy A. Weitz, and Diana Greene Foster. 2014. "Out-of-Pocket Costs and Insurance Coverage for Abortion in the United States." *Women's Health Issues*, 24(2): e211–e218.

Rossen, Lauren M., Brady E. Hamilton, Joyce C. Abma, Elizabeth C. W. Gregory, Vladislav Beresovsky, Adriana V. Resendez, Anjani Chandra, and Joyce A. Martin. 2023. "Updated Methodology to Estimate Overall and Unintended Pregnancy Rates in the United States." National Center for Health Statistics Vital and Health Statistics Series 2, No. 201.

Sabbath, Erika L., Samantha M. McKetchnie, Kavita S. Arora, and Mara Buchbinder. 2024. "US Obstetrician-Gynecologists' Perceived Impacts of Post–Dobbs v Jackson State Abortion Bans." *JAMA Network*

Open, 7(1): e2352109.

Salganicoff, Alina, Laurie Sobel, Ivette Gomez, and Amrutha Ramaswamy. 2024. "The Hyde Amendment and Coverage for Abortion Services under Medicaid in the Post-Roe Era." *KFF Issue Brief.* https://www.kff.org/womens-health-policy/the-hyde-amendment-and-coverage-for-abortion-services-under-medicaid-in-the-post-roe-era/.

Sarayani, Amir, Xi Wang, Thuy Nhu Thai, Yasser Albogami, Nakyung Jeon, and Almut G. Winterstein. 2020. "Impact of the Transition from ICD-9-CM to ICD-10-CM on the Identification of Pregnancy Episodes in US Health Insurance Claims Data." *Clinical Epidemiology*, 12: 1129–1138.

Seitz, Amanda. 2024. "Over 100 Pregnant Women, Some Bleeding or in Labor, Turned Away from Emergency Rooms since Dobbs Ruling." *Associated Press, via Mississippi Free Press, August 12.* https://www.mississippifreepress.org/over-100-pregnant-women-some-bleeding-or-in-labor-turned-away-from-emergency-rooms-since-dobbs-ruling/.

Simmons-Duffin, Selena. 2022. "Doctors Who Want to Defy Abortion Laws Say It's Too Risky." *NPR, November 23.* https://www.npr.org/sections/health-shots/2022/11/23/1137756183/doctors-who-want-to-defy-abortion-laws-say-its-too-risky.

Singh, Parvati, and Maria F. Gallo. 2024. "National Trends in Infant Mortality in the US After Dobbs." *JAMA Pediatrics*, 178(12): 1364–1366.

SisterSong v. Governor of Georgia. 2022. *SisterSong Women of Color Reproductive Justice Collective v. Governor of the State of Georgia, 40 F.4th 1320 (11th Cir. 2022).*

Smith, Mikaela H., Zoe Muzyczka, Payal Chakraborty, Elaina Johns-Wolfe, Jenny Higgins, Danielle Bessett, and Alison H. Norris. 2022. "Abortion Travel within the United States: An Observational Study of Cross-State Movement to Obtain Abortion Care in 2017." *The Lancet Regional Health - Americas*, 10: 100214.

Society of Family Planning. 2025. "#WeCount Report: April 2022 through June 2025." https://doi.org/10.46621/750591yxmcwh.

Stevenson, Amanda Jean, and Leslie Root. 2024. "Trends in Maternal Death Post-Dobbs v Jackson Women's Health." *JAMA Network Open*, 7(8): e2430035.

Sun, Liyang, and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics*, 225(2): 175–199.

Texas State Law Library. 2026*a*. "Abortion Laws: Criminal Penalties; Civil Penalties." *Texas State Law Library research guide.* https://guides.sll.texas.gov/abortion-laws/criminal-penalties; https://guides.sll.texas.gov/abortion-laws/civil-penalties.

Texas State Law Library. 2026*b*. "Timeline of Texas Abortion Laws." *Texas State Law Library research guide.* https://guides.sll.texas.gov/abortion-laws/history-of-abortion-laws.

Thompson, Terri-Ann. 2019. "Insured but Still Uncovered: A Review of Private Insurance Coverage of Abortion Care." *Ibis Reproductive Health.* https://www.ibisreproductivehealth.org/publications/insured-still-uncovered-review-private-insurance-coverage-abortion-care.

Tolbert, Jennifer, and Meghana Ammula. 2023. "10 Things to Know About the Unwinding of the Medicaid Continuous Enrollment Provision." *KFF.* https://www.kff.org/medicaid/10-things-to-know-about-the-unwinding-of-the-medicaid-continuous-enrollment-provision/.

Upadhyay, Ushma D., Chris Ahlbach, Shelly Kaller, Clara Cook, and Isabel Muñoz. 2022. "Trends in Self-Pay Charges and Insurance Acceptance for Abortion in the United States, 2017–20." *Health Affairs*, 41(4): 507–515.

Venator, Joanna, and Jason Fletcher. 2021. "Undue Burden beyond Texas: An Analysis of Abortion Clinic Closures, Births, and Abortions in Wisconsin." *Journal of Policy Analysis and Management*, 40(3): 774–813.

Whole Woman's Health v. Jackson. 2021. *Order denying application for injunctive relief, No. 21A24, September 1, 2021 (Supreme Court of the United States).* https://www.supremecourt.gov/opinions/20pdf/21a24_8759.pdf.

Wing, Coady, Seth M. Freedman, and Alex Hollingsworth. 2024. "Stacked Difference-in-Differences." National Bureau of Economic Research Working Paper 32054.

Xu, Xiao, Ling Chen, Vrunda B. Desai, Cary P. Gross, Craig Evan Pollack, Peter E. Schwartz, and Jason D. Wright. 2024. "Tubal Sterilization Rates by State Abortion Laws after the Dobbs Decision." *JAMA*, 332(14): 1204–1206.

Supplemental Appendix

It Was Already Dobbs: Abortion Bans and Maternal Health

Appendix

This appendix supports the main text. Section A1 documents the timing of the bans, how each state is classified, and the legal and policy sources. Section A2 describes how pregnancy episodes are built and dated. Section A3 describes the samples: their construction and pre-period balance. Section A4 compares the continuous-enrollment sample with the episode-enrolled sample. Section A5 reports the subgroup estimates, and Section A6 breaks the adverse-outcome composite into its components. Section A7 reports the fertility estimates and the composition of the pregnant population. Section A8 documents the stacked design and compares it with alternative estimators, two-way fixed effects, other reference periods, and synthetic difference-in-differences, and Section A9 presents the close-to-care design. Section A10 defines the outcomes.

A1. Ban Timing, State Classification, and Policy Sources

Policy sources

Table A1 lists the statute behind each ban. The *Dobbs* decision (*Dobbs v. Jackson Women's Health Organization*, 2022) overturned *Roe v. Wade* (*Roe v. Wade*, 1973). In Texas, SB8 (Hughes, 2021) took effect on September 1, 2021, and because private lawsuits rather than state officials enforced it, there was no official a federal court could enjoin to stop it (*Whole Woman's Health v. Jackson*, 2021). The state's trigger law (Texas Legislature, 2021) took effect on August 25, 2022 (Texas State Law Library, 2026). After *Dobbs*, federal courts allowed the previously enjoined bans of Alabama (*Robinson v. Marshall*, 2022) and Georgia (*SisterSong v. Governor of Georgia*, 2022) to take effect. The author of the Texas law and Kentucky Senate Republicans have each described the reach of their state's health exceptions (Hughes, 2024; Kentucky Senate GOP, 2025), and reporting on Kate Cox's case in Texas documents women denied terminations despite serious health risks (Klibanoff, 2023; Goldhill, 2023). Further investigative reporting documents pregnant women in Texas who died after emergency care was delayed (Surana and Presser, 2024; Surana and Jaramillo, 2024). The Texas sepsis figures in Section I.B come from hospital discharge records analyzed by

Presser et al. (2025), whose method is described in Suozzo, Chou and Presser (2025). Among protected states, Illinois and California widened protections and coverage after *Dobbs* (Office of Governor JB Pritzker, 2023; Office of Governor Gavin Newsom, 2023).

A2. Constructing and Dating Pregnancy Episodes

Endings

To identify the ending of a pregnancy, I identify every claim carrying a diagnosis, procedure, or diagnosis-related-group code for a live birth, stillbirth, molar pregnancy, ectopic pregnancy, induced abortion, or spontaneous abortion, each as a separate ending type. Codes that indicate the ending of a pregnancy may span several days, for example several days of labor and the postpartum care that follows, so I link ending codes that fall within 30 days of the previous one into a single ending, dated at the earliest claim. Where an ending carries codes for more than one type, I apply the hierarchy that Moll et al. (2021) validated against physician-adjudicated records, which orders endings by how reliably each is coded, from live birth, stillbirth, molar pregnancy, ectopic pregnancy, and induced abortion to spontaneous abortion (miscarriage). Because the abortion codes include the billing codes for mifepristone and misoprostol, which also treat miscarriage, and the hierarchy ranks induced abortion above miscarriage, a miscarriage managed with these medications can be counted as an abortion, so the classification errs toward counting too many abortions rather than too few. Additionally, because women may use the health care and insurance systems intermittently throughout pregnancy, especially for an early miscarriage or an abortion paid outside insurance, some endings imply a pregnancy that began implausibly soon after the woman's previous one ended. To account for this, I treat an ending whose estimated start falls within the recovery period after her previous ending, 24 weeks after a birth or stillbirth and 60 days after any other ending, as part of the earlier pregnancy, and I exclude it.

Pregnancy start

I date each pregnancy from the first day of the estimated last menstrual period, the clinical convention from which gestational age is counted (American College of Obstetricians and Gynecologists, 2017), and call this date the pregnancy start. I largely rely on the Z3A diagnosis code, which records the gestational week at a visit (American Academy of Professional Coders, 2024).[1] For each Z3A code in the 300 days before the ending, I subtract the recorded weeks from the date of service, and I take the median of these estimates, assigning four weeks to codes below eight weeks,

[1] Where an embryo transfer is observed, I set the start 14 days before it.

as the validated algorithm does (Chomistek et al., 2023). In my analysis sample, Z3A codes date 87 percent of ban-state and 75 percent of protected-state pregnancies (Table A3). Against last-period dates that an epidemiologist adjudicated from medical records, this method has a median absolute error of four days, with 65 percent of estimates within a week (Chomistek et al., 2023). However, its accuracy varies by ending. For example, 89.7 percent of premature deliveries fall within seven days, but only 34.3 percent of spontaneous abortions do, because early losses often end before the first visit at which a Z3A code is recorded. These facts give the first of two reasons I measure event time in quarters rather than months (Section IV), as miscarriages alone end about 16 percent of ban-state pregnancies with an observed ending before the bans, so monthly event time would add noise from dating error. Second, pooling to quarters estimates each event-time coefficient on more pregnancies, at the cost of making changes within a quarter harder to detect.

I date episodes without a Z3A code from the earliest test performed at a known gestational age, adapting the priority order in Moll et al. (2021).[2] Failing all of these, I assign the ending-type defaults validated by Chomistek et al. (2023) (39 weeks for a singleton live birth, 36 for a multiple birth, 28 for a stillbirth, 10 for an abortion of either type, and 8 for ectopic and molar pregnancies). Table A3 reports the share of episodes dated by each source.

Pregnancies without an observed ending

Some pregnancies carry pregnancy care but no ending code. In my main analysis sample, they are 6.2 percent of ban-state pregnancies conceived in the six quarters before each state's ban (Section II.D). To account for these pregnancies, I build episodes with no observed ending from pregnancy-indicating claims (for example, prenatal visits, gestational-age codes, and diagnoses of adverse pregnancy outcomes) that fall outside every other observed pregnancy for a woman,[3] as Sarayani et al. (2020) do for pregnancies with unknown outcomes. I group these claims into clusters, starting a new cluster whenever more than 60 days separate two claims, similar to the two-month separation between pregnancies that Ailes et al. (2016) require, and a cluster with claims on at least two dates becomes an episode. Because nothing in the claims marks how such a pregnancy ended, I end the episode on the date of its last pregnancy-indicating claim. I date its start from its latest Z3A code or, if it has none, eight weeks before its first visit, close to the median time from the last menstrual period to the first pregnancy-care claim among deliveries in insurance claims

[2]The order is nuchal translucency ultrasound, cell-free DNA screening, and maternal serum screening, then the detailed fetal anatomy ultrasound and glucose tolerance testing. I exclude hCG tests from the serum-screening tier, because they are drawn early in pregnancy to work up a suspected ectopic pregnancy or loss (American College of Obstetricians and Gynecologists, 2018*a*,*b*) rather than at the gestational age that tier assumes.

[3]An observed pregnancy runs from 90 days before its start through its 42-day postpartum window.

(Sarayani et al., 2020). Finally, a cluster whose last claim falls within 200 days before an observed ending is likely early care for that same pregnancy rather than a separate one, so I exclude it.[4]

A3. Sample Construction and Descriptives

Additional restrictions

The pregnancy episode construction described in Section II.C and Appendix A2 yields 10,635,041 pregnancies for women aged 15 to 45, 1,848,589 of them to continuously enrolled women. Among those, I keep pregnancies in the 15 ban states (the 13 treated states, Indiana, and South Carolina) and 25 protected jurisdictions (1,576,819) and pregnancies that started in 2020 or later (1,135,994), because some pregnancies observed in my window started in 2019 and continued into 2020. I require the pregnancy to start no later than February 2024 so that a typical 40-week pregnancy can be followed to its ending (996,163). I exclude observed endings with an implied gestation shorter than 28 or longer than 315 days, about 0.1 percent of the remaining pregnancies, which likely reflect coding or dating errors. The continuous-enrollment sample contains 995,377 pregnancies. Table A4 reports counts after each step.

The pre-pregnancy year

I exclude pregnancies that start in 2019 (Section II.D). For pregnancies that start in 2020, part of the pre-pregnancy year falls in 2019, before the enrollment requirement begins, so their baseline health measures draw on whatever 2019 claims the woman has. Of these 2020 pregnancies, 81 percent were also enrolled for the whole year before the pregnancy, from 72 percent of those that started in the first quarter to 91 percent in the fourth. The share is similar in ban states (83 percent) and protected states (81 percent), so a shorter look-back does not fall more on one side of the comparison of baseline health.

Provider locations

Claims locate care only through the registered address of the provider's NPI, which is why in my core analysis I assign treatment by the woman's state of residence in the enrollment file. Of all claim lines, 25.5 percent name no provider, and 24.3 percent are billed under an organizational NPI, whose registered address may be a billing office or a health system's headquarters rather than the place of care.

[4]This 200-day window is comparable to the 168 to 182 days that Matcho et al. (2018) require between an earlier outcome and a live birth.

Pre-pregnancy comorbidity score

The conditions of the obstetric comorbidity index (Section II.D) include pulmonary hypertension, cardiac and renal disease, chronic hypertension, pre-existing diabetes, autoimmune and bleeding disorders, anemia, asthma, and mental health and substance use disorders, each measured in the year before conception. I omit placenta accreta, which is diagnosed during pregnancy. The score weights each pre-pregnancy condition by the points Leonard et al. (2020) estimated for severe maternal morbidity (Table A22). I also report the number of conditions and an indicator for any condition, which do not depend on the weights (Section VI.B).

A4. Comparison with the Episode-Enrolled Sample

The episode-enrolled sample requires enrollment from twelve months before the pregnancy start through three months after the ending, so it keeps women who move between insurers or plans outside their pregnancies; it contains 2,416,491 pregnancies. The main text does not use it, because its composition changes over the period for reasons the bans likely did not cause. First, insurers enter and leave the data, and because plans can be regional, each entry or exit can shift who is in the sample. Second, its requirement of coverage through three months after each pregnancy ends means that its membership after mid-2022 is set by the end of the pandemic-era Medicaid continuous-enrollment provision on March 31, 2023, after which states began removing women from Medicaid (Tolbert and Ammula, 2023), and in this sample the resulting loss of members is concentrated in ban states. Between conceptions in the second quarter of 2022 and the third quarter of 2023, the number of episode-enrolled pregnancies changes by -42 percent in ban states and -22 percent in protected states, and the Medicaid share of ban-state pregnancies in that sample falls from 61.5 to 45.9 percent, while the protected-state share holds near 60.4 percent. Over the same quarters the continuous-enrollment sample changes by 2 and -2 percent. Figure A1 shows the birth and pregnancy rates in the episode-enrolled sample, where the ban-state birth rate falls to the protected-state rate after 2022, as the end of Medicaid continuous enrollment would predict in a sample whose membership it sets. Table A8 reports the difference-in-differences estimates in both samples under the Controls specification, and Figure A2 overlays their event studies. The maternal health estimates are similar in the two samples. In the episode-enrolled sample, the adverse-outcome composite changes by 1.3 percent and severe maternal morbidity by 0.5 percent, ruling out increases larger than 4.9 and 6.4 percent. The pregnancy endings differ. There the shift runs toward miscarriages rather than live births, as the miscarriage share rises 10.1 percent ($p = 0.03$) and the live-birth share changes by 1.0 percent ($p = 0.59$). Because the end of Medicaid

continuous enrollment changes who remains in this sample, less is known about how much of that difference reflects the bans rather than its changing composition.

A5. Subgroup Estimates

Table A9 reports the stacked estimates behind Figure 3 for each subgroup of Section V.D. Table A10 repeats the maternal health and care outcomes within live births and within losses. Because the bans changed how pregnancies ended, these within-ending estimates describe the pregnancies in each group rather than isolate effects of the bans.

A6. Components of the Adverse-Outcome Composite

The adverse-outcome composite equals one if a pregnancy has any of five diagnoses (Section II.E). Table A11 estimates each on its own and Figure A3 shows their event studies. Hypertensive disorders, reported in Table 2, are not part of the composite.

A7. Fertility and the Composition of the Pregnant Population

Table A12 reports the birth and pregnancy rates of Section VI.A for all enrolled women and for each subgroup, and Table A13 reports the stacked estimates for the pre-pregnancy health and demographic measures discussed in Section VI.B.

A8. Alternative Estimators

The stacked design and two-way fixed effects

The event-study sample is the continuous-enrollment pregnancies conceived in the eleven quarters around each cohort's ban, from the second quarter of 2020 through the fourth quarter of 2022 for Texas, from the first quarter of 2021 through the third quarter of 2023 for the *Dobbs* cohort, and from the second quarter of 2021 through the fourth quarter of 2023 for West Virginia. Table A4 reports the number of distinct pregnancies in this event study sample, 818,245, and Table A14 the pregnancies in each sub-experiment. Because a control pregnancy enters every sub-experiment whose window covers it, the stacked regression has more observations than distinct pregnancies.

Reference quarter

The pooled estimates in Table 2 compare the post-ban quarters with the average of quarters -6 to -1 (Section IV). Table A15 repeats them against quarter -1 alone, as in Wing, Freedman and Hollingsworth (2024), where the pre-period interactions stay in the pooled regression, and

against quarter −2, and Figure A5 repeats the event studies with quarter −1 omitted. The post-ban quarters are the same, so each estimate moves only by the difference between the reference periods. Against quarter −1, the adverse-outcome composite rises 3.7 percent ($p = 0.06$), and severe maternal morbidity changes by 1.7 percent, ruling out increases larger than 15.4 percent. Against quarter −2, the suggestion of harm in the composite weakens. It falls from 2.8 to 1.8 percent of the pre-ban mean ($p = 0.40$), with an interval that rules out increases larger than 5.9 percent, and its pre-period coefficients do not jointly differ from zero ($p = 0.98$). Severe maternal morbidity changes by 1.2 percent, with an interval that rules out increases larger than 12.1 percent, and preterm birth by −1.1 percent. Hypertensive disorders move the other way, to 2.1 percent ($p = 0.06$), and the fall in prolonged delivery stays shrinks to 2.6 percent ($p = 0.21$). The number of visits rises 4.0 percent ($p = 0.001$), although its pre-period coefficients jointly differ from zero.

Synthetic difference-in-differences

If protected states are a poor comparison for ban states, the stacked estimates could reflect differences in trends rather than the bans. Synthetic difference-in-differences (Arkhangelsky et al., 2021) addresses this by choosing weights on protected states and on pre-ban quarters so that the weighted protected-state path matches the ban states' path before the bans. I estimate it on state by conception-quarter means, with 38 states, 13 of them treated, and 16 quarters, and placebo standard errors (Table A16). Insured abortion falls 213.9 percent of the pre-ban mean ($p = 0.01$), and no maternal health estimate points toward harm. Severe maternal morbidity changes by −4.3 percent (95 percent confidence interval −32.6 to 24.0 percent) and the adverse-outcome composite by 2.0 percent (−7.7 to 11.7 percent). Because it uses state means rather than pregnancies, its intervals are wider than the stacked ones, and it does not distinguish the rises in live births and in the composite from zero.

Table A17 repeats the fertility estimates of Table A12 against quarter −1 and with synthetic difference-in-differences on state by quarter rates, and Figure A6 compares the event studies. Synthetic difference-in-differences puts the change in the birth rate at 6.2 percent (95 percent confidence interval −2.8 to 15.2 percent) and in the pregnancy rate at 1.9 percent (−3.7 to 7.5 percent).

A9. Close-to-Care Zip Code Design

As a check that does not rely on protected states, I implement a close-to-care zip-code design that restricts the sample to pregnancies in ban-state zip codes and compares women by their distance to emergency care in protected states. A pregnancy is close to care, the untreated group, when the

centroid of its assigned zip code lies within 50 miles of the zip-code centroid of the nearest general acute care hospital or emergency medicine provider (an organizational, Type 2, NPI in NPPES) in a protected state. Restrictions on abortion care likely bind less for women who live closer to care outside their state, because crossing the border for care costs them less, especially in an emergency; pregnancies farther than 50 miles from it form the treated group.

The first step is assigning pregnancies to zip codes, because the data do not record a woman's five-digit zip code. For each pregnancy I take the modal practice zip code, from NPPES, of the individual (Type 1) providers the woman saw in the 365 days before her pregnancy started, and use it as her assigned zip code. Assigning the zip code pregnancy by pregnancy rather than once per woman avoids a stale location for women who move between pregnancies, and using individual providers avoids placing a woman at a hospital system's headquarters. Treatment timing and the state fixed effects follow the state of the assigned zip code, and standard errors are clustered by assigned zip code, the unit at which treatment is assigned. The design only works if the assigned zip code is a reasonable proxy for residence. Table A18 reports placement diagnostics by sub-sample. In every sub-sample, a majority of the visits with a provider zip code fall in the modal zip code, and most fall within its three-digit zip area, so the assigned zip code likely describes where the woman gets care. In the border file as a whole, before the continuous-enrollment restriction, 12.1 percent of pregnancies lie within 50 miles of protected-state care, and the mean distance from the assigned zip code to that care is 248 miles. The assigned zip code is not an exact residence but a stable location that supports a close-to-care versus interior contrast.

Figure A7 maps the close-to-care classification across all lower-48 zip codes. Ban-state zip codes within 50 miles of protected-state care identify the close-to-care (untreated) group, ban-state zip codes farther than 50 miles form the treated group, and protected-state zip codes near the ban states form the natural set of substitution destinations. Zip codes in states whose abortion policy was ambiguous during the analysis window are white and excluded from both groups. The continuous-enrollment border sample contains 235,305 pregnancies, and the border file as a whole spans 5,462 assigned ban-state zip codes. Sample descriptives for the close-to-care groups appear in Table A19.

Table A20 reports the close-to-care estimates and Figure A8 their event studies, and Table A21 repeats the estimates against quarter -1, with and without the covariates. The design is less precise than the main one, and no maternal health estimate points toward harm. The adverse-outcome composite changes by -3.0 percent far from protected-state care relative to near it, severe maternal morbidity by -26.6 percent, and preterm birth by -9.0 percent, none distinguishable from zero. Insured abortion, already rare on both sides of the 50-mile line, changes by -56.9 percent, with

an interval from −141.4 to 27.5 percent. Only the share of pregnancies with no observed ending moves clearly, falling 32.4 percent ($p = 0.02$). These estimates are consistent with the main results in a design that does not rely on protected states, although their intervals are too wide to rule out effects that the main design rules out. The design does not show the shift toward live births (−0.4 percent), but it compares women within ban states, all of whom faced a ban, so it measures what distance from protected-state care added rather than the effect of the ban itself.

A10. Outcome Definitions

Severe maternal morbidity uses the 16 diagnosis indicators of the definition of the Centers for Disease Control and Prevention (Section II.E). The definition adds five procedure indicators: transfusion, hysterectomy, ventilation, temporary tracheostomy, and conversion of cardiac rhythm. I do not use them, because procedure codes are recorded mainly on facility claims and their completeness varies by payer and setting, and because transfusion, the most common of them, is also the most common false positive in chart validation (Main et al., 2016). Preterm birth is measured from gestational age at delivery for pregnancies dated by Z3A codes or embryo transfer (Leonard et al., 2020, Appendix 2), and otherwise from the preterm-labor diagnosis. Insurance-billed abortion counts an ending with any abortion-related code. The codes are ICD-10-CM Z33.2 (encounter for elective termination), O04, and O07; CPT 59840, 59841, 59850 to 59852, and 59855 to 59857; ICD-10-PCS 10A0; and HCPCS S0190, S0191, S0199, S2260, and S2265 to S2267.

Table A1—State Abortion Policy Classification

State	Ban in Force	Law	First Treated Quarter
Ban States (N = 15)			
Texas	September 1, 2021	Six-week ban (SB8)	Fourth quarter of 2021
Alabama	June 24, 2022	Total ban (HB 314, 2019)	Third quarter of 2022
Arkansas	June 24, 2022	Trigger ban	Third quarter of 2022
Georgia	July 20, 2022	Six-week ban (HB 481)	Third quarter of 2022
Idaho	August 12, 2022	Six-week ban; trigger ban August 25	Third quarter of 2022
Kentucky	June 24, 2022	Trigger ban	Third quarter of 2022
Louisiana	June 24, 2022	Trigger ban	Third quarter of 2022
Mississippi	July 7, 2022	Trigger ban	Third quarter of 2022
Missouri	June 24, 2022	Trigger ban	Third quarter of 2022
Oklahoma	May 3, 2022	Civil bans; trigger ban June 24	Third quarter of 2022
South Dakota	June 24, 2022	Trigger ban	Third quarter of 2022
Tennessee	June 28, 2022	Six-week ban; total ban August 25	Third quarter of 2022
West Virginia	September 16, 2022	Total ban (HB 302)	Fourth quarter of 2022
Indiana	August 21, 2023	Total ban (SEA 1)	Fourth quarter of 2023
South Carolina	August 23, 2023	Six-week ban (S. 474)	Fourth quarter of 2023
Protected States and DC (N = 25)			
Alaska, California, Colorado, Connecticut, Delaware, District of Columbia, Hawaii, Illinois, Kansas, Maine, Maryland, Massachusetts, Michigan, Minnesota, Montana, Nevada, New Hampshire, New Jersey, New Mexico, New York, Oregon, Rhode Island, Vermont, Virginia, Washington			
Excluded States (N = 11)			
Arizona, Florida, Iowa, Nebraska, North Carolina, North Dakota, Ohio, Pennsylvania, Utah, Wisconsin, Wyoming			

Notes: The date is the first date a ban at six weeks of gestation or earlier was in force. The analysis gives every state in a cohort the cohort's first fully treated quarter (last column). Table A2 gives the reasons for each state whose legal date differs from its cohort's and for each excluded state. Protected states kept abortion legal through at least 22 weeks of gestation throughout the window. Excluded states had bans that were enjoined, partial, or changing during the window. Classification follows Dench, Pineda-Torres and Myers (2024). Sources: Nash and Guarnieri (2022), Curhan (2026), KFF, the Center for Reproductive Rights, state statutes, and court orders.

Table A2—Classification Decisions for States with Contested Timing or Status

State	Group and Timing in the Analysis	Reason
Ban States		
Texas	Ban; first treated in the fourth quarter of 2021	SB8 took effect September 1, 2021, and was never enjoined, because its private enforcement left no official to sue; clinics stopped providing abortion after about six weeks that day. The August 2022 trigger ban extended a ban already in force on the ground.
Oklahoma	Ban, *Dobbs* cohort; third quarter of 2022	Civilly enforced bans took effect in May 2022, weeks before *Dobbs*, but the ban regime was incomplete until then: enforcement rested on private suits, and the criminal trigger ban took effect only with *Dobbs*. Coding Oklahoma with the *Dobbs* cohort places May and June 2022 in its reference quarter, so any early effect raises the reference level and pulls its estimate toward zero rather than away from it.
Georgia, Idaho, Mississippi	Ban, *Dobbs* cohort; third quarter of 2022	Each ban took effect between July 7 and August 25, 2022, inside the cohort's first post quarter. That quarter is only partly treated for these states, which attenuates the estimate at event time 0 (Section IV); every later quarter is fully treated, apart from November 15 to 23, 2022, when a state court blocked Georgia's ban.
Kentucky, Louisiana	Ban, *Dobbs* cohort; third quarter of 2022	State courts blocked the trigger bans for parts of July 2022, but each ban was in force from late June, again from August 2022, and for the rest of the window. The interruptions fall inside the partly treated first post quarter.
West Virginia	Ban, own cohort; fourth quarter of 2022	Providers stopped under the 1849 statute from June 24 until a court enjoined it on July 18, 2022, and resumed until the total ban (HB 302) took effect September 16, 2022. The analysis dates the ban from the statute that remained in force; the three-week pause falls in West Virginia's reference quarter and, if it mattered, biases its estimate toward zero.
Indiana	Ban from the fourth quarter of 2023; not in the stacked pregnancy estimates	SEA 1 was in force for one week in September 2022 and then enjoined, and abortion remained available until August 21, 2023. Indiana has no complete post-ban window in the pregnancy data, so it does not enter the stacked pregnancy estimates.
South Carolina	Ban from the fourth quarter of 2023; not in the stacked pregnancy estimates	A six-week ban was in force from June 27 to August 17, 2022, before the state supreme court blocked it; abortion was then legal to about 22 weeks until August 23, 2023. South Carolina has no complete post-ban window in the pregnancy data, so it does not enter the stacked pregnancy estimates.
Protected States		
Kansas	Protected	Abortion is legal to 22 weeks, before viability, but the state constitution protects it and voters rejected an amendment to remove that protection in August 2022. The 22-week limit binds fewer than 1 percent of abortions (Kortsmit et al., 2023), and more than three-quarters of abortions in Kansas in 2023 were to nonresidents, most from neighboring ban states (Conlon, 2024).
Michigan	Protected	The 1931 ban was enjoined on May 17, 2022, before *Dobbs*, and was never enforced afterward; Proposal 3 added the right to the state constitution in November 2022.
Excluded States		
Wisconsin	Excluded	Providers stopped under the 1849 statute from June 2022, a ban on the ground, but they resumed in September 2023 after a court ruling. Treating Wisconsin as a ban state would place a reversal inside its post-period, and treating it as protected would place fifteen months of no access inside a control.
Ohio, Utah, Arizona	Excluded	Each had a near-total or six-week ban in force for days to months in 2022 and a gestational limit of 15 to 18 weeks, or a later reversal, for the rest of the window, so neither group describes it.
Florida, Nebraska, North Carolina, Iowa	Excluded	Each moved from legal abortion to a 15-week, 12-week, or six-week limit during the window, a partial restriction that changed within the period rather than a ban or protected access throughout.
North Dakota	Excluded	The trigger ban was blocked by a state court, but the state's only clinic moved to Minnesota in August 2022, so legal status and access on the ground diverged; a near-total ban (SB 2150) then took effect on April 24, 2023.
Wyoming	Excluded	The trigger ban was in effect for a day in July 2022 and then enjoined, with further bans litigated through November 2024.
Pennsylvania	Excluded	Abortion was legal to 24 weeks throughout, but the state kept one of the most restrictive regimes among states where it remained legal: a 24-hour waiting period with state-directed counseling, parental consent for minors, no public coverage beyond the federal Hyde exceptions, and a requirement that clinics meet ambulatory surgical facility standards, a targeted regulation of abortion providers. Placing it among the protected states would put a restrictive regime in the comparison group, which would bias the estimates toward zero.

Notes: The classification in Table A1 is fixed in the analysis code. This table gives, for each state whose placement is not obvious from its statute, the reason for its group and the direction in which any misplacement would bias the estimates. Where a ban's legal status and access on the ground diverged, the analysis follows the statute in force for the rest of the window and notes the gap here. Sources as in Table A1.

Table A3—Source of the Pregnancy Start Date

Dating Source	**Ban States, %**	**Protected States, %**
Gestational-week diagnosis code (Z3A)	87.051	75.345
Ending-type default	9.112	19.579
First prenatal visit default	2.804	2.862
Embryo transfer	0.711	1.647
Cell-free DNA screening	0.175	0.205
Glucose tolerance test	0.063	0.115
Ending-type default, multiple birth	0.028	0.048
Maternal serum screening	0.023	0.068
Nuchal translucency ultrasound	0.021	0.102
Fetal anatomy ultrasound	0.012	0.029

Notes: Percent of pregnancy episodes in the continuous-enrollment sample dated by each source, under the priority order in Appendix A2; rows are sorted by the ban-state share.

Table A4—Sample Construction

Restriction	Continuous Enrollment	Episode Enrollment
All pregnancy episodes meeting the enrollment requirement	1,848,589	3,021,280
Ban or protected state at pregnancy start	1,576,819	2,495,494
Pregnancy start in 2020 or later*	1,135,994	2,495,494
Pregnancy start on or before February 2024 (40 weeks observable)	996,163	2,418,056
Exclude ended episodes with implausible gestation	995,377	2,416,491
Analysis sample	**995,377**	**2,416,491**
Ban states	262,367	798,269
Protected states	733,010	1,618,222
Event-study sample†	**818,245**	**2,000,319**
Ban states, in own cohort's window	155,502	485,714
Protected states	662,743	1,514,605

Notes: Pregnancy episodes remaining after each restriction in the continuous-enrollment and episode-enrolled samples (Section II.D and Appendix A4). *This step excludes no episode-enrolled pregnancy. That sample requires enrollment from twelve months before the pregnancy start, and the enrollment data begin in January 2019, so no pregnancy that started in 2019 can qualify; every episode-enrolled pregnancy already starts in 2020 or later. †Distinct pregnancies in each sample used by the event studies and pooled estimates (Section IV): the ban-state pregnancies conceived in their own cohort's eleven-quarter window (Table A5, quarters −6 to 4, excluding Indiana and South Carolina) and the protected-state pregnancies conceived from the second quarter of 2020 through the fourth quarter of 2023. Because a control pregnancy enters every sub-experiment whose window covers it, the stacked regression has more observations than these counts (1,607,569 in the continuous-enrollment sample for outcomes measured on all pregnancies).

Table A5—Pregnancies in Each Ban Cohort by Quarter Relative to the Ban

Quarter from Ban	Banned at *Dobbs*	Texas	West Virginia	Indiana and South Carolina	All Ban States
First treated quarter	2022Q3	2021Q4	2022Q4	2023Q4	
Pregnancies Conceived in the Quarter					
−6	7,376	6,484	241	1,555	15,656
−5	7,050	6,798	226	1,571	15,645
−4	6,810	7,517	238	1,625	16,190
−3	7,588	7,138	211	1,590	16,527
−2	7,050	7,176	253	1,643	16,122
−1	6,900	6,929	217	1,481	15,527
0	6,706	7,185	262	1,539	15,692
1	7,231	6,781	236	1,001	15,249
2	7,076	6,248	239	–	13,563
3	7,392	6,187	198	–	13,777
4	6,866	6,485	208	–	13,559
Total, quarters −6 to 4	**78,045**	**74,928**	**2,529**	**12,005**	**167,507**
Total, every quarter observed	**118,684**	**113,751**	**3,854**	**26,078**	**262,367**

Notes: Ban-state pregnancies in the continuous-enrollment sample by quarter of conception relative to each cohort's first fully treated quarter, the series behind Figure 1. The *Dobbs* cohort is Alabama, Arkansas, Georgia, Idaho, Kentucky, Louisiana, Mississippi, Missouri, Oklahoma, South Dakota, and Tennessee. – marks a quarter whose pregnancies would be conceived after the data end, which is also why Indiana and South Carolina are not a sub-experiment of the stacked design.

Table A6—Pre-Period Balance Between Ban and Protected States

	Ban States Mean	Protected States Mean	Normalized Difference
Characteristics Fixed Before Pregnancy			
Age (years)	29.014	30.352	−0.21
Under 25	0.267	0.193	0.18
Race: White	0.416	0.459	−0.09
Race: Black	0.192	0.140	0.14
Race: Asian	0.019	0.039	−0.12
Race: unknown	0.334	0.215	0.27
Ethnicity: Hispanic	0.087	0.238	−0.42
Ethnicity: unknown	0.406	0.260	0.31
Medicaid	0.424	0.534	−0.22
Commercial insurance	0.575	0.464	0.22
First observed pregnancy	0.257	0.257	0.00
Multiple gestation	0.020	0.018	0.01
Obstetric comorbidity score	8.744	8.372	0.03
Number of comorbid conditions	0.776	0.719	0.05
Any comorbid condition	0.437	0.420	0.03
Pre-existing hypertension	0.067	0.051	0.07
Pre-existing obesity	0.169	0.180	−0.03
Pre-existing drug or alcohol use disorder	0.048	0.049	−0.01
Any of hypertension, obesity, or substance use	0.236	0.240	−0.01
Pre-existing pulmonary hypertension	0.000	0.001	−0.01
Maternal Health			
Any adverse pregnancy outcome	0.286	0.261	0.06
Severe maternal morbidity	0.028	0.027	0.01
Hypertensive disorder	0.213	0.157	0.14
Preterm birth (live births)	0.173	0.149	0.06
Timing and Intensity of Care			
Length of pregnancy, days	225.000	201.728	0.26
Days to first visit	45.992	45.261	0.02
Prolonged delivery stay (live births)	0.233	0.243	−0.02
Number of visits	16.203	16.316	−0.01
Pregnancy Endings			
Live birth	0.793	0.671	0.28
Miscarriage	0.157	0.142	0.04
Ending not observed	0.064	0.049	0.06
Insurance-billed abortion	0.012	0.152	−0.53

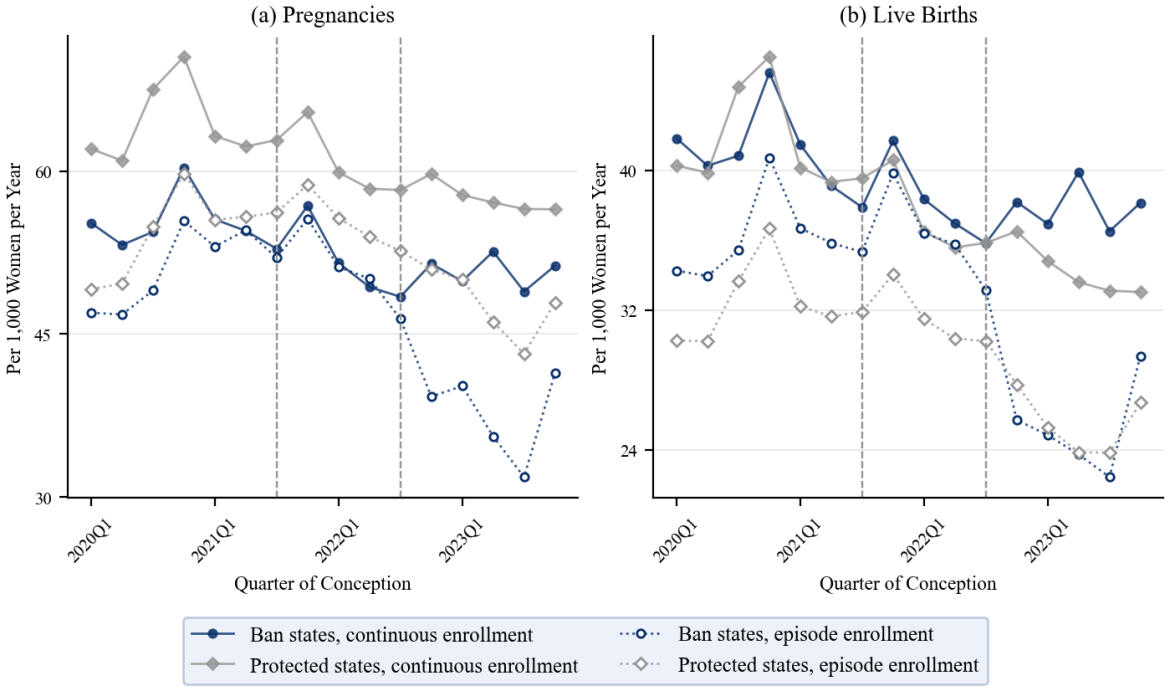
Notes: Pre-ban means in the continuous-enrollment sample, as proportions except for age, the comorbidity score and count, and the timing and intensity outcomes (in days and visits), and normalized differences (the difference in means divided by the square root of the average of the two variances). Live birth, miscarriage, insurance-billed abortion, the length of pregnancy, and the days to the first visit are among pregnancies with an observed ending, preterm birth and a prolonged delivery stay among live births, and the other rows among all pregnancies. Normalized differences above 0.25 in absolute value, for Hispanic ethnicity, unknown race and ethnicity, the length of pregnancy, live birth, and insurance-billed abortion, are level differences that the state fixed effects absorb.

Table A7—The Continuous-Enrollment Sample and U.S. Births

	Continuous-Enrollment Sample	**U.S. Births, 2023**
Pregnancies, %		
Mean age of mother, years	29.9	29.6
Medicaid	50.7	41.5
Commercial or private insurance	49.1	51.0
Mother under 25	22.7	21.1
White	44.9	49.7
Black	15.1	13.7
Asian	3.3	6.0
Hispanic	21.3	26.3
Race unknown	24.4	–
Live Births, %		
Cesarean delivery	31.0	32.3
Gestational diabetes	11.4	8.3

Notes: Column 1 pools ban and protected states in the continuous-enrollment sample for pregnancies started January 2020 to February 2024, weighting by pregnancies; its upper rows are percentages of all pregnancies and its lower rows percentages of live births. Column 2 is every U.S. birth in 2023 from the final natality data (Osterman et al., 2025), with mean age from Brown et al. (2025); race and Hispanic origin refer to non-Hispanic single-race mothers and Hispanic mothers of any race, and payer is the source of payment for the delivery. Mean age is in years, and – marks a row with no counterpart in the natality data. Race is unknown for many pregnancies in the claims (the race unknown row), so the race rows are not strictly comparable, and birth certificates miss many complications that claims record (Bell, Anjur-Dietrich and Gemmill, 2026), so the gestational diabetes rows also differ in how they are measured.

Figure A1. Pregnancy and Birth Rates in the Two Enrollment Samples



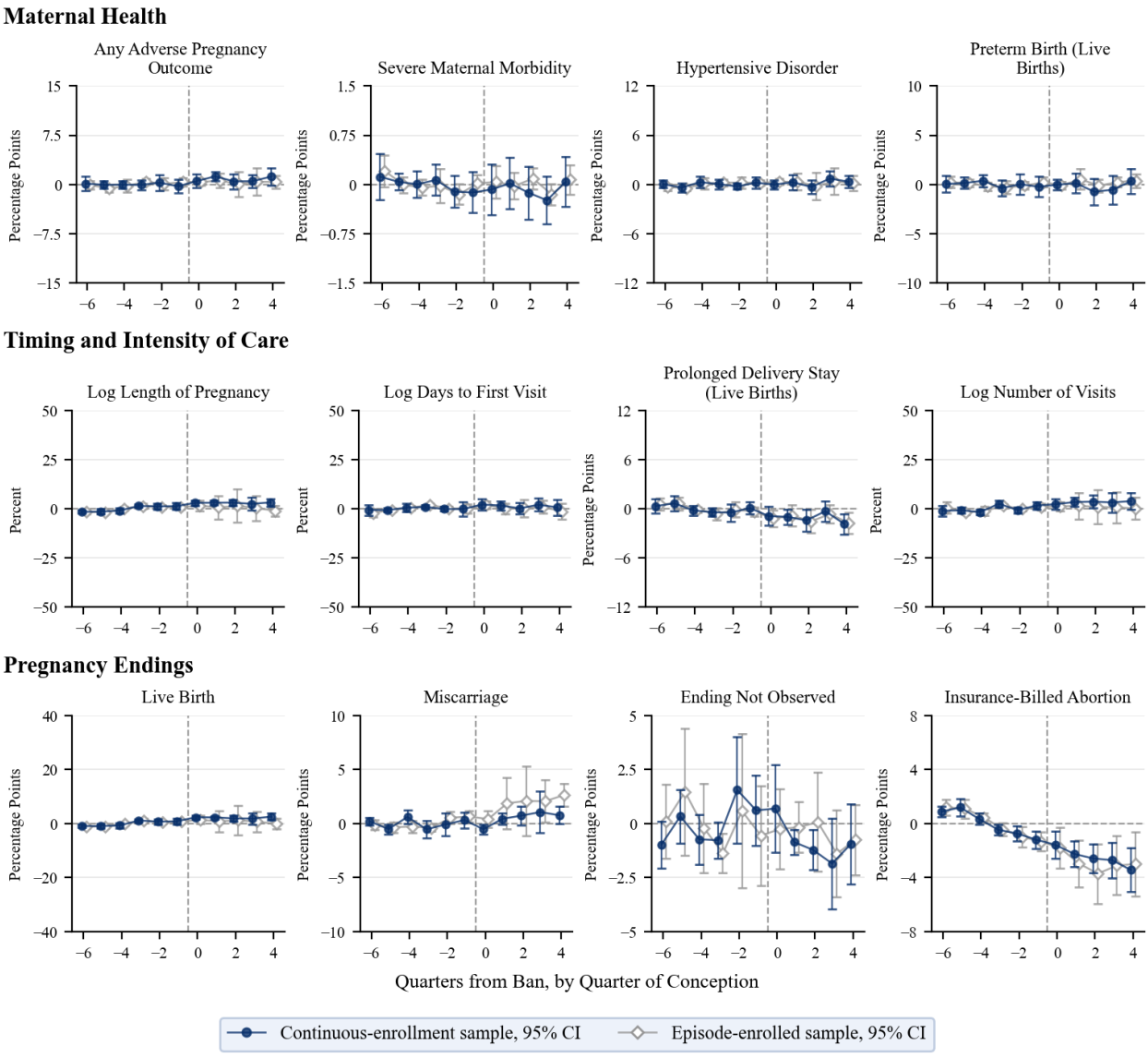
Notes: Pregnancies and live births per 1,000 enrolled women aged 15 to 45 per year, by quarter of conception, in ban and protected states, in the continuous-enrollment sample (solid lines, filled markers, as in Figure 4) and the episode-enrolled sample (dotted lines, hollow markers). Monthly rates are averaged to quarters. Ban states are circles and protected states diamonds. Dashed lines mark the third quarters of 2021 (Texas Senate Bill 8) and 2022 (*Dobbs*).

Table A8—Estimates by Enrollment Sample

	Episode-Enrolled Sample				Continuous-Enrollment Sample			
Outcome	**Pre-Ban Mean (1)**	**Estimate (SE) (2)**	**Percent of Mean (3)**	**95% CI, Percent of Mean (4)**	**Pre-Ban Mean (5)**	**Estimate (SE) (6)**	**Percent of Mean (7)**	**95% CI, Percent of Mean (8)**
Maternal Health								
Any adverse pregnancy outcome	0.278	0.004 (0.005)	1.3	[−2.4, 4.9]	0.285	0.008 (0.003)	2.8	[0.6, 4.9]
Severe maternal morbidity	0.027	0.000 (0.001)	0.5	[−5.4, 6.4]	0.026	−0.001 (0.001)	−2.9	[−11.4, 5.6]
Hypertensive disorder	0.202	0.002 (0.004)	0.9	[−3.5, 5.4]	0.208	0.002 (0.003)	1.0	[−1.5, 3.4]
Preterm birth (live births)	0.182	0.002 (0.002)	1.0	[−1.3, 3.4]	0.171	−0.002 (0.003)	−0.9	[−4.5, 2.6]
Timing and Intensity of Care								
Log length of pregnancy	5.242	0.006 (0.023)	0.6	[−4.0, 5.3]	5.272	0.030 (0.006)	3.0	[1.8, 4.2]
Log days to first visit	3.676	0.004 (0.014)	0.4	[−2.3, 3.2]	3.616	0.014 (0.012)	1.4	[−1.1, 3.8]
Prolonged delivery stay (live births)	0.246	−0.011 (0.005)	−4.6	[−9.1, −0.1]	0.233	−0.011 (0.005)	−4.6	[−8.6, −0.6]
Log number of visits	2.397	0.009 (0.025)	0.9	[−4.2, 6.1]	2.496	0.033 (0.014)	3.3	[0.4, 6.2]
Pregnancy Endings								
Live birth	0.772	0.008 (0.014)	1.0	[−2.8, 4.8]	0.789	0.022 (0.004)	2.7	[1.7, 3.8]
Miscarriage	0.170	0.017 (0.007)	10.1	[1.2, 19.0]	0.162	0.005 (0.003)	3.0	[−0.9, 6.9]
Ending not observed	0.097	−0.004 (0.005)	−4.4	[−15.1, 6.3]	0.062	−0.008 (0.005)	−13.3	[−28.3, 1.6]
Insurance-billed abortion	0.012	−0.028 (0.009)	−225.1	[−377.7, −72.5]	0.011	−0.025 (0.006)	−237.1	[−344.5, −129.6]

Notes: Stacked difference-in-differences estimates in the Controls specification, with standard errors (SE), clustered by state, in parentheses. Columns 1 to 4 use the episode-enrolled sample (Appendix A4) and columns 5 to 8 the continuous-enrollment sample of Table 2. Columns 1 and 5 are each sample's ban-state mean in event quarters −6 to −1; columns 1, 2, 5, and 6 are in each outcome's units. Columns 3 and 7 are the estimate as a percent of that sample's pre-ban mean (100 times the coefficient for outcomes in logs), and columns 4 and 8 its 95 percent confidence interval (CI).

Figure A2. Event Studies, Continuous-Enrollment and Episode-Enrolled Samples



Notes: Stacked event-study coefficients in the continuous-enrollment sample (filled markers, as in Figure 2) and the episode-enrolled sample (hollow markers), in the Controls specification, with 95 percent confidence intervals from standard errors clustered by state. The pre-ban coefficients average zero, and the dashed vertical line falls between quarters −1 and 0. Vertical axes are drawn as in Figure 2.

Table A9—Subgroup Estimates for the Pregnancy Outcomes

Outcome	All Pregnancies (1)	Medicaid (2)	Minority (3)	High Risk (4)	Under 25 (5)	Excluding Texas (6)
Maternal Health						
Any adverse pregnancy outcome	2.8 (1.1)	4.2 (1.7)	2.9 (2.0)	4.0 (1.9)	6.3 (1.8)	3.4 (1.9)
	[0.6, 4.9]	[0.7, 7.7]	[−1.0, 6.9]	[0.1, 7.8]	[2.8, 9.9]	[−0.5, 7.3]
Pre-ban mean	0.285	0.322	0.313	0.315	0.286	0.295
Severe maternal morbidity	−2.9 (4.2)	−6.3 (8.0)	−7.5 (6.3)	−2.2 (4.6)	−6.6 (8.5)	−2.1 (6.8)
	[−11.4, 5.6]	[−22.6, 10.0]	[−20.2, 5.2]	[−11.4, 7.1]	[−23.7, 10.6]	[−15.9, 11.8]
Pre-ban mean	0.026	0.035	0.036	0.037	0.026	0.029
Hypertensive disorder	1.0 (1.2)	−1.6 (2.0)	0.7 (2.0)	0.1 (1.4)	2.0 (2.7)	−0.9 (1.6)
	[−1.5, 3.4]	[−5.7, 2.5]	[−3.3, 4.8]	[−2.8, 3.1]	[−3.5, 7.5]	[−4.1, 2.3]
Pre-ban mean	0.208	0.247	0.238	0.264	0.203	0.234
Preterm birth (live births)	−0.9 (1.8)	−2.3 (2.6)	−3.2 (3.3)	−1.7 (1.8)	0.0 (2.7)	−1.8 (2.6)
	[−4.5, 2.6]	[−7.5, 2.9]	[−9.8, 3.5]	[−5.4, 2.0]	[−5.6, 5.6]	[−7.1, 3.5]
Pre-ban mean	0.171	0.223	0.218	0.220	0.172	0.195
Timing and Intensity of Care						
Log length of pregnancy	3.0 (0.6)	3.1 (1.1)	4.1 (1.3)	3.0 (0.8)	3.9 (1.0)	2.5 (1.0)
	[1.8, 4.2]	[0.7, 5.4]	[1.5, 6.6]	[1.5, 4.5]	[1.9, 5.9]	[0.5, 4.5]
Pre-ban mean	5.272	5.255	5.223	5.234	5.315	5.262
Log days to first visit	1.4 (1.2)	−0.4 (1.7)	0.2 (1.2)	0.7 (1.4)	0.5 (1.3)	0.0 (1.8)
	[−1.1, 3.8]	[−3.9, 3.1]	[−2.2, 2.5]	[−2.1, 3.5]	[−2.2, 3.2]	[−3.5, 3.6]
Pre-ban mean	3.616	3.616	3.643	3.490	3.635	3.604
Prolonged delivery stay (live births)	−4.6 (2.0)	−2.4 (3.2)	−5.6 (2.6)	−4.0 (2.9)	−4.6 (2.6)	−1.8 (2.6)
	[−8.6, −0.6]	[−8.8, 4.1]	[−10.8, −0.4]	[−9.9, 1.9]	[−9.9, 0.6]	[−7.1, 3.5]
Pre-ban mean	0.233	0.261	0.265	0.246	0.303	0.247
Log number of visits	3.3 (1.4)	4.1 (2.1)	4.3 (1.8)	3.1 (1.6)	5.1 (1.9)	3.8 (1.8)
	[0.4, 6.2]	[−0.1, 8.3]	[0.7, 7.8]	[−0.1, 6.4]	[1.3, 8.9]	[0.1, 7.5]
Pre-ban mean	2.496	2.566	2.465	2.621	2.535	2.505
Pregnancy Endings						
Live birth	2.7 (0.5)	3.0 (0.9)	3.8 (1.1)	2.6 (0.6)	3.6 (0.9)	2.3 (0.8)
	[1.7, 3.8]	[1.2, 4.8]	[1.6, 6.0]	[1.4, 3.8]	[1.8, 5.5]	[0.7, 3.8]
Pre-ban mean	0.789	0.785	0.763	0.771	0.819	0.787
Miscarriage	3.0 (1.9)	0.5 (4.2)	4.3 (4.0)	2.2 (1.9)	2.9 (3.6)	3.1 (3.4)
	[−0.9, 6.9]	[−8.1, 9.0]	[−3.9, 12.5]	[−1.6, 6.0]	[−4.4, 10.1]	[−3.8, 10.1]
Pre-ban mean	0.162	0.155	0.171	0.172	0.133	0.157
Ending not observed	−13.3 (7.4)	−25.6 (16.9)	−16.7 (9.4)	−15.6 (9.0)	−22.6 (16.2)	−20.1 (12.2)
	[−28.3, 1.6]	[−59.8, 8.7]	[−35.7, 2.4]	[−33.8, 2.6]	[−55.4, 10.2]	[−44.9, 4.7]
Pre-ban mean	0.062	0.067	0.071	0.076	0.055	0.067
Insurance-billed abortion	−237.1 (53.0)	−192.5 (63.1)	−237.8 (50.7)	−179.4 (43.2)	−285.1 (75.3)	−176.8 (47.9)
	[−344.5, −129.6]	[−320.5, −64.5]	[−340.7, −134.8]	[−267.0, −91.9]	[−437.7, −132.6]	[−274.0, −79.6]
Pre-ban mean	0.011	0.011	0.015	0.011	0.012	0.012

Notes: Stacked difference-in-differences estimates in the Controls specification and the continuous-enrollment sample, each as a percent of the subgroup's pre-ban mean (100 times the coefficient for outcomes in logs), with the standard error, clustered by state, in parentheses, the 95 percent confidence interval in brackets, and the group's ban-state mean in event quarters −6 to −1, in the outcome's units, beneath. The All Pregnancies column repeats Table 2, and the subgroups are defined as in Figure 3.

Table A10—Maternal Health and Care within Live Births and within Losses

Outcome	Pre-Ban Mean (1)	Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)
A. Live Births				
Any adverse pregnancy outcome	0.369	−0.005 (0.003)	−1.2	[−3.1, 0.6]
Severe maternal morbidity	0.030	−0.003 (0.001)	−9.5	[−17.6, −1.3]
Hypertensive disorder	0.262	−0.008 (0.003)	−3.0	[−5.7, −0.3]
Preterm birth (live births)	0.171	−0.002 (0.003)	−0.9	[−4.5, 2.6]
Log length of pregnancy	5.579	−0.001 (0.001)	−0.1	[−0.4, 0.1]
Log days to first visit	3.653	0.013 (0.013)	1.3	[−1.4, 4.0]
Prolonged delivery stay (live births)	0.233	−0.011 (0.005)	−4.6	[−8.6, −0.6]
Log number of visits	2.864	−0.020 (0.011)	−2.0	[−4.3, 0.3]
B. Losses				
Any adverse pregnancy outcome	0.022	0.001 (0.002)	5.5	[−10.2, 21.2]
Severe maternal morbidity	0.014	0.002 (0.002)	11.3	[−10.7, 33.4]
Hypertensive disorder	0.035	−0.003 (0.001)	−8.6	[−16.0, −1.3]
Log length of pregnancy	4.113	0.010 (0.005)	1.0	[0.1, 2.0]
Log days to first visit	3.464	0.026 (0.012)	2.6	[0.0, 5.1]
Log number of visits	1.338	−0.013 (0.013)	−1.3	[−3.9, 1.4]

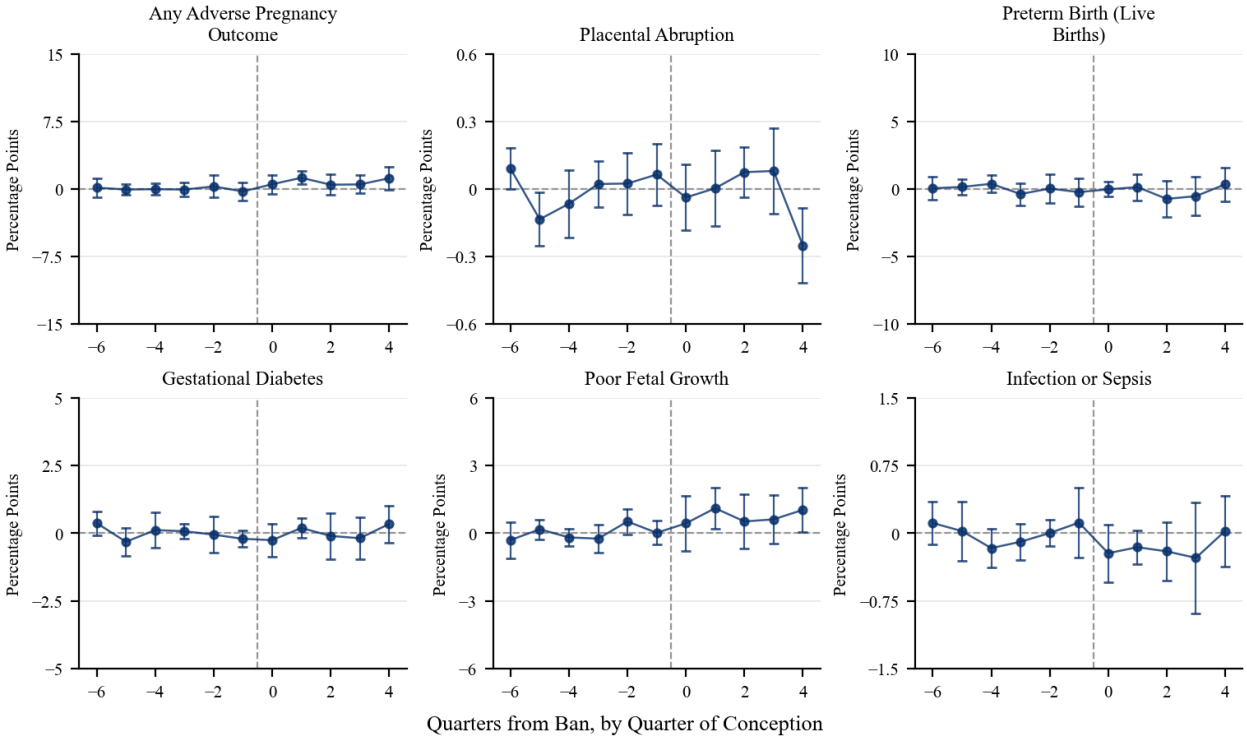
Notes: Stacked difference-in-differences estimates in the Controls specification and the continuous-enrollment sample, among pregnancies that ended in a live birth (Panel A) and among pregnancies that ended in a miscarriage, stillbirth, or ectopic pregnancy (Panel B). Column 1 is the group's ban-state mean in event quarters −6 to −1 and column 2 the estimate, both in each outcome's units, with its standard error (SE), clustered by state, in parentheses. Column 3 is the estimate as a percent of that mean (100 times the coefficient for outcomes in logs) and column 4 its 95 percent confidence interval (CI). Preterm birth and a prolonged delivery stay are defined only for live births, and the ending outcomes are not estimated, because each group fixes the ending. Because the bans changed how pregnancies ended, these estimates condition on an outcome of the bans, so they describe how the pooled estimates divide across endings rather than identify effects within each.

Table A11—The Adverse-Outcome Composite and Its Components

Outcome	Pre-Ban Mean (1)	Controls			Raw	
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Estimate (SE) (5)	Percent of Mean (6)
Composite						
Any adverse pregnancy outcome	0.285	0.008 (0.003)	2.8	[0.6, 4.9]	0.009 (0.003)	3.2
Components						
Placental abruption	0.0117	−0.0003 (0.0004)	−2.2	[−9.0, 4.7]	−0.0002 (0.0004)	−2.0
Preterm birth (live births)	0.171	−0.002 (0.003)	−0.9	[−4.5, 2.6]	−0.001 (0.003)	−0.8
Gestational diabetes	0.094	0.000 (0.002)	0.0	[−5.2, 5.2]	−0.001 (0.002)	−1.3
Poor fetal growth	0.101	0.007 (0.005)	7.4	[−2.1, 16.8]	0.009 (0.005)	8.7
Infection or sepsis	0.0240	−0.0016 (0.0012)	−6.8	[−17.0, 3.4]	−0.0011 (0.0012)	−4.6

Notes: Stacked difference-in-differences estimates in the continuous-enrollment sample, with fixed effects and weights as in Table 2 and laid out as that table without its pre-trend test. The composite equals one if a pregnancy has any of the five components below it. Preterm birth is measured among live births. Poor fetal growth is diagnosed only in pregnancies that continue, and in one validation study about a third of claims diagnoses of a small-for-gestational-age infant were confirmed in medical records (Chomistek et al., 2023).

Figure A3. Event Studies for the Adverse-Outcome Composite and Its Components



Notes: Stacked difference-in-differences event-study coefficients by quarter of conception relative to the ban, with 95 percent confidence intervals from standard errors clustered by state, in the Controls specification and the continuous-enrollment sample, for the composite (first panel, as in Figure 2) and its five components. Coefficients are in percentage points (of live births for preterm birth), the pre-ban coefficients average zero, and the dashed vertical line falls between quarters −1 and 0. Vertical axes are drawn as in Figure 2.

Table A12—Fertility: Pregnancy and Live-Birth Rates among Enrolled Women

Group	Pre-Ban Mean (1)	State Fixed Effects: Estimate (SE) (2)	State Fixed Effects: Percent of Mean (3)	State Fixed Effects: 95% CI, Percent of Mean (4)	Raw: Estimate (SE) (5)	Raw: Percent of Mean (6)
Pregnancy Rate						
All	52.09	0.11 (0.61)	0.2	[−2.2, 2.6]	−0.31 (0.74)	−0.6
Medicaid	67.82	0.11 (0.98)	0.2	[−2.8, 3.1]	−1.63 (1.86)	−2.4
Minority	63.12	−2.25 (0.65)	−3.6	[−5.6, −1.5]	−3.54 (1.04)	−5.6
High risk[a]	53.17	0.85 (0.66)	1.6	[−0.9, 4.1]	0.10 (1.00)	0.2
Under 25	41.06	1.16 (1.59)	2.8	[−5.0, 10.6]	0.04 (1.27)	0.1
Excluding Texas	52.30	0.39 (0.63)	0.8	[−1.7, 3.2]	0.57 (1.04)	1.1
Live-Birth Rate						
All	38.50	1.41 (0.57)	3.7	[0.7, 6.7]	1.11 (0.71)	2.9
Medicaid	49.04	2.28 (1.03)	4.7	[0.4, 8.9]	0.81 (1.68)	1.7
Minority	43.73	0.87 (0.87)	2.0	[−2.0, 6.0]	−0.09 (1.17)	−0.2
High risk[a]	38.11	2.10 (0.62)	5.5	[2.2, 8.8]	1.45 (0.87)	3.8
Under 25	31.57	2.12 (1.20)	6.7	[−1.0, 14.4]	1.22 (0.98)	3.9
Excluding Texas	38.62	1.56 (0.74)	4.0	[0.2, 7.9]	1.70 (0.93)	4.4

Notes: Stacked difference-in-differences (DiD) estimates on the panel of enrolled women aged 15 to 45 by state and quarter, weighted by enrolled women, with standard errors (SE), clustered by state, in parentheses. State Fixed Effects has event-time and sub-experiment by state fixed effects; Raw has event-time fixed effects and a treated-state indicator. Rates are pregnancies or live births per 1,000 enrolled women per year (conceptions in the state and quarter divided by woman-months of enrollment, times 12,000), so they do not condition on who becomes pregnant. Column 1 is the mean in treated states before the ban; columns 1, 2, and 5 are in rates. Column 3 is the estimate as a percent of that mean, and column 4 its 95 percent confidence interval (CI). Unlike the pregnancy estimates, this stack keeps Indiana and South Carolina as a fourth sub-experiment, so it has 40 clusters. Because a pregnancy that started after February 2024 cannot be followed to its ending, the rates stop with the fourth quarter of 2023, so this sub-experiment adds its pre-ban quarters and only its first post-ban quarter. Each estimate compares the post-ban quarters with the average of quarters −6 to −1. Minority women have a Black race code or a Hispanic ethnicity code. [a]High-risk women have any condition of the obstetric comorbidity index in the preceding 12 months.

Table A13—Composition of the Pregnant Population by Pre-Pregnancy Health

Characteristic	Pre-Ban Mean (1)	Raw: Estimate (SE) (2)	Raw: Percent of Mean (3)	Raw: 95% CI, Percent of Mean (4)
Obstetric comorbidity score	8.387	0.033 (0.139)	0.4	[−3.0, 3.7]
Number of obstetric comorbidities	0.741	0.005 (0.012)	0.7	[−2.5, 3.9]
Any obstetric comorbidity	0.423	0.002 (0.005)	0.4	[−1.8, 2.6]

Notes: Stacked difference-in-differences (DiD) estimates in the continuous-enrollment sample for pre-pregnancy health measures, with standard errors (SE), clustered by state, in parentheses. The columns are as under Controls in Table 2 without the pre-trend test, but the estimates use the Raw specification because the comorbidity score is itself a control. A ban cannot change these measures for any one woman, so a nonzero estimate means that the set of women who became pregnant changed. The comorbidity score weights conditions by the points of Leonard et al. (2020), as in the regression controls and Table 1. The Medicaid and under-25 shares are in Section VI.B.

Table A14—Sub-Experiments in the Stacked Design

First Treated Quarter	Treated States	Treated Pregnancies	Control Pregnancies	Share of Treated Pregnancies, %
2021Q4	TX	74,928	493,967	48.2
2022Q3	AL, AR, GA, ID, KY, LA, MO, MS, OK, SD, TN	78,045	480,762	50.2
2022Q4	WV	2,529	477,338	1.6

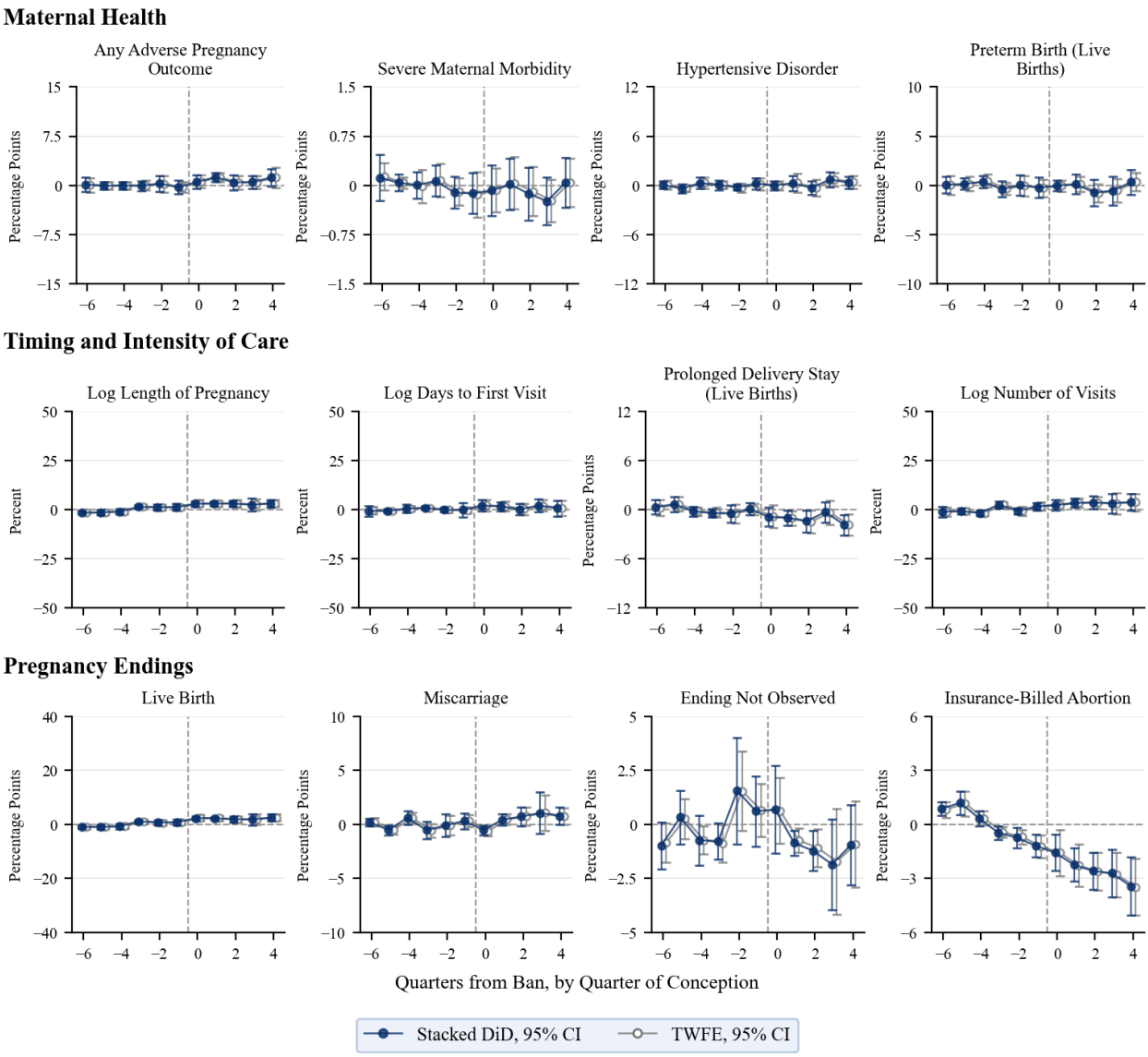
Notes: One row per ban cohort, defined by its first fully treated quarter, in the continuous-enrollment sample. Each sub-experiment pairs the cohort with the protected states over its event window, quarters −6 to 4. Treated pregnancies are the cohort's pregnancies conceived in the window, and control pregnancies are the protected-state pregnancies conceived in it, so a protected-state pregnancy is counted in every sub-experiment whose window covers it. The last column is each cohort's share of the treated pregnancies in the stacked design. The window of Indiana and South Carolina runs past the data, so they are not a sub-experiment (Table A5).

Table A15—Stacked Estimates by Reference Period

		Quarter −1			Quarter −2		Pre-Ban Average	
Outcome	Pre-Ban Mean (1)	Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Estimate (SE) (5)	Percent of Mean (6)	Estimate (SE) (7)	Percent of Mean (8)
Maternal Health								
Any adverse pregnancy outcome	0.285	0.011 (0.005)	3.7	[−0.2, 7.6]	0.005 (0.006)	1.8	0.008 (0.003)	2.8
Severe maternal morbidity	0.026	0.000 (0.002)	1.7	[−12.0, 15.4]	0.000 (0.001)	1.2	−0.001 (0.001)	−2.9
Hypertensive disorder	0.208	0.000 (0.005)	−0.1	[−5.1, 5.0]	0.004 (0.002)	2.1	0.002 (0.003)	1.0
Preterm birth (live births)	0.171	0.001 (0.007)	0.4	[−8.3, 9.2]	−0.002 (0.003)	−1.1	−0.002 (0.003)	−0.9
Timing and Intensity of Care								
Log length of pregnancy	5.272	0.017 (0.009)	1.7	[−0.1, 3.5]	0.017 (0.006)	1.7	0.030 (0.006)	3.0
Log days to first visit	3.616	0.015 (0.022)	1.5	[−2.9, 5.9]	0.015 (0.016)	1.5	0.014 (0.012)	1.4
Prolonged delivery stay (live births)	0.233	−0.012 (0.005)	−5.0	[−9.6, −0.4]	−0.006 (0.005)	−2.6	−0.011 (0.005)	−4.6
Log number of visits	2.496	0.016 (0.014)	1.6	[−1.2, 4.5]	0.040 (0.011)	4.0	0.033 (0.014)	3.3
Pregnancy Endings								
Live birth	0.789	0.014 (0.006)	1.7	[0.2, 3.2]	0.015 (0.004)	1.9	0.022 (0.004)	2.7
Miscarriage	0.162	0.002 (0.005)	1.0	[−5.1, 7.0]	0.006 (0.005)	3.6	0.005 (0.003)	3.0
Ending not observed	0.062	−0.014 (0.008)	−23.3	[−49.7, 3.2]	−0.024 (0.012)	−38.6	−0.008 (0.005)	−13.3
Insurance-billed abortion	0.011	−0.013 (0.003)	−123.6	[−188.3, −58.9]	−0.018 (0.005)	−166.9	−0.025 (0.006)	−237.1

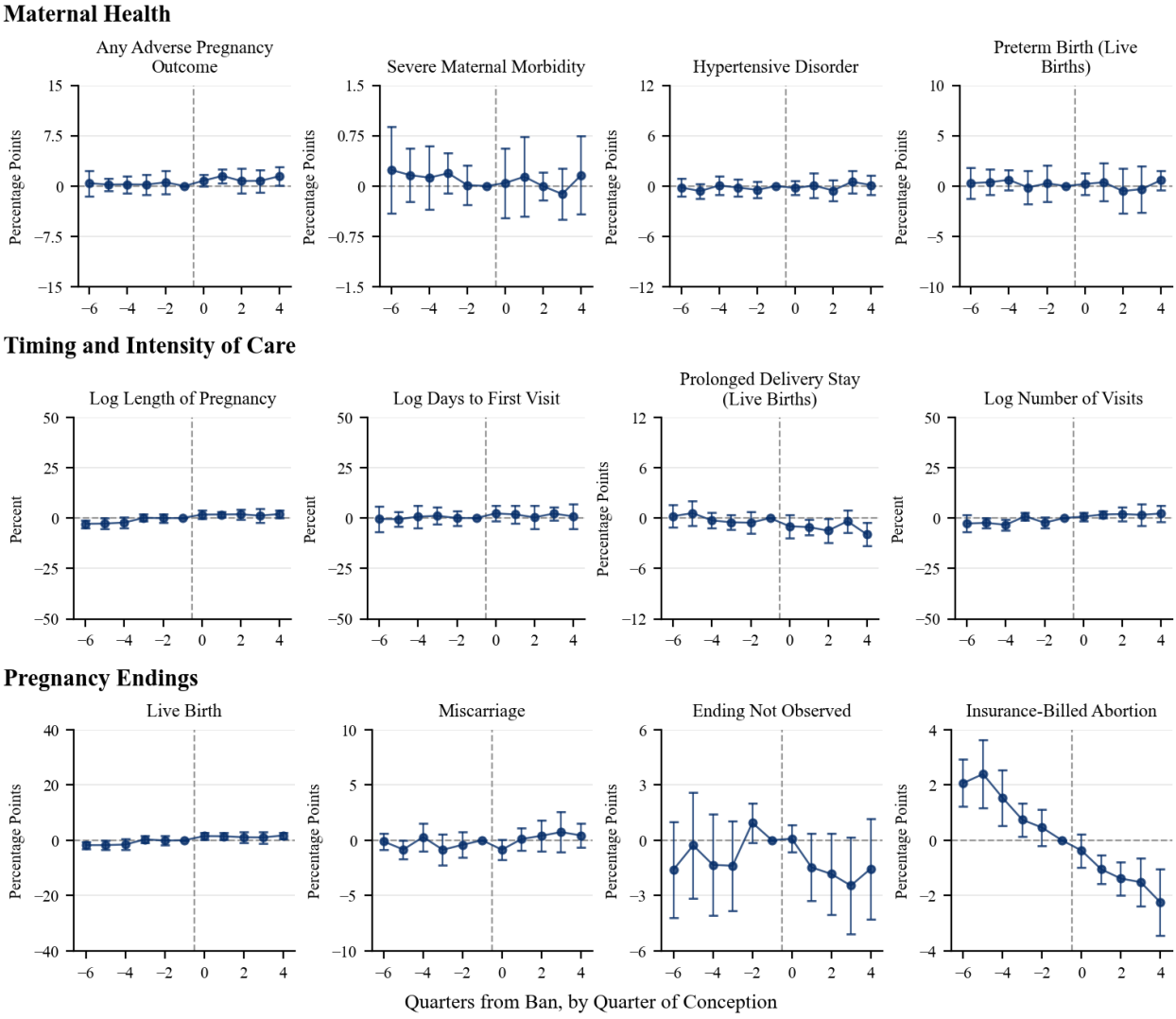
Notes: Stacked difference-in-differences estimates in the Controls specification and the continuous-enrollment sample, laid out as Table 2 without its pre-trend test, with standard errors (SE), clustered by state, in parentheses. Columns 2 to 4 compare the post-ban quarters with quarter −1 alone, as in Wing, Freedman and Hollingsworth (2024), and columns 5 and 6 with quarter −2. Columns 7 and 8 repeat columns 2 and 3 of Table 2, which compare them with the average of quarters −6 to −1. The post-ban quarters are the same in all three.

Figure A4. Event Studies, Stacked and Two-Way Fixed Effects



Notes: Stacked difference-in-differences event-study coefficients from Figure 2 (filled markers) and two-way fixed effects (TWFE) coefficients for the same outcome (hollow markers), in the Controls specification and the continuous-enrollment sample, with 95 percent confidence intervals from standard errors clustered by state. The TWFE regression uses the pregnancies of the stacked design, with state and quarter-of-conception fixed effects and the Controls covariates, each pregnancy entering once and unweighted. In both, the pre-ban coefficients average zero, and the dashed vertical line falls between quarters −1 and 0. Vertical axes are drawn as in Figure 2.

Figure A5. Event Studies with Quarter −1 as the Reference



Notes: Stacked difference-in-differences event-study coefficients as in Figure 2, with quarter −1 omitted as the reference period (Wing, Freedman and Hollingsworth, 2024), so its coefficient is zero. The dashed vertical line falls between quarters −1 and 0, and vertical axes are drawn as in Figure 2.

Table A16—Stacked and Synthetic Difference-in-Differences Estimates

Outcome	Pre-Ban Mean (1)	Synthetic DiD, Controls: Estimate (SE) (2)	Synthetic DiD, Controls: Percent of Mean (3)	Synthetic DiD, Controls: 95% CI, Percent of Mean (4)	Synthetic DiD, Raw: Estimate (SE) (5)	Synthetic DiD, Raw: Percent of Mean (6)	Stacked DiD, Controls, Percent of Mean (7)
Maternal Health							
Any adverse pregnancy outcome	0.285	0.006 (0.014)	2.0	[−7.7, 11.7]	0.007 (0.014)	2.4	2.8
Severe maternal morbidity	0.026	−0.001 (0.004)	−4.3	[−32.6, 24.0]	−0.001 (0.004)	−4.1	−2.9
Hypertensive disorder	0.208	0.001 (0.011)	0.3	[−9.9, 10.6]	0.002 (0.011)	1.0	1.0
Preterm birth (live births)	0.171	−0.012 (0.013)	−7.1	[−21.5, 7.4]	−0.008 (0.011)	−4.9	−0.9
Timing and Intensity of Care							
Log length of pregnancy	5.272	0.009 (0.019)	0.9	[−2.9, 4.7]	0.011 (0.020)	1.1	3.0
Log days to first visit	3.616	0.011 (0.026)	1.1	[−4.0, 6.2]	0.011 (0.026)	1.1	1.4
Prolonged delivery stay (live births)	0.233	−0.019 (0.022)	−8.0	[−26.5, 10.6]	−0.017 (0.023)	−7.2	−4.6
Log number of visits	2.496	0.011 (0.034)	1.1	[−5.5, 7.6]	0.002 (0.032)	0.2	3.3
Pregnancy Endings							
Live birth	0.789	0.005 (0.013)	0.6	[−2.5, 3.7]	0.010 (0.015)	1.2	2.7
Miscarriage	0.162	0.021 (0.013)	13.1	[−2.4, 28.7]	0.017 (0.012)	10.2	3.0
Ending not observed	0.062	−0.009 (0.018)	−13.9	[−70.0, 42.1]	−0.008 (0.018)	−12.5	−13.3
Insurance-billed abortion	0.011	−0.023 (0.009)	−213.9	[−382.4, −45.4]	−0.023 (0.009)	−217.4	−237.1

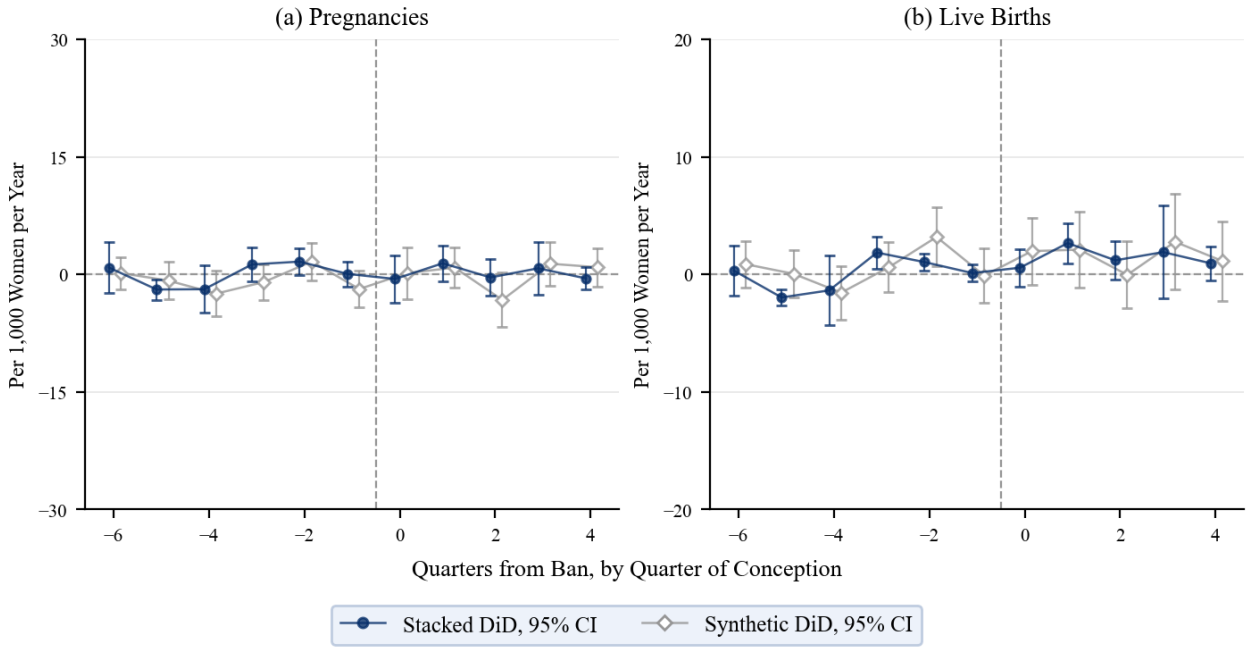
Notes: Column 1 is the ban-state mean in event quarters −6 to −1 of the stacked design, which scales every percent (100 times the coefficient for outcomes in logs). Columns 2 to 6 are synthetic difference-in-differences (DiD) estimates (Arkhangelsky et al., 2021) on state by quarter-of-conception means in the continuous-enrollment sample, conceptions from the first quarter of 2020 through the fourth quarter of 2023, excluding Indiana and South Carolina. Columns 2 to 4 average the outcome net of the Controls covariates, fitted on pregnancies with state and quarter fixed effects, and columns 5 and 6 the outcome itself. Standard errors (SE) come from 200 placebo replications (seed 20261013), and the 95 percent confidence interval (CI) in column 4 uses the normal critical value. Synthetic DiD has no pre-trend test. Column 7 repeats column 3 of Table 2.

Table A17—Fertility Estimates by Estimator

Group	Pre-Ban Mean (1)	Stacked, Pre-Ban Average: Percent of Mean (2)	Stacked, Pre-Ban Average: 95% CI, Percent of Mean (3)	Stacked, Quarter −1: Percent of Mean (4)	Stacked, Quarter −1: 95% CI, Percent of Mean (5)	Synthetic DiD: Percent of Mean (6)	Synthetic DiD: 95% CI, Percent of Mean (7)
Pregnancy Rate							
All	52.09	0.2	[−2.2, 2.6]	0.1	[−4.2, 4.4]	1.9	[−3.7, 7.5]
Medicaid	67.82	0.2	[−2.8, 3.1]	1.2	[−2.7, 5.1]	−6.0	[−30.7, 18.8]
Minority	63.12	−3.6	[−5.6, −1.5]	−2.2	[−6.4, 2.1]	−18.3	[−25.8, −10.8]
High risk[a]	53.17	1.6	[−0.9, 4.1]	−0.7	[−4.8, 3.5]	4.4	[−2.8, 11.5]
Under 25	41.06	2.8	[−5.0, 10.6]	2.5	[−1.6, 6.5]	3.4	[−6.9, 13.8]
Excluding Texas	52.30	0.8	[−1.7, 3.2]	1.7	[−0.7, 4.1]	1.9	[−3.5, 7.3]
Live-Birth Rate							
All	38.50	3.7	[0.7, 6.7]	3.4	[−0.5, 7.3]	6.2	[−2.8, 15.2]
Medicaid	49.04	4.7	[0.4, 8.9]	4.8	[−1.1, 10.8]	1.9	[−36.1, 39.9]
Minority	43.73	2.0	[−2.0, 6.0]	2.4	[−2.7, 7.4]	−10.3	[−27.4, 6.8]
High risk[a]	38.11	5.5	[2.2, 8.8]	3.4	[−0.4, 7.3]	4.8	[−5.7, 15.4]
Under 25	31.57	6.7	[−1.0, 14.4]	5.4	[0.0, 10.7]	4.5	[−7.2, 16.3]
Excluding Texas	38.62	4.0	[0.2, 7.9]	3.3	[−1.5, 8.1]	5.9	[−3.5, 15.3]

Notes: Pregnancy and live-birth rates per 1,000 enrolled women aged 15 to 45 per year in the continuous-enrollment sample, as in Table A12. Column 1 is the mean in treated states before the ban in the stacked design, which scales every percent and 95 percent confidence interval (CI). Columns 2 and 3 repeat columns 3 and 4 of Table A12, which compare the post-ban quarters with the average of quarters −6 to −1, and columns 4 and 5 compare them with quarter −1 alone (Wing, Freedman and Hollingsworth, 2024). Columns 6 and 7 are synthetic difference-in-differences (DiD) estimates (Arkhangelsky et al., 2021) on state by quarter rates, each state unweighted, from the first quarter of 2020 through the fourth quarter of 2023, excluding Indiana and South Carolina, with standard errors from 200 placebo replications and the normal critical value. [a]High-risk women have any condition of the obstetric comorbidity index in the preceding 12 months.

Figure A6. Fertility Event Studies, Stacked and Synthetic Difference-in-Differences



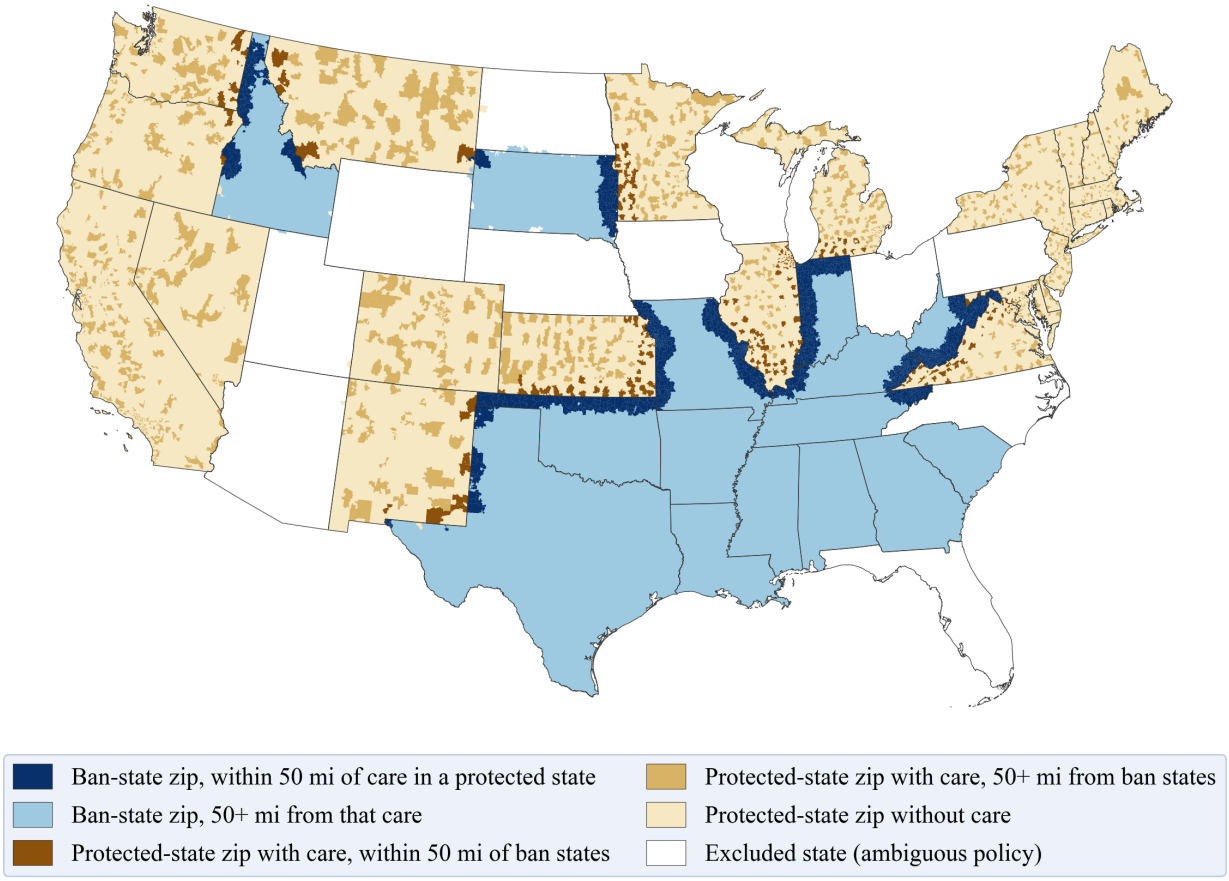
Notes: Pregnancies and live births per 1,000 enrolled women aged 15 to 45 per year, all women. Filled markers repeat panels (a) and (b) of Figure 4, whose pre-ban coefficients average zero. Hollow markers are synthetic difference-in-differences coefficients on the state by quarter rates of Table A17: in each quarter, treated states less their synthetic control, less the pre-ban gap under the synthetic time weights, averaged over ban cohorts by treated states, so their pre-ban coefficients need not average zero. Their 95 percent confidence intervals use standard errors from 200 bootstrap draws of states, while the pooled estimates in Table A17 use placebo standard errors. The dashed vertical line falls between quarters −1 and 0, and vertical axes are drawn as in Figure 4.

Table A18—Zip-Code Placement Diagnostics

	Within 50 Miles	**Beyond 50 Miles, by Ban Cohort**		
	of Protected Care	**Texas**	***Dobbs* Cohort**	**Late Adopters**
Visits in the year before pregnancy	21.09	17.26	22.11	20.01
in the modal zip code	7.20	5.94	7.03	6.88
in the second most common zip code	3.04	2.64	3.01	2.91
in the modal three-digit zip area	9.02	8.10	9.42	9.11
with no provider zip code	7.54	5.81	8.42	7.11
Unique zip codes visited	4.41	4.21	4.66	4.25
Share of visits with a provider zip in the modal zip code	**0.53**	**0.52**	**0.51**	**0.53**
in the second most common zip code	0.22	0.23	0.22	0.23
in the modal three-digit zip area	0.67	0.71	0.69	0.71
Pregnancies	28,285	95,685	94,265	17,070

Notes: Means across women in the year before their first observed pregnancy, in the continuous-enrollment sample. Column 1 is the close-to-care sample, pregnancies whose assigned ban-state zip code lies within 50 miles of protected-state emergency care (Section A9); the other columns are assigned zip codes farther away, by ban cohort, with the late adopters West Virginia, Indiana, and South Carolina. A visit is located by the NPPES practice zip code of its rendering NPI, and the modal three-digit zip area counts visits to any zip code with the same first three digits.

Figure A7. Close-to-Care Classification of Lower-48 Zip Codes



Notes: Each polygon is a 2020 Census ZIP Code Tabulation Area (ZCTA), with states classified as in Table A1 and excluded states in white. In ban states, ZCTAs within 50 miles of a ZCTA with observed emergency care (a general acute care hospital or emergency medicine provider) in a protected state are navy and the rest light blue, the close-to-care classification, drawn here from each ZCTA's interior point rather than the centroid the estimation uses. In protected states, ZCTAs with observed care within 50 miles of a ban-state ZCTA are dark brown, those with care farther away are tan, and those without observed care are cream. Distances use the Conus Albers equal-area projection (EPSG 5070). As the data-use terms require, the map shows every eligible zip code, not the zip codes where pregnancies are observed.

Table A19—Close-to-Care Sample Characteristics

	Beyond 50 Miles		Within 50 Miles	
	Before	**After**	**Before**	**After**
Pregnancies	107,453	99,567	18,678	9,607
Demographics, % Unless Noted				
Age, mean (SD)	29.10 (6.40)	28.99 (6.83)	28.30 (6.41)	27.85 (6.71)
Under 25	26.1	30.0	31.4	36.6
Medicaid	42.2	36.1	53.6	55.2
Commercial	57.7	63.8	46.3	44.8
White	41.6	41.1	52.6	57.6
Black	20.0	16.5	14.9	11.6
Asian	1.9	1.9	0.9	0.8
Hispanic	9.3	14.4	9.3	12.8
First observed pregnancy	25.7	26.8	25.5	29.5
Health Before Pregnancy, % Unless Noted				
Hypertension, obesity, or substance use	25.8	27.4	28.0	33.2
Obstetric comorbidity score, mean (SD)	9.49 (14.68)	9.84 (15.14)	10.63 (15.68)	11.85 (16.42)
Maternal Health, %				
Any adverse pregnancy outcome	28.7	29.2	28.4	28.8
Adjusted for characteristics	[28.9]	[29.5]	[28.4]	[28.4]
Severe maternal morbidity	2.8	2.6	2.9	3.0
Adjusted for characteristics	[2.8]	[2.6]	[2.8]	[2.8]
Hypertensive disorder	21.6	22.1	21.1	23.6
Adjusted for characteristics	[21.3]	[22.1]	[20.8]	[22.9]
Preterm birth (live births)	17.7	17.6	17.7	19.2
Adjusted for characteristics	[17.8]	[17.9]	[17.6]	[19.0]
Timing and Intensity of Care, % Unless Noted				
Length of pregnancy, days, mean (SD)	223.9 (84.2)	223.2 (84.8)	221.4 (86.2)	222.7 (84.9)
Adjusted for characteristics	[225.1]	[224.9]	[221.4]	[221.1]
Days to first visit, mean (SD)	43.2 (29.4)	43.0 (29.1)	44.1 (32.0)	42.9 (30.0)
Adjusted for characteristics	[44.6]	[44.4]	[44.9]	[43.8]
Prolonged delivery stay (live births)	23.5	24.1	23.2	23.5
Adjusted for characteristics	[23.8]	[24.0]	[23.5]	[23.0]
Number of visits, mean (SD)	16.6 (11.0)	16.7 (10.9)	17.1 (11.6)	18.0 (11.8)
Adjusted for characteristics	[16.6]	[16.8]	[16.8]	[17.3]
Pregnancy Endings, % of Observed Endings Unless Noted				
Live birth	78.8	78.5	77.5	78.1
Adjusted for characteristics	[79.5]	[79.4]	[77.4]	[77.3]
Miscarriage	16.0	16.6	15.5	15.4
Adjusted for characteristics	[15.3]	[15.5]	[15.3]	[15.8]
Ending not observed (percent of all pregnancies)	6.2	5.6	6.7	6.9
Adjusted for characteristics	[6.3]	[6.1]	[6.8]	[7.1]
Insurance-billed abortion	1.3	1.1	3.5	2.7
Adjusted for characteristics	[1.3]	[1.2]	[3.6]	[2.9]

Notes: Pregnancies in ban-state zip codes in the continuous-enrollment sample, laid out as Table 1. Beyond 50 miles are assigned zip codes farther than 50 miles from protected-state care (the treated group in Table A20); within 50 miles are the close-to-care zip codes. Before and after are relative to the ban in each pregnancy's state.

Table A20—Close-to-Care Design Estimates

Outcome	Pre-Ban Mean (1)	Controls		
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)
Maternal Health				
Any adverse pregnancy outcome	0.285	−0.009 (0.011)	−3.0	[−10.4, 4.4]
Severe maternal morbidity	0.027	−0.007 (0.004)	−26.6	[−56.9, 3.6]
Hypertensive disorder	0.210	−0.007 (0.010)	−3.2	[−12.8, 6.5]
Preterm birth (live births)	0.174	−0.016 (0.012)	−9.0	[−22.1, 4.0]
Timing and Intensity of Care				
Log length of pregnancy	5.265	−0.009 (0.015)	−0.9	[−3.8, 2.0]
Log days to first visit	3.573	−0.025 (0.035)	−2.5	[−9.3, 4.3]
Prolonged delivery stay (live births)	0.230	0.016 (0.011)	6.8	[−2.8, 16.5]
Log number of visits	2.522	0.008 (0.025)	0.8	[−4.0, 5.6]
Pregnancy Endings				
Live birth	0.785	−0.003 (0.009)	−0.4	[−2.7, 1.9]
Miscarriage	0.165	0.007 (0.008)	4.1	[−5.2, 13.3]
Ending not observed	0.062	−0.020 (0.009)	−32.4	[−59.5, −5.3]
Insurance-billed abortion	0.012	−0.007 (0.005)	−56.9	[−141.4, 27.5]

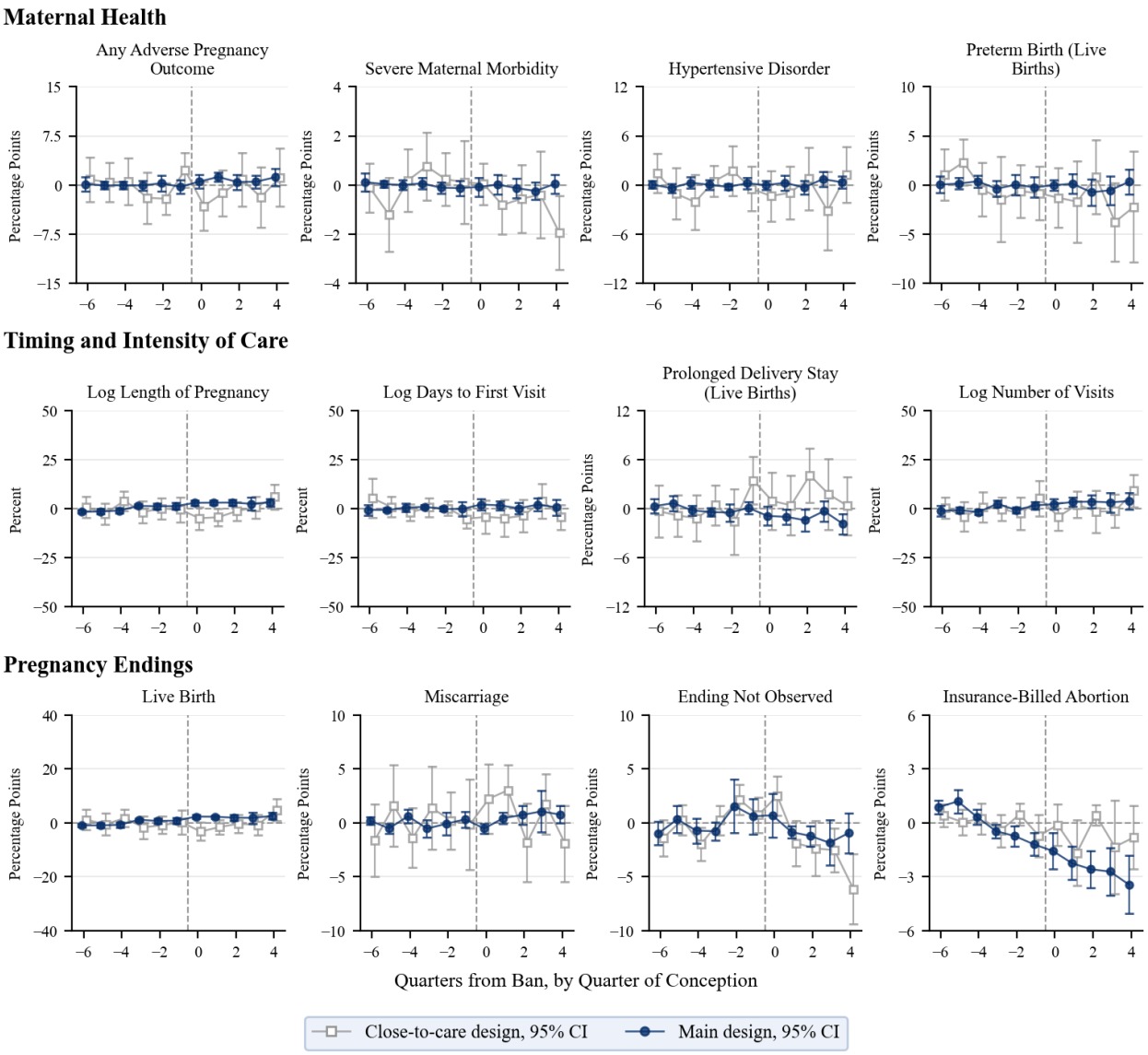
Notes: Pregnancies in ban-state zip codes in the continuous-enrollment sample. Treated pregnancies are in assigned zip codes farther than 50 miles from protected-state emergency care (Section A9), and comparison pregnancies are within 50 miles. Column 1 is the treated mean in event quarters −6 to −1 in each outcome's units. Column 2 is the stacked difference-in-differences (DiD) estimate of the post-ban quarters against the average of quarters −6 to −1, comparing each cohort's treated zip codes only with close-to-care zip codes in states banned on the same date, with the Controls covariates, the weights of Wing, Freedman and Hollingsworth (2024), and standard errors (SE), clustered by assigned zip code, in parentheses. Column 3 is the estimate as a percent of the pre-ban mean (100 times the coefficient for outcomes in logs), and column 4 its 95 percent confidence interval (CI). The estimates without covariates were run only against quarter −1 (Table A21).

Table A21—Close-to-Care Design Estimates against Quarter −1

Outcome	Pre-Ban Mean (1)	Controls			Raw	
		Estimate (SE) (2)	Percent of Mean (3)	95% CI, Percent of Mean (4)	Estimate (SE) (5)	Percent of Mean (6)
Maternal Health						
Any adverse pregnancy outcome	0.285	−0.032 (0.019)	−11.1	[−24.4, 2.3]	−0.036 (0.020)	−12.5
Severe maternal morbidity	0.027	−0.008 (0.008)	−30.8	[−86.4, 24.8]	−0.009 (0.008)	−33.2
Hypertensive disorder	0.210	−0.002 (0.021)	−1.1	[−20.5, 18.3]	−0.005 (0.022)	−2.3
Preterm birth (live births)	0.174	−0.007 (0.018)	−4.2	[−24.7, 16.2]	−0.005 (0.019)	−3.0
Timing and Intensity of Care						
Log length of pregnancy	5.265	−0.004 (0.027)	−0.4	[−5.6, 4.8]	−0.007 (0.031)	−0.7
Log days to first visit	3.573	0.027 (0.042)	2.7	[−5.5, 10.8]	0.022 (0.041)	2.2
Prolonged delivery stay (live births)	0.230	−0.019 (0.017)	−8.2	[−22.6, 6.1]	−0.018 (0.018)	−7.7
Log number of visits	2.522	−0.045 (0.051)	−4.5	[−14.5, 5.6]	−0.053 (0.051)	−5.3
Pregnancy Endings						
Live birth	0.785	−0.006 (0.019)	−0.7	[−5.6, 4.1]	−0.008 (0.022)	−1.0
Miscarriage	0.165	0.009 (0.024)	5.2	[−23.7, 34.1]	0.010 (0.027)	6.1
Ending not observed	0.062	−0.027 (0.012)	−44.3	[−81.5, −7.1]	−0.028 (0.012)	−44.6
Insurance-billed abortion	0.012	0.000 (0.008)	2.9	[−124.2, 130.0]	0.001 (0.008)	4.8

Notes: Laid out as Table A20, with each estimate comparing the post-ban quarters with quarter −1 alone, as in Wing, Freedman and Hollingsworth (2024). Columns 5 and 6 drop the covariates.

Figure A8. Event Studies, Close-to-Care and Main Designs



Notes: Stacked event-study coefficients of the close-to-care design (hollow markers), comparing ban-state zip codes farther than 50 miles from protected-state care with close-to-care zip codes in states whose ban took effect on the same date, beside the main design's coefficients from Figure 2 (filled markers), in the Controls specification and the continuous-enrollment sample. The 95 percent confidence intervals use standard errors clustered by assigned zip code in the close-to-care design and by state in the main design. The pre-ban coefficients average zero, and the dashed vertical line falls between quarters −1 and 0. Vertical axes are drawn as in Figure 2.

Table A22—Outcome Construction

Outcome	Definition	Codes	Form
Pregnancy Endings			
Live birth	Ending type after the validated hierarchy (live birth, stillbirth, molar, ectopic, induced abortion, miscarriage), among observed endings.	CM Z37.0, Z37.2, Z37.3, Z37.5, Z37.6, O80, O82; PCS 10D00Z0–10D00Z2, 10D07Z3–10D07Z8, 10E0XZZ; CPT 59400–59622 (delivery); DRG 765–768, 774, 775, 783–788, 796–798, 805–807	Indicator
Miscarriage	Ending type after the same hierarchy, among observed endings.	CM O02.0, O02.1, O02.8, O02.9, O03; CPT 59812, 59820, 59821; PCS 10D17Z; DRG 770, 779	Indicator
Ending not observed	Two or more dates with prenatal, adverse-outcome, or Z3A codes, no gap over 60 days, and no ending code.	CM Z34, Z36, O09, Z3A; CPT 59425, 59426, 76801–76817, 59000–59015; adverse-outcome codes below	Indicator
Insurance-billed abortion	Ending type after the same hierarchy, among observed endings.	CM Z33.2, O04, O07; CPT 59840, 59841, 59850–59852, 59855–59857; PCS 10A0; HCPCS S0190, S0191, S0199, S2260, S2265–S2267	Indicator
Maternal Health (window: pregnancy start through 42 days after the ending for SMM; pregnancy otherwise)			
Any adverse pregnancy outcome	Any of infection or sepsis, preterm birth, gestational diabetes, poor fetal growth, or placental abruption. Preterm birth: live births under 259 days from start to delivery if dated by Z3A codes or embryo transfer, otherwise a preterm-delivery diagnosis.	Infection CM A40, A41, A32.7, R65.20, O85, O86, O41.12, T80.211A, T81.4XXA, T81.44; preterm dating CM Z3A, CPT 58974, 58976, 89258, diagnosis CM O60.1, O60.2; gestational diabetes CM O24.4; fetal growth restriction CM O36.5; abruption CM O45	Indicator
Severe maternal morbidity	Any of the 16 CDC diagnosis indicators on any claim; the five procedure indicators (transfusion, hysterectomy, ventilation, temporary tracheostomy, conversion of cardiac rhythm) are not used.	Leonard et al. (2020, Appendix 1)	Indicator
Hypertensive disorder	Diagnosis.	CM O10–O16	Indicator
Preterm birth	Live births only: under 259 days from start to delivery if dated by Z3A codes or embryo transfer, otherwise a preterm-delivery diagnosis.	CM Z3A; CPT 58974, 58976, 89258; CM O60.1, O60.2	Indicator
Timing and Intensity of Care (window: pregnancy)			
Log length of pregnancy	Pregnancy start to ending, among observed endings of 28 to 315 days.	Start dated from CM Z3A, embryo transfer, or screening tests	ln
Log days to first visit	Pregnancy start to the first claim of any kind, among observed endings; values under 10 days set to missing.	All claims in the window	Wins. p99; ln
Prolonged delivery stay	Live birth whose delivery admission lasts more than 2 days after a vaginal delivery or more than 4 after a cesarean, the federal minimum stays (42 U.S.C. 300gg-25)	Delivery admission, delivery-mode codes	Live births
Log number of visits	Days with any claim.	All claims in the window.	Wins. p99; ln

Table A22—Outcome Construction (Continued)

Outcome	Definition	Codes	Form
Controls and Subgroups (window: 365 days before the pregnancy start unless noted)			
Obstetric comorbidity score	Leonard conditions present before pregnancy, weighted by the severe maternal morbidity points of Leonard et al. (2020): pulmonary hypertension 50; chronic renal disease 38; bleeding disorder 34; cardiac disease 31; HIV 30; anemia 20; gastrointestinal disease 12; asthma 11; substance use, autoimmune disease, chronic hypertension 10; pre-existing diabetes, neuromuscular disease 9; mental health 7; thyrotoxicosis 6; BMI 40 or more 5; prior cesarean 4; bariatric surgery 0. High risk: any of these conditions.	Leonard et al. (2020) list, 18 conditions, e.g. chronic hypertension CM I10, O10, O11; pre-existing diabetes CM E08–E13, O24.0, O24.1, O24.3, O24.8, O24.9, Z79.4	Score; indicator
Under 25; Medicaid	Enrollment-file age under 25; Medicaid payer group, each in the enrollment month nearest the pregnancy start.	Enrollment file	Indicator
Minority	A Black race code or a Hispanic ethnicity code, each counted once.	Enrollment file	Indicator

Notes: CM denotes ICD-10-CM diagnosis codes and PCS ICD-10-PCS procedure codes; CPT, Current Procedural Terminology; HCPCS, Healthcare Common Procedure Coding System Level II; DRG, Medicare Severity Diagnosis Related Group. Codes match by prefix, so a three-character code covers every code that begins with it. Pregnancy and ending codes draw on the Maternal and Infant Health code lists of Centers for Medicare & Medicaid Services (2023), the mapping of Ailes et al. (2016), and the Z3A codes of American Academy of Professional Coders (2024).

REFERENCES

Ailes, Elizabeth C., Regina M. Simeone, April L. Dawson, Emily E. Petersen, and Suzanne M. Gilboa. 2016. "Using Insurance Claims Data to Identify and Estimate Critical Periods in Pregnancy: An Application to Antidepressants." *Birth Defects Research Part A: Clinical and Molecular Teratology*, 106(11): 927–934.

American Academy of Professional Coders. 2024. "ICD-10 Code for Weeks of Gestation." *Codify by AAPC.* https://www.aapc.com/codes/icd-10-codes/Z3A.

American College of Obstetricians and Gynecologists. 2017. "Committee Opinion No. 700: Methods for Estimating the Due Date." *Obstetrics and Gynecology*, 129(5): e150–e154.

American College of Obstetricians and Gynecologists. 2018*a*. "ACOG Practice Bulletin No. 193: Tubal Ectopic Pregnancy." *Obstetrics and Gynecology*, 131(3): e91–e103.

American College of Obstetricians and Gynecologists. 2018*b*. "ACOG Practice Bulletin No. 200: Early Pregnancy Loss." *Obstetrics and Gynecology*, 132(5): e197–e207.

Arkhangelsky, Dmitry, Susan Athey, David A. Hirshberg, Guido W. Imbens, and Stefan Wager. 2021. "Synthetic Difference-in-Differences." *American Economic Review*, 111(12): 4088–4118.

Bell, Suzanne O., Selena Anjur-Dietrich, and Alison Gemmill. 2026. "Birth Certificate Measures of Maternal Morbidity and Their Limits in Abortion Policy Research." *American Journal of Epidemiology*, kwag159.

Brown, Andrea D., Brady E. Hamilton, Dmitry M. Kissin, and Joyce A. Martin. 2025. "Trends in Mean Age of Mothers: United States, 2016–2023." *National Vital Statistics Reports*, 74(9).

Centers for Medicare & Medicaid Services. 2023. "Identifying Pregnant and Postpartum Beneficiaries in Medicaid and CHIP Administrative Data: Technical Specifications." *Maternal and Infant Health (MIH) Technical Specifications and Reference Codes, Center for Medicaid and CHIP Services.* https://www.medicaid.gov/medicaid/data-and-systems/downloads/macbis/mih_techspecs.pdf.

Chomistek, Andrea K., Kelesitse Phiri, Michael C. Doherty, Jenna F. Calderbank, Stephanie E. Chiuve, Brenda Hinman McIlroy, Michael C. Snabes, Cheryl Enger, and John D. Seeger. 2023. "Development and Validation of ICD-10-CM-Based Algorithms for Date of Last Menstrual Period, Pregnancy Outcomes, and Infant Outcomes." *Drug Safety*, 46(2): 209–222.

Conlon, Rose. 2024. "Kansas Provided More Abortions in 2023 than Ever Before, Mostly for Out-of-State Patients." *KCUR.* https://www.kcur.org/2024-12-13/kansas-provided-more-abortions-in-2023-than-ever-before-mostly-for-out-of-state-patients.

Curhan, Talia. 2026. "State Bans on Abortion Throughout Pregnancy." *Guttmacher Institute.* https://www.guttmacher.org/state-policy/explore/state-policies-abortion-bans.

Dench, Daniel, Mayra Pineda-Torres, and Caitlin Myers. 2024. "The Effects of Post-Dobbs Abortion Bans on Fertility." *Journal of Public Economics*, 234: 105124.

Dobbs v. Jackson Women's Health Organization. 2022. *597 U.S. 215 (Supreme Court of the United States).*

Goldhill, Olivia. 2023. "Kate Cox Is One of Hundreds in Texas Denied Abortions despite Serious Health Risks, Data Show." *STAT, December 15.* https://www.statnews.com/2023/12/15/abortion-kate-cox-texas-health-risks-trisomy-18/.

Hughes, Bryan. 2021. "Texas Heartbeat Act." *Texas Senate Bill 8, 87th Legislature, Regular Session.*

Hughes, Bryan. 2024. "I Wrote Texas' Abortion Law. It's Clear about Medical Emergencies." *Houston Chronicle.*

https://www.houstonchronicle.com/opinion/outlook/article/texas-abortion-medical-emergencies-19900172.php.

Kentucky Senate GOP. 2025. "Senate Advances Clarifying Language in House Bill 90 to Protect Women, Doctors." https://kysenaterepublicans.com/press-releases/2025/3/26/senate-advances-clarifying-language-in-house-bill-90-to-protect-women-doctors.

Klibanoff, Eleanor. 2023. "Kate Cox's Case Reveals How Far Texas Intends to Go to Enforce Abortion Laws." *The Texas Tribune.* https://www.texastribune.org/2023/12/13/texas-abortion-lawsuit/.

Kortsmit, Katherine, Antoinette T. Nguyen, Michele G. Mandel, Lisa M. Hollier, Stephanie Ramer, Jessica Rodenhizer, and Maura K. Whiteman. 2023. "Abortion Surveillance: United States, 2021." *MMWR Surveillance Summaries*, 72(9): 1–29.

Leonard, Stephanie A., Chris J. Kennedy, Suzan L. Carmichael, Deirdre J. Lyell, and Elliott K. Main. 2020. "An Expanded Obstetric Comorbidity Scoring System for Predicting Severe Maternal Morbidity." *Obstetrics and Gynecology*, 136(3): 440–449.

Main, Elliott K., Anisha Abreo, Jennifer McNulty, William Gilbert, Colleen McNally, Debra Poeltler, Katarina Lanner-Cusin, Douglas Fenton, Theresa Gipps, Kathryn Melsop, Naomi Greene, Jeffrey B. Gould, and Sarah Kilpatrick. 2016. "Measuring Severe Maternal Morbidity: Validation of Potential Measures." *American Journal of Obstetrics and Gynecology*, 214(5): 643.e1–643.e10.

Matcho, Amy, Patrick Ryan, Daniel Fife, Dina Gifkins, Chris Knoll, and Andrew Friedman. 2018. "Inferring Pregnancy Episodes and Outcomes within a Network of Observational Databases." *PLOS ONE*, 13(2): e0192033.

Moll, Keran, Hui Lee Wong, Kathryn Fingar, Shayan Hobbi, Minya Sheng, Timothy A. Burrell, Linda O. Eckert, Flor M. Munoz, Bethany Baer, Azadeh Shoaibi, and Steven Anderson. 2021. "Validating Claims-Based Algorithms Determining Pregnancy Outcomes and Gestational Age Using a Linked Claims-Electronic Medical Record Database." *Drug Safety*, 44(11): 1151–1164.

Nash, Elizabeth, and Isabel Guarnieri. 2022. "13 States Have Abortion Trigger Bans: Here's What Happens When Roe Is Overturned." *Guttmacher Institute.* https://www.guttmacher.org/article/2022/06/13-states-have-abortion-trigger-bans-heres-what-happens-when-roe-overturned.

Office of Governor Gavin Newsom. 2023. "California Expands Access and Protections for Reproductive Health Care." *Press release, Office of the Governor of California, September 27.* https://www.gov.ca.gov/2023/09/27/california-expands-access-and-protections-for-reproductive-health-care/.

Office of Governor JB Pritzker. 2023. "Gov. Pritzker Signs Sweeping Reproductive Rights Protections Into Law." *Press release, State of Illinois.* https://www.illinois.gov/news/release.html?releaseid=25906.

Osterman, Michelle J. K., Brady E. Hamilton, Joyce A. Martin, Anne K. Driscoll, and Claudia P. Valenzuela. 2025. "Births: Final Data for 2023." National Center for Health Statistics National Vital Statistics Reports 74(1).

Presser, Lizzie, Andrea Suozzo, Sophie Chou, and Kavitha Surana. 2025. "Texas Banned Abortion. Then Sepsis Rates Soared." *ProPublica, February 20.* https://www.propublica.org/article/texas-abortion-ban-sepsis-maternal-mortality-analysis.

Robinson v. Marshall. 2022. *Order dissolving the preliminary injunction against Alabama HB 314, No. 2:19-cv-365-MHT (M.D. Ala. June 24, 2022).*

Roe v. Wade. 1973. *410 U.S. 113 (Supreme Court of the United States).*

Sarayani, Amir, Xi Wang, Thuy Nhu Thai, Yasser Albogami, Nakyung Jeon, and Almut G. Winterstein. 2020. "Impact of the Transition from ICD-9-CM to ICD-10-CM on the Identification of Pregnancy Episodes in US Health Insurance Claims Data." *Clinical Epidemiology*, 12: 1129–1138.

SisterSong v. Governor of Georgia. 2022. *SisterSong Women of Color Reproductive Justice Collective v. Governor of the State of Georgia, 40 F.4th 1320 (11th Cir. 2022).*

Suozzo, Andrea, Sophie Chou, and Lizzie Presser. 2025. "Texas Won't Study How Its Abortion Ban Impacts Women, So We Did." *ProPublica methodology note.* https://www.propublica.org/article/texas-maternal-mortality-analysis-methodology.

Surana, Kavitha, and Cassandra Jaramillo. 2024. "A Texas Woman Died After Waiting 40 Hours for Miscarriage Care." *ProPublica.* https://www.propublica.org/article/josseli-barnica-death-miscarriage-texas-abortion-ban.

Surana, Kavitha, and Lizzie Presser. 2024. "A Pregnant Teenager Died After Trying to Get Care in Three Visits to Texas Emergency Rooms." *ProPublica.* https://www.propublica.org/article/nevaeh-crain-death-texas-abortion-ban-emtala.

Texas Legislature. 2021. "House Bill 1280, Human Life Protection Act of 2021." *87th Legislature, Regular Session.*

Texas State Law Library. 2026. "Timeline of Texas Abortion Laws." *Texas State Law Library research guide.* https://guides.sll.texas.gov/abortion-laws/history-of-abortion-laws.

Tolbert, Jennifer, and Meghana Ammula. 2023. "10 Things to Know About the Unwinding of the Medicaid Continuous Enrollment Provision." *KFF.* https://www.kff.org/medicaid/10-things-to-know-about-the-unwinding-of-the-medicaid-continuous-enrollment-provision/.

Whole Woman's Health v. Jackson. 2021. *595 U.S. 30 (Supreme Court of the United States).*

Wing, Coady, Seth M. Freedman, and Alex Hollingsworth. 2024. "Stacked Difference-in-Differences." National Bureau of Economic Research Working Paper 32054.